

**DRAFT PERFORMANCE WORK STATEMENT (PWS)
FOR
REAGAN TEST SITE (RTS)
OPERATIONS, SUSTAINMENT, IMPROVEMENT,
and ENGINEERING (ROSIE)**

**24 November 2025
(REV00)**

Prepared by:
U.S. Army Space and Missile Defense Command (USASMDC)
Redstone Arsenal, AL

Table of Contents

This Page Left Blank Intentionally	10
1. SCOPE.....	11
2. BACKGROUND.....	12
3. PERFORMANCE REQUIREMENTS	15
3.1 Test Mission Support	15
3.1.1 Program Planning Support	15
3.1.2 Customer Mission Estimating	16
3.1.3 Mission Test Planning.....	17
3.1.4 Instrumentation/Systems Operations	18
3.1.5 Post Mission Support.....	20
3.1.6 Mission Operations Support Functions	21
3.2 Sustainment / Maintenance (Standard Technical Services).....	22
3.2.1 Space Operations	23
3.2.1.1 Huntsville Space Surveillance Center	Error! Bookmark not defined.
3.2.2 Radar Extended Maintenance Periods.....	30
3.2.3 Optical Sensor	30
3.2.4 Telemetry	31
3.2.5 Command and Control Center	31
3.2.6 Range Safety System	31
3.2.7 U.S. Army Vessel Worthy	31
3.2.8 Kwajalein Missile Impact Scoring System Maintenance	31
3.2.9 Ground Release of Calibration Spheres	32
3.2.10 Update Instrumentation and Support Facilities Manual	32
3.2.11 Mean Time Between Failures and To Repair Record Maintenance	32
3.2.12 Sensor and Network Availability Report	32
3.2.13 Geodetic Coordinate Surveys and Coordinates Manual	32
3.2.14 Instrumentation Status and Weekly Activity Reports.....	33
3.2.15 Wake Island and Off Atoll Missions	33
3.3 Improvements & Modernization Engineering Services	33
3.3.1 I&M Studies / Task / Engineering Projects / Instrumentation Modifications	33
3.3.1.1 Studies / Tasks.....	33
3.3.1.2 Engineering Projects	34
3.3.1.3 Instrumentation Modifications	34
3.3.2 Systems Engineering Integration and Test Support	34
3.3.2.1 Systems Engineering Functions	34
3.3.2.2 Integration and Test	35
3.3.3 Project Management.....	36

3.3.3.1 TD Development and Execution	36
3.3.3.2 Project Planning / WBS	36
3.3.3.3 Engineering Project Estimate	37
3.3.3.4 Engineering Change Proposal.....	37
3.3.3.5 Monthly Status	37
3.3.3.6 Schedule	37
3.3.3.7 Expenditures	37
3.3.3.8 Studies / Tasks.....	38
3.3.3.9 Final Report.....	38
3.3.3.10 Subcontracting / Procurement	38
3.3.4 Engineering Support.....	38
3.3.4.1 Project Execution	38
3.3.4.2 Technical Interchange Meetings.....	38
3.3.4.3 Subject Matter Expertise	38
3.4 Cyber Security General Requirements	39
3.4.1 Cyber Operations.....	41
3.4.1.1 User Account Management.....	41
3.4.1.2 User Account Analysis.....	42
3.4.1.3 Audit Records Analysis and Archive	42
3.4.1.4 Configuration Item Baseline	43
3.4.1.5 Incident Response and Contingency Plan Support.....	43
3.4.1.6 Backup of User and System Level Information	44
3.4.1.7 Password Management.....	45
3.4.1.8 Multifactor Authentication	45
3.4.1.9 Vulnerability Management Plan and Scans	46
3.4.1.10 Application Static Code Testing and Analysis	46
3.4.1.11 STIG/SRG Scans	47
3.4.1.12 Incident Monitoring and Event Management.....	47
3.4.1.13 Antivirus / Antispyware / Intrusion Detection and Prevention	48
3.4.1.14 Assessment and Authorization	48
3.4.1.15 Continuous Monitoring.....	49
3.4.1.16 Data Calls and Mandate Responses	49
3.5 Communications/Networks/Information Technology Services.....	49
3.5.1 Communications/Networks/IT Operations and Sustainment	51
3.5.1.1 IT Support Service Desk.....	52
3.5.1.2 Restoration of IT Services and Response Time.....	53
3.5.1.3 Scheduled and Unscheduled Outages	54
3.5.1.4 Problem Reporting, Tracking, and Monitoring.....	54

3.5.1.5 Mission Specific Knowledge Management via Government Enterprise.....	55
3.5.1.6 Endpoint Management Operations and Sustainment.....	55
3.5.1.7 IT Server Management Operations and Sustainment.....	56
3.5.1.8 Application Engineering Support for Networks.....	57
3.5.1.9 Warranty/Software License Compliance.....	58
3.5.1.10 Networks Operation, Maintenance, and Sustainment.....	59
3.5.1.11 Voice, Video and Mission Communications Operations and Maintenance.....	60
3.5.1.12 Optical Transport Communications System.....	61
3.5.1.13 Outside Plant, Inside Plant, and Fiber Optics.....	62
3.5.1.14 Technical Control Facility.....	63
3.5.1.15 End User Video Telecommunications Support.....	64
3.5.1.16 Satellite Communications Facility.....	64
3.5.1.17 Radio and Pagers.....	65
3.5.2 Network Support for Joint Worldwide Intelligence Communications System.....	65
3.5.3 IT Support for NIPR and SIPR.....	65
3.5.4 Mission Business Network.....	66
3.5.5 Development Network Maintenance (RTS-D).....	67
3.5.6 Network/IT Performance, Reliability and Maintainability.....	67
3.5.7 5G Support.....	68
3.6 Support Services Functions.....	68
3.6.1 Work Center for Technical Facility Maintenance.....	68
3.6.1.1 Projects and USAG-KA Annual Work Plan Process.....	69
3.6.1.2 Design and Engineering Support.....	69
3.6.1.3 Air Conditioning Systems Supporting Instrumentation.....	69
3.6.1.4 Electrical Systems at All Range Technical Facilities.....	69
3.6.1.5 Technical Facilities and Equipment Maintenance.....	70
3.6.1.6 Technical Facilities and Equipment Manual.....	70
3.6.1.7 Obsolescence Surveillance.....	70
3.6.2 Calibration Lab Support.....	71
3.6.2.1 TMDE Calibration and Repair.....	71
3.6.2.2 TMDE Shipment.....	71
3.6.3 RF Safety for Instrumentation.....	71
3.6.3.1 Safety Engineering Support.....	71
3.6.3.2 Risk Assessments and Safety Evaluations.....	71
3.6.3.3 RF Surveys.....	72
3.6.3.4 RF Emitters Procedures.....	72
3.6.3.5 RF Exposure Safety Training.....	72
3.6.3.6 RF Sensor Safety Oversight.....	72

3.6.3.7 RF Radiation Monitoring	72
3.6.3.8 RF Incident Analysis	72
3.6.3.9 RF Incident Investigation	73
3.6.4 Environmental Support.....	73
3.6.4.1 Spill Response Material / Kit Provision for RTS Missions	73
3.6.4.2 National Environmental Policy and Records	73
3.6.4.3 Environmental Protection Compliance	73
3.6.4.4 Environmental Management System/Quality Control Committee	73
3.6.4.5 Solid and Hazardous Waste.....	74
3.6.4.6 Hazardous Material Use and Storage Oversight	74
3.6.5 Mission Specific Kwajalein Installation Logistics Support	74
3.6.6 Government Property Management	74
3.6.6.1 Accountability Records Maintenance	74
3.6.6.2 Government Property Investigation and Reporting (Issued to the Contractor)	74
3.6.6.3 Repair or Replacement of Government Property (Issued to Contractor)	74
3.6.6.4 Excess Property Identification and Turn-In.....	75
3.6.7 Range User's Manual	75
3.6.8 Data Item Description Manual	75
3.6.9 State of the Range Briefing	76
3.7 Transition	76
3.7.1 Phase-In	76
3.7.2 Phase-Out.....	76
4. GENERAL REQUIREMENTS.....	78
4.1 Contract Administration and Management	78
4.2 Subcontract Management and Procurements	78
4.3 Key Contractor Personnel, Disciplines, Specialties	78
4.3.1 Program Manager.....	79
4.3.2 Instrumentation Operations Manager	79
4.3.3 Chief Engineer.....	79
4.3.4 Safety Manager	80
4.3.5 Information Systems Manager.....	80
4.4 Location and Hours of Work	81
4.5 Travel / Temporary Duty	81
4.6 Program Management	81
4.6.1 Integrated Master Schedule.....	81
4.7 Financial Resource Management	81
4.7.1 Annual Operating Budget.....	82
4.7.2 Contract Work Breakdown Structure	82
4.7.3 Budget Cost Variance Reporting	82

4.7.4 Contract Funds Status Report.....	82
4.7.5 Cost Reporting for RTS Customer Reimbursable Support.....	82
4.8 Configuration Management.....	82
4.8.1 Configuration Identification.....	83
4.8.2 Change Control and Status Accounting.....	83
4.8.3 Configuration Audits.....	83
4.8.4 Configuration Control.....	83
4.8.5 Technical Library.....	83
4.9 Agile Software Development and Management.....	84
4.10 Risk Management.....	84
4.10.1 Identification.....	85
4.10.2 Risk Assessment.....	85
4.10.2.1 Risk Analysis.....	85
4.10.2.2 Risk Ranking.....	85
4.10.3 Risk Mitigation.....	85
4.10.3.1 Risk Mitigation Planning.....	85
4.10.4 Risk Monitoring.....	86
4.10.5 Risk Reporting.....	86
4.11 Supply Services.....	86
4.11.1 Mission Specific Stock Management.....	86
4.11.1.1 Annual Supply Budget Management.....	87
4.11.1.2 Military Standard Requisitioning and Issuances Management.....	87
4.11.1.3 Military Standard Requisitioning and Issuances Priority.....	87
4.11.2 Government-Provided Warehouse(s) Supplies Management.....	87
4.11.2.1 Warehouse Stock Level Surveys.....	87
4.11.2.2 Sensitive Items Inventories.....	88
4.11.2.3 Shelf-Life Program Management.....	88
4.11.3 Hazardous Materials Management.....	88
4.11.3.1 Material Safety Data Sheet Maintenance.....	88
4.11.3.2 Shelf-Life Management in Storage.....	88
4.11.3.3 Turn-In.....	88
4.12 Quality.....	88
5. SPECIAL REQUIREMENTS.....	89
5.1 Security.....	89
5.1.1 Management.....	89
5.1.1.1 Common Access Card Issuance.....	89
5.1.1.2 Non-Common Access Card Access to DoD Facility or Installation.....	89
5.1.1.3 Formal Operations Security Program.....	90
5.1.1.4 General Protection/Security Policies and Procedures.....	90

5.1.1.5 Information Assurance/Information Technology Certification	90
5.1.1.6 Contractor Access to Classified Information	90
5.1.2 Classification	90
5.1.2.1 Facility Classification	91
5.1.2.2 Personnel Classification	91
5.1.2.3 Documentation Markings	91
5.1.3 Operations	91
5.1.3.1 Command Security Procedures	91
5.1.3.2 Physical Security Assets and Infrastructure Security Policies.....	91
5.1.3.3 Security-Related Mandatory Training.....	92
5.1.3.4 Kwajalein Atoll Facility Access Control.....	92
5.1.3.5 Document Control.....	92
5.1.3.6 Courier Authorizations	92
5.1.3.7 Security Self-inspections.....	92
5.1.3.8 Security Container Management.....	93
5.1.3.9 Sensitive Compartmented Information / Special Access Program Management	93
5.1.3.10 Contractor Personnel Security Management.....	93
5.1.3.11 Operations Security	94
5.1.3.12 Safe and Vault Technician and Locksmith (RTS Kwajalein Atoll Only)	94
5.1.3.13 Destruction of Information.....	94
5.1.3.14 Communications Security Management	95
5.2 Training	95
5.2.1 Anti-Terrorism Level I.....	95
5.2.2 Anti-Terrorism Awareness for Contractor Personnel Traveling Overseas	95
5.2.3 iWATCH	96
5.2.4 Army Training Certification Tracking	96
5.2.5 Operations Security	96
5.2.6 Information Assurance/Information Technology	96
5.2.7 Threat Awareness Reporting Program	96
5.3 Industrial Safety.....	97
5.3.1 Contractor Compliance.....	97
5.3.2 Accident/Incident Investigations Coordination	97
5.3.3 Mishap Notification and Investigation	97
5.3.4 Inspections	98
5.3.5 Safety Plans, Reports and Procedures.....	98
5.3.6 Rescue Services for RTS	98
5.4 Environmental Compliance	98
5.5 Data Rights Assertions / Inventions / Proprietary Information	99

5.5.1 Contract Data Requirement List.....	99
5.5.2 Assertions.....	99
5.5.3 Reporting Subject Inventions	99
5.5.4 Proprietary Markings on Non-Technical Data.....	99
5.5.5 Special Works.....	99
5.5.6 Deferred Ordering.....	100
5.6 Government Access/Use	100
5.7 Contract Performance or Delivery in a Foreign Country	100
6. DELIVERABLES	102
7. GOVERNMENT FURNISHED ASSETS	106
8. CONTRACT MANPOWER REPORTING	107
8.1 Reporting Process.....	107
8.2 Reporting Details.....	107
9. APPLICABLE DIRECTIVES	108
10. IDENTIFICATION OF CONTRACTOR EMPLOYEES.....	109
11. CONTRACTOR TELEWORK	110
12. ACRONYMS AND ABBREVIATIONS LIST	111

This Page Left Blank Intentionally

1. SCOPE

This PWS identifies all Performance, General, and Special requirements the contractor shall perform in support of operations, sustainment, engineering, and other supporting tasks required for successful contract performance. The intent is for the contractor to provide all required qualified personnel (to include [but not limited to] all needed training, certifications, licenses, experience, and education) and required labor, materials, and equipment necessary to execute the requirements specified herein, except for those items explicitly designated as Government-Furnished Property (GFP). The contractor shall perform all necessary actions, including travel, to ensure timely, complete, and compliant execution of the requirements. Access to standard Garrison services will be provided to contractor personnel when authorized by the Government.

Performance requirements are categorized as (but are not limited to):

- Operations
- Sustainment/Maintenance
- Improvements & Modernization Engineering Services (I&M)
- Cyber Security
- Mission Communications and Information Technology (IT) Services
- Support Services
- Transition

General requirements are categorized as (but are not limited to):

- Contract Management
- Program Management
- Financial Management
- Configuration Management
- Risk Management
- Quality Management
- Supply Services

Specialty requirements are categorized as (but are not limited to):

- Security Management
- Training
- Safety
- Environmental

The Government reserves the right to modify, reduce, or withdraw such services at any time. The Government also reserves the right to modify the requirements under in-scope change to address future requirements.

The contractor shall ensure all activities are performed in accordance with applicable laws, regulations, directives, standards, and Government-approved procedures.

2. BACKGROUND

RTS is an activity of the Department of Defense (DoD) Major Range and Test Facility Base (MRTFB) located on Kwajalein Atoll, in the Republic of the Marshall Islands (RMI) approximately 2,136 nautical miles west-southwest of Honolulu, Hawaii. Kwajalein Atoll is a crescent shaped loop of coral islands with an attendant reef that encloses an area of 1,100 square miles - one of the largest lagoons in the world. The distance from Kwajalein Island on the southern reef to Ebaddon Island on the northwest corner of the reef is approximately 75 miles. The lagoon is 15 miles across at its widest point. RTS consists of technical and support facilities located on 11, low-lying tropical islands, utilized by the U.S. under the terms of the Compact of Free Association with the RMI.

RTS performs two major activities: Space Operations and Test and Evaluation of DoD weapons systems (including offensive missiles, defensive missiles, hypersonic, and space test). RTS also supports unique or non-recurring scientific research and applied experiments. Execution of these operations/tests is performed from RTS or the RTS Operations Center in Huntsville, Alabama (ROC-H), using a wide array of highly sophisticated radar, telemetry, and optical sensors to meet the needs of the DoD in defending the Nation and its interests. ROC-H provides remote operations capability, mission control, and technical oversight for RTS activities.

RTS is also a United States (U.S.) Anti-Ballistic Missile (ABM) Test Range and a Defense Site pursuant to the Compact of Free Association, Public Law 99-239, with the RMI.

The RTS operational area of responsibility (AOR) during test operations spans a radius of approximately 500 nautical miles of Kwajalein Island and Wake Island. Wake Island is located approximately 700 miles north of Kwajalein Atoll and contains a limited suite of instrumentation used to support testing activities on a campaign basis. Refer to the Government Furnished Equipment (GFE) list for this equipment and RTS Instrumentation and Support Facilities Manual (ISFM) for Wake Island support system description.

RTS is the Primary Organizational Element (POE) for technical support services provided by separate contractors for ROSIE, Range Safety Support (RSS) services and Meteorological Support Services (MSS). RTS is a USASMDC organization consisting of the Office of RTS Director (which includes the Deputy Director, Chief Engineer, and Technical Director); the Mission Operations Division; Mission Safety Division; Space Operations Division; and program management support. RTS is headquartered in Huntsville, Alabama, at ROC-H. RTS on-island presence consists of the Office of the Range Director, an engineering and maintenance element, and a Communications and Networking Division.

RTS receives scientific advice from a Federally Funded Research and Development Center (FFRDC) contracted through the U.S. Air Force. The PWS requirements are necessary to meet a continuing need for research, development, test and evaluation (RDT&E) efforts for on-site and off-site engineering, technical support services, and materials required to operate, maintain, analyze, and modify the Kwajalein range instrumentation and related support systems. RTS instrumentation includes an array of sensors, systems, and centers (but not limited to):

- a. Highly sophisticated one-of-a-kind, state-of-the-art radar systems designed for collection of precise metric, imaging, and signature data.
- b. Additional radar metric and impact location assets.
- c. Telemetry data acquisition systems.
- d. Optics capabilities for highly accurate metrics, spectral characteristics, documentary video, photography, scoring data.
- e. Command and control centers.

hereafter referred to as “RTS Instrumentation” (note that a detailed listing is provided in the RTS ISFM). The Kiernan Reentry Measurements Site (KREMS) was initiated as a reentry-vehicle data gathering facility for the Advanced Research Projects Agency (ARPA) ballistic missile defense program. The KREMS includes 4 highly sophisticated one-of-a-kind, state-of-the-art radar systems designed for collection of precise metric, imaging, and signature data (described above). These four radars are:

- a. Target Ranging and Discrimination Experiment (TRADEX)
- b. ARPA Long Range Tracking and Instrumentation Radar (ALTAIR)
- c. ARPA Lincoln C-Band Observables Radar (ALCOR)
- d. Millimeter Wave Radar (MMW)

These PWS requirements include a variety of technical support services for RTS and customers (customers can include both DoD users [including DoD training users] and non-DoD users such as U.S. Government Agencies, State and local Governments, allied foreign governments, and commercial entities as defined by DoD Directive [DoDD] 3200.11). These technical support services can include (but are not limited to):

- a. Operation and maintenance of communications systems that provide data, voice, timing, and control circuits for conducting test programs at RTS, along with routine administrative communication needs.
- b. Electro-magnetic frequency control, surveillance, and analysis.
- c. Timing reference, synchronization, and control.
- d. Photographic and video services.
- e. Ordnance receipt, inventory, handling, and storage.
- f. Communications Security (COMSEC) equipment receipt, inventory, and control.
- g. Calibration, repair, and maintenance of Test, Measurement, and Diagnostic Equipment (TMDE).
- h. Mission planning and preparation include evaluation of mission requirements,

development of mission support plans and operational planning documents, preparation of operations schedules, and the adaptation of procedures as required to successfully support mission requirements.

- i. Design, implementation, testing, and documentation of hardware and software modifications to the instrumentation and technical support systems to maintain the state-of-the-art capability required by the complex RDT&E missions supported by RTS.

The contractor shall perform all work in support of these RTS missions in a manner that maintains operational readiness, protects mission continuity, and complies with applicable DoD, Army, MRTFB, and host-nation requirements. NOTE: All software development/modification shall be in full compliance with Army Directive 2024-02.

3. PERFORMANCE REQUIREMENTS

As applicable, this specifies the Performance Objectives, Metrics, and Data Requirements for the ROSIE contract.

3.1 Test Mission Support

The contractor shall provide RTS Test Mission Support (TMS) consisting of program planning support, performing accurate estimating and billing functions relative to customer missions, complete mission test planning, test mission execution, post-mission support, and mission support at extended areas of operations.

3.1.1 Program Planning Support

The contractor shall perform test planning and develop mission documentation for each test program. The contractor shall ensure that all customer requirements are identified and satisfied through comprehensive support plans documented In Accordance With (IAW) the Universal Documentation System (UDS); the UDS consists of standardized procedures governed by the DoD Range Commanders Council (RCC) (see Attachment 05). The contractor shall provide timely, innovative, cost-effective program and mission support concepts that meet all documented requirements. These functions are to be performed in coordination with the Governmental RTS mission planning and operations organization(s) located primarily in Huntsville, Alabama.

3.1.1.1 Universal Documentation System

Customer-provided UDS requirements/documents include (but not limited to) the Program Introduction (PI), Program Requirements Document (PRD) and Operations Requirement (OR) and associated operation extracts / supplements / addendums / Memorandums of Agreement (MOAs) / Memorandums of Understanding (MOUs).

At a minimum, the contractor shall:

- a. Evaluate requirements identified in customer-provided documentation and identify all RTS resources required to collect, process, and deliver data necessary to meet customer requirements.
- b. Prepare UDS response documentation (including but not limited to the Statement of Capabilities (SC), Program Support Plan (PSP), and Operations Directive (OD) based on validated requirements.
- c. The contractor shall deliver UDS response documentation to RTS

Government personnel, and other Government-authorized agencies as directed.

- d. Prepare and submit schedules for the review and preparation of UDS documentation that meet Government-approved timelines and

- format.
- e. Deliver accurate, complete, and Government- compliant UDS documents IAW approved schedules.

3.1.1.2 Mission Test Plan

The contractor shall deliver a mission test plan IAW Contract Requirement Data List (CDRL) A045. The contractor shall ensure all package data is accurate, complete, and timely per CDRL timelines.

3.1.1.3 Mission Briefing Package

The contractor shall deliver the mission briefing package IAW CDRL A014.

3.1.1.4 Operations Supported Document

The contractor shall prepare and provide the RTS Operations Supported Document IAW CDRL A017.

3.1.1.5 Range Safety System Mission Planning Support

The contractor shall prepare and provide all safety mission test planning, develop mission execution documentation, perform special studies, and provide configuration recommendations and readiness reporting for RSS necessary for successful and safe execution of missions performed at RTS or by RTS personnel. This includes preparation of an RSS Mission Support Plan IAW CDRL A046, RSS Readiness Report CDRL A047, and RSS Countdown Procedures CDRL A048.

3.1.2 Customer Mission Estimating

The contractor shall support Government cost estimation process by providing mission resource estimate input. These efforts are performed in support of RTS in Huntsville, Alabama. NOTE: As part of the MRTFB, RTS is required to collect test reimbursable costs from test customers.

3.1.2.1 New Program Initial Estimate Input

The contractor shall support initial requirements discussions and support initial requirements discussions inputs in support of establishing a new Technical Directive (TD) for a new program.

3.1.2.2 Customer Resource Estimates Inputs

The contractor shall determine mission support resource estimate inputs to include work hours, material costs, other direct costs, logistics support costs for RTS assets (e.g.,

provisioning and fuel for the Worthy), and a funding execution plan. Estimates shall be based on requirements provided by range customers documented with the UDS system or other requirements type documentation such as letters and/or MOAs and MOUs. The contractor shall document and deliver resource estimates using the Program Cost Estimates document IAW CDRL A018. Note that some projects may require separate estimates for different areas of support.

3.1.2.3 Resource Loaded Mission Schedule

The contractor shall provide a resource-loaded schedule for all missions within the overarching Integrated Master Schedule (IMS) IAW CDRL A059.

3.1.2.4 Mission Estimate Closeout

The contractor shall provide a summary of actual costs compared to the estimated costs for all complete missions. Initial submittal of this summary shall be completed no later than 90 calendar days after mission execution and updated as required if new data is received.

3.1.3 Mission Test Planning

The contractor shall provide all mission test planning, documentation development and support to Integrated Product Teams (IPTs) to ensure all customer requirements are satisfied through a comprehensive mission testing process. The contractor shall provide timely, innovative and cost-effective mission test planning concepts that meet all documented technical requirements.

3.1.3.1 Mission Schedule

The contractor shall provide the planning and operations schedule for near-term missions IAW CDRL A019 requirements. The operations schedule shall be a part of CDRL A059.

3.1.3.2 Mission Execution Files

The contractor shall develop mission execution files for RTS Instrumentation. The contractor shall prepare, test and modify mission execution files as follows:

- a. Radar mission profile, consisting of a radar configuration file, classifier initialization file and auto tasker file for each of the RTS radars.
- b. Simulation files that define the target scene and the radar parameters for each of the radars.
- c. A mission planning workstation "recall" file that permits changes to be made to the files as necessary.
- d. Range command and control software operation files (nominal (expected mission profile), off-nominals, graphics initializations, Radar Cross Section

- (RCS) profiles and auto-ID information).
- e. Optics mission profile to collect all required data.
- f. Telemetry mission profile to collect all required data.
- g. Off nominal (deviations from the nominal) for mission preparation/practice.

3.1.3.3 Mission Test Documentation

The contractor shall prepare and modify all mission test documentation as required so all documentation accurately and completely identifies mission essential instrumentation and the roles of that instrumentation in support of RTS missions. The contractor shall include updated mission test plans and schedules necessary to ensure all customer requirements are met IAW CDRL A045 and A019.

3.1.3.4 Working Groups

The contractor shall provide technical expertise for, and participation in, Government established working groups as necessary to identify requirements, resolve mission planning issues, develop lessons learned, apply lessons learned from past missions for future missions, and determine root cause for failure to meet mission requirements.

3.1.4 Instrumentation/Systems Operations

The contractor shall operate all RTS range instrumentation according to mission test plans and schedules and collect the required data for each mission. During mission freeze, the contractor shall also notify the Government range control officer within 24 hours of any changes in status or functionality of range instrumentation scheduled to support a mission.

3.1.4.1 Command and Control Center Operations

The contractor shall support operations at all required command and control centers (including but not limited to Huntsville Mission Control Center [HMCC] and Kwajalein Mission Control Center [KMCC]), according to mission test plans and schedules and collect the required data for each mission.

3.1.4.2 Radar Operations

The contractor shall operate all required RTS range radars according to mission test plans and schedules and collect the required data for each mission. All radars with a declared mission status (Mandatory, Required, or Desired) shall be available and operational for all scheduled activities.

3.1.4.3 Optical Sensor Operations

The contractor shall operate all required RTS range optical sensors according to mission

test plans and schedules and collect the required data for each mission.

3.1.4.4 Telemetry Operations

The contractor shall operate all required RTS telemetry systems, to include on atoll systems, Kwajalein Missile Impact Scoring System (KMISS) and Kwajalein Mobile Range Safety System (KMRSS), according to mission test plans and schedules and collect the required data for each mission. All telemetry systems with a declared mission status (Mandatory, Required, or Desired) shall be available and operational for all scheduled activities.

3.1.4.5 Mission Data Networks Operations

The contractor shall operate the RTS Mission Data Network (MDN) backbone for RTS Instrumentation systems according to mission test plans and schedules. The MDN backbone shall include all network routers, switches, cabling and ancillary equipment which transports data between RTS Instrumentation/systems/sensors as defined in the ISFM and RTS Technical Facilities and Equipment Manual (TFEM). The MDN backbone is not a part of the baseline Comms/IT Information Management System (IMGTS) / networks (i.e. not part of Non-secure Internet Protocol Router Network [NIPR]/ Secure Internet Protocol Router Network [SIPR]). RTS Mission Data Network infrastructure includes both local and long-haul circuits, physical network media, virtual networks, and the like, including wireless, Satellite Communications (SATCOM), outside plant, local undersea fiber optic cables, and management of long-haul undersea cables.

3.1.4.6 Range Safety System Operations

The contractor shall operate all required RTS range safety systems (such as the Range Safety Control Center [RSCC]), which include Kwajalein and Roi Namur Command Destruct sub-systems, and the KMRSS U.S. Army Vessel (USAV) Worthy, according to mission test plans and schedules and collect the required data for each mission. The only exception will be that the Missile Fight Control Officer(s) (MFCO) and Flight Termination System Officer (FTSO) range safety console positions during mission support activities will be manned by Government personnel.

The contractor shall ensure implementation of all safety system algorithms adequately satisfies all mission requirements and allows for safe mission execution.

3.1.4.7 USAV Worthy Operations

The contractor shall conduct USAV Worthy operations in support of RTS mission requirements IAW most current U.S. Coast Guard standards, Code of Federal Regulations (CFR) 49, Occupational Safety and Health Administration (OSHA) 29 CFR 1917, OSHA 29 CFR 1917, Army Techniques Publication (ATP) 4-15, RTS TFEM, and host nation standards.

Crew required above non-mission staffing, and as required by test mission requirements, shall be provided as needed for the duration necessary, in support of RTS mission operations.

3.1.5 Post Mission Support

The contractor shall support test mission data preparation, transmission, and After-Action Reviews (AARs).

3.1.5.1 On-Site Data Preparation and Data Transmission

The contractor shall perform any necessary data processing to prepare the data for transmission and transmit to RTS Data Analysis Center (RDAC) or directly to customer agencies as determined by the mission documentation. The contractor shall provide mission data reports to RTS to confirm data transmission IAW CDRL A022.

The contractor shall ensure data required to support RTS customers is collected / analyzed and provided to customers in accordance with documented customer requirements.

3.1.5.2 “Quick Look” Data Analysis, Reports and Presentations

The contractor shall prepare and transmit data to RDAC, or customers, perform Quick Look analysis, provide Quick Look Reports/Presentations as requested. The contractor shall perform any necessary data processing and analysis to support a Quick Look analysis IAW CDRL A021.

3.1.5.3 After Action Reviews

The contractor shall review data collected and assess the performance of the RTS Instrumentation in its data collection role, document any lessons learned because of execution of the mission, and present a post mission brief that summarizes RTS's support of the mission. The contractor shall prepare and deliver the Mission Assessment Report and Range Safety Mission Assessment Report IAW CDRLs A049 and A050.

The contractor shall ensure Mission Assessment and Range Safety Mission Assessment Reports are:

- a. Accurate.
- b. Complete IAW requirements of the CDRL and the applicable Data Item Description.
- c. Submitted for all missions.
- d. Includes all identified lessons learned and AAR action items, to include a plan for closure.

The contractor shall ensure implementation of all safety system algorithms adequately satisfies all mission requirements and allows for safe mission execution.

3.1.5.4 Lessons Learned Database

The contractor shall create and maintain a lesson learned database for all lessons identified during the mission planning and execution process. The contractor shall review the lessons learned database at the start of the mission planning process to identify applicable lessons learned from prior missions and apply them to the current mission.

3.1.6 Mission Operations Support Functions

The contractor shall provide TMS services in direct support of RTS test missions. TMS support services include (but are not limited to) environmental; ground release of calibration spheres for calibration of instrumentation in direct support of test missions; explosive handling and launch ordnance; logistics and baseline Command, Control, Communications, Computers, and Information Management (C4IM) coordination; and mission support at extended areas of operations.

3.1.6.1 Ground Release of Calibration Spheres Mission Requirements

The contractor shall prepare and launch spheres in support of RTS test mission customer requirements. The contractor shall ensure calibration spheres operations/ launches are successfully conducted to meet schedule and requirements.

3.1.6.2 Explosive Handling Operations

The contractor shall conduct all operations involving receipt, transportation, storage and shipment of explosive materials (including rocket motors) IAW applicable Federal, DoD, Army, and RTS explosive safety standards, and all approved mission documentation. The contractor shall provide an Ordnance Inventory Report IAW CDRL A030. The contractor shall ensure the following:

- a. All required certified personnel, equipment, facilities, and safeguards are available to conduct all explosive handling operations.
- b. Zero explosive incidents in the conduct of these operations.
- c. Operations conducted IAW Government- approved explosive handling and launch procedures.

All explosives work shall be performed by qualified/certified personnel whose qualifications are documented and maintained.

3.1.6.3 Compliance with Environmental Standards and Procedures

The contractor shall support and assist RTS and RTS customers in complying with United

States Army Garrison Kwajalein Atoll (USAG-KA) Environmental Standards (UES) and associated requirements (e.g., Kwajalein Environmental Emergency Plan [KEEP]); Final Historic Preservation Plan, United States Army Kwajalein Atoll [HPP]; and associated Army and DoD regulations). The contractor shall ensure RTS operations maintain full compliance with UES.

3.1.6.4 Installation Logistics, Baseline Command Information Management Support

The contractor shall serve as liaison to USAG-KA installation support and NETCOM elements. As liaison, contractor shall ensure RTS mission specific logistics and baseline C4IM support requirement identification, coordination, consolidation and execution. The contractor shall review customer-provided UDS documentation to identify test mission installation support requirements and provide input for the requirements package created by RTS government personnel for installation support contractors. The contractor shall collect each installation support element's support position and consolidate it into a single UDS response document. Data from this task shall be integrated into CDRLs A045 and A046 as appropriate.

3.1.6.5 Off Atoll Missions

The contractor shall operate, maintain, and calibrate RTS systems and associated equipment at locations outside Kwajalein Atoll, to include Wake Island (land based) and USAV Worthy, in support of approved mission requirements. All movement of RTS equipment outside of Kwajalein Atoll shall be reported to designated RTS Government personnel. The contractor shall provide an Extended Areas of Support Report IAW CDRL A032.

3.2 Sustainment / Maintenance (Standard Technical Services)

The contractor shall perform all necessary actions to maintain all the RTS Instrumentation and systems to include hardware, software, ancillary equipment and facilities on-site or in the Continental US (CONUS). The contractor shall meet the capabilities and accuracies published in the ISFM and use the guidelines of the Army's overarching Army Material Maintenance Policy (AR 750-1) and the Army's overarching Army Test, Measurement, and Diagnostic Equipment (AR 750-43). This requirement includes (but is not limited to) maintaining instrumentation/ systems published accuracy and performance, as well as maintaining availability, Reliability and Maintainability (R&M), integrity, supportability, and ease of use. Conduct any necessary testing to collect the appropriate data and present this data (as required) to prove that the maintenance requirements have been met. The contractor shall maintain all maintenance personnel qualifications.

The contractor shall ensure that all RTS Instrumentation listed in the RTS ISFM and TFEM are maintained and all RTS Instrumentation maintenance on systems listed in

the ISFM and TFEM is recorded via an Army-approved standard Logistics Information System (LIS). The contractor shall provide maintenance documentation for quality - codes, parts, time to repair, and corrective action plans.

3.2.1 Space Operations

The contractor shall conduct Space Surveillance Operations (SOPS) at RTS as a non-organic sensor to the Space Surveillance Network, as tasked by United States Space Command (USSPACECOM). Space surveillance operations are conducted at the Huntsville Space and Surveillance Center (HSSC) as a mandatory part of space support activities. The contractor shall provide the following HSSC support 24 hours a day, 7 days a week, and 365 days per year.

3.2.1.1 Space Domain Awareness

The contractor shall conduct Space Domain Awareness (SDA) Operations utilizing the Advanced Research Projects Agency (ARPA) Long Range Tracking and Instrumentation Radar (ALTAIR) and/or Target Resolution and Discrimination Experiment (TRADEX), in support of U.S. Space Command (USSPACECOM), as a contributing non-organic sensor.

Standard:

- a. The contractor shall provide 24/7/365 metric sensor coverage IAW the most current PRD Space Surveillance Network operations at the Reagan Test Site (latest dated 26 Jun 2014); the most current Memorandum for Record Reagan Test Site Space Surveillance Operations to USSPACECOM (latest dated 19 Mar 2025); the most current 18th Space Defense Squadron (SDS) SDA Operating Instructions (OI) (latest dated 23 Feb 2023).
- b. The contractor shall perform detection and tracking, and object discrimination in the geosynchronous belt, high-rate revisits and persistent coverage as tasked, or as directed.
- c. The contractor shall provide **at least one metric sensor** capable of detecting and tracking all launches within RTS coverage, to collect data required to determine launch vector for sub-orbital, earth-orbital and earth-escape launches.
- d. The contractor shall utilize Tactical Observation Processor (TOP) in support of object detection in low earth orbit and the geosynchronous belt. All data shall be provided to the Combined Space Operations Center (CSpOC) via the 18th SDS.
- e. The contractor shall complete all taskings received from the CSpOC including, but not limited to, special taskings and all new launch coverage within RTS visibility upon the earliest available object revolution.
- f. The contractor shall provide optics and passive-RF sensors for SDA operations when requested for special taskings in support of SDA operations.
- g. To determine mission capability, the contractor must maintain the sensors to operate in accordance with the values in Table 1. A sensor is deemed **partially mission capable** when it demonstrates **threshold power levels** in all radar frequency bands.

A sensor is deemed **fully mission capable** when it provides **nominal power levels** in all radar frequency bands. The objective power level represents the designed sensor performance for the specific radar band. If a sensor cannot meet threshold parameters, it is considered non-mission capable and is unavailable to conduct SDA operations. (Note: The RTS Director may approve exceptions in writing.)

- i. ALTAIR VHF in mono-static operations shall maintain a minimum threshold power level of 1.5 MW (Objective 5 MW*) as measured at the output of the combined high power amplifiers to achieve detection [acceptable SNR = 15] of new foreign launch targets [$\sim 1 \text{ m}^2$ or 0 dBsm] at LEO ranges [no greater than 4,700 km**] with minimal integration to achieve space launch data collection.
- ii. ALTAIR UHF in mono-static operations shall maintain a minimum threshold power level of 3.7 MW (Objective 5 MW*) as measured at the output of the combiner network to achieve detection [minimum acceptable SNR = 15] of small targets [$\sim 0.4 \text{ m}^2$ or -4 dBsm] at GEO ranges [$\sim 36,500 \text{ km}$] with minimal integration constrained to no more than 100 sec to achieve operationally relevant search strategies.
- iii. TRADEX L-Band in mono-static operations shall maintain a minimum threshold power level of 1.2 MW (Objective 2 MW*) as measured in the RF box to achieve detection [minimum acceptable SNR = 15] of small targets [$\sim 0.4 \text{ m}^2$ or -4 dBsm] at GEO ranges [$\sim 36,500 \text{ km}$] with minimal integration constrained to no more than 100 sec to achieve operationally relevant search strategies.
- iv. TRADEX S-Band in mono-static operations shall maintain a minimum threshold power level of 1.2 MW (Objective 2 MW*) as measured in the RF box to conduct tracking and wideband imaging [minimum acceptable SNR = 15] of medium to large size targets [$\sim 1.5 \text{ m}^2$ or 1.76 dBsm] at LEO/HEO ranges [$\sim 15,000 \text{ km}$] with minimal integration to achieve space MASINT data collection.

Sensor		ALTAIR VHF	ALTAIR UHF	TRADEX L-Band	TRADEX S-Band	ALCOR C-Band	MMW Ka-Band
Waveform		V.05 - 600*	U.05 - 1000*	L.05 - 565*	S280 - 94**	C512 - 10**	K4000 -100**
Target/Mission		NFL	Small GEO	Small GEO	LEO/HEO MASINT	LEO MASINT	LEO MASINT
Acceptable SNR		15	15	15	15	15	15
Slant Range		4,700 km	36,500 km	36,500 km	15,000 km	6,200 km	3,000 km
Max Int. Time		0.8 s	92 s	98.5 s	1.4 s	0.8 s	0.13 s
Threshold	Power Level	1.5 - 2.9 MW	3.7 - 4.3 MW	1.2 – 1.4 MW	1.2 – 1.7 MW	2.2 - 2.6 MW	13 - 24 kW
	Target Size	1 m^2 / 0 dBsm	0.4 m^2 or -3.98 dBsm	0.4 m^2 or -3.98 dBsm	1.5 m^2 or 1.76 dBsm	1 m^2 or 0 dBsm	0.2 m^2 or -6.99 dBsm
Nominal	Power Level	3 – 4.9 MW	4.4 – 4.9 MW	1.5 – 1.9 MW	1.8 – 1.9 MW	2.7 – 2.9 MW	25 -29 kW
	Target Size	0.5 m^2 or -3 dBsm	0.33 m^2 or -4.8 dBsm	0.33 m^2 or -4.8 dBsm	1 m^2 or 0 dBsm	0.75 m^2 or -1.25 dBsm	0.1 m^2 or -10 dBsm
Objective	Power Level	5 MW	5 MW	2 MW	2 MW	3 MW	30 kW
	Target Size	0.32 m^2 or -5 dBsm	0.29 m^2 or -5.36 dBsm	0.24 m^2 or -6.27 dBsm	0.89 m^2 or -0.5 dBsm	0.65 m^2 or -1.9 dBsm	0.08 m^2 or -10.75 dBsm

*Typical waveforms for target detection **Typical waveforms for target imaging

Table 1. KREMS Sensors Power Levels and Sensitivity

- h. The contractor shall ensure the KREMS sensors are calibrated to meet the following requirements to be operational and available to conduct SDA operations. Sensors not calibrated to the specifications listed below in Table 2 will be considered non mission capable.
- i. ALTAIR VHF absolute 2 sigma metric errors on a 15 dBsm RCS target in geosynchronous orbit shall not exceed 8-meter range error, 0.4 mrad angle error and a 1 cm/s range rate error.
 - ii. ALTAIR UHF absolute 2 sigma metric errors on a 15 dBsm RCS target in geosynchronous orbit shall not exceed 5-meter range error, 0.2 mrad angle error and a 1 cm/s range rate error.
 - iii. TRADEX L-band absolute 2 sigma metric errors on a 15 dBsm RCS target in geosynchronous orbit shall not exceed 2-meter range error, 0.1 mrad angle error and a 0.3 cm/s range rate error.
 - iv. TRADEX S-band absolute 2 sigma metric errors on a 15 dBsm RCS target in geosynchronous orbit shall not exceed 2-meter range error, 0.1 mrad angle error and a 0.3 cm/s range rate error.

Sensor	Absolute Metric Error	RCS Target	Orbit	Range Error (m)	Angle Error (mrad)	Range Rate Error (cm/s)
ALTAIR VHF	2 sigma	15 dBsm	geosynchronous	8	0.4	1
ALTAIR UHF	2 sigma	15 dBsm	geosynchronous	5	0.2	1
TRADEX L	2 sigma	15 dBsm	geosynchronous	2	0.1	0.3
TRADEX S	2 sigma	15 dBsm	geosynchronous	2	0.1	0.3

Table 2. Sensor Calibration Specifications for SDA

- i. The contractor shall report operational information to RTS Space Operations Division IAW CDRL A020. The contractor shall notify the Government within 15 minutes of any anomaly degradation, or sensor non-availability affecting SDA coverage.

3.2.1.2 Space Measurement and Signature Data

The contractor shall provide Sensor Imagery utilizing Millimeter Wave (MMW) and/or the Advanced Research Projects Agency (ARPA) Lincoln C-Band Observables Radar (ALCOR) Sensor coverage for the in support of Space Measurement and Signature Intelligence (MASINT) operations for the National Space Defense Center (NSDC) via the 18th SDS.

Standard:

- a. The contractor shall provide Millimeter Wave (MMW) and/or the Advanced Research Projects Agency (ARPA) Lincoln C-Band Observables Radar (ALCOR) Sensor coverage IAW the 18th SDS SDA OI, (latest dated 23 Feb 2023).
- b. The contractor shall provide one imaging sensor to perform complex/payload characterization for all launches within RTS coverage.
- c. The contractor shall complete a minimum of 98% of monthly taskings from the responsible tasking authorities, to include, but not limited to, all new launches, planned and unplanned, within RTS coverage upon first viable opportunity, and/or as directed. The contractor shall recall MMW and/or ALCOR if a planned launch occurs outside of the scheduled operating hours.
- d. The contractor shall provide quick-look assessment data IAW RTS SOPS Government guidance CDRL A021.
- e. To determine mission capability, the contractor must maintain the sensors to operate in accordance with the values in Table 1. A sensor is deemed partially mission capable when it demonstrates threshold power levels in all radar frequency bands. A sensor is deemed fully mission capable when it provides nominal power levels in all radar frequency bands. The objective power level represents the designed sensor performance for the specific radar band. If a sensor cannot meet threshold parameters, it is considered non-mission capable and is unavailable to conduct MASINT operations. (Note: The RTS Director may approve exceptions in writing.)
 - i. ALCOR C-Band in mono-static operations shall maintain a minimum threshold power level of 2.2 MW (Objective 3 MW*) as measured in the RF box to conduct tracking and wideband imaging [minimum acceptable SNR = 15] of medium to large size targets [$\sim 1 \text{ m}^2$ or 0 dBsm] at LEO ranges [$\sim 6,000 \text{ km}$] with minimal integration to achieve space MASINT data collection.
 - ii. MMW Ka-Band in mono-static, single tube operations shall maintain a minimum threshold power level of 13 kW (Objective 30 kW*) as measured at the output of a single high power amplifier to conduct tracking and wideband imaging [minimum acceptable SNR = 15] of small size targets [$\sim 0.2 \text{ m}^2$ or -7 dBsm] at LEO ranges [$\sim 3,000 \text{ km}$] with minimal integration to achieve space MASINT data collection.
- f. The contractor shall ensure KREMS imaging sensors are calibrated to meet the following requirements to be operational and available to conduct Space MASINT operations. Sensors not calibrated to the specifications listed below in Table 3 will be considered non mission capable:

- i. ALCOR C-band absolute 2 sigma metric errors shall not exceed 0.2-meter range error, 0.1 mrad angle error.
- ii. MMW Ka-band absolute 2 sigma metric errors shall not exceed 0.2-meter range error, 0.035 mrad angle error.
- iii. ALCOR and MMW absolute 2 sigma RCS errors shall not exceed 1 dB.

Sensor	Absolute Metric Error	RCS Error (dB)	Range Error (m)	Angle Error (mrad)
ALCOR C	2 sigma	1	0.2	0.1
MMW Ka	2 sigma	1	0.2	0.035

Table 3. Sensor Calibration Specifications for MASINT

- g. The contractor shall ensure use of the Tactical Imaging Processor (TIP) when conducting Space MASINT operations. The contractor shall ensure the data is successfully delivered to the MASINT Rapid Analysis Cell (MRAC) in the USSPACECOM J2 via the approved paths.
- h. The contractor shall report operational information to RTS SOPS Division weekly IAW CDRL A020.

3.2.1.3 Sensor Availability and Reporting

The contractor shall provide/report sensor availability for ALTAIR, TRADEX, ALCOR and MMW to RTS SOPS, 18th SDS, CSpOC and USASMDC Operations Center. For reporting purposes, the threshold values listed above are minimum requirements and shall be reported as AMBER. Values above the threshold shall be reported as GREEN. Values below the threshold shall be reported as RED.

Standard:

- a. The contractor shall provide no less than 95% of the required operating hours per week. The contractor shall provide one metric tracking radar 24/7. If a second metric or imaging radar is required for tasking, the contractor shall coordinate with RTS SOPS and RTS Range Operations within seven days of execution to reduce the operational availability requirement and account for lost maintenance time.
- b. The contractor shall not accumulate operational hours concurrently when multiple sensors are operated simultaneously (two sensors operating for 2 hours at the same time does not equal 4 hours of credit towards availability).
- c. The contractor shall submit RTS-approved routine and extended maintenance periods to the 18th SDS via the Operational Data Integrated Node (ODIN) for all

KREMS sensors. The contractor shall submit individual sensor scheduled non-availability to RTS SOPS IAW CDRL A051.

- d. The contractor shall coordinate with the Test Mission Operations Division to submit the required test mission support into ODIN upon mission execution times becoming available.
- e. The contractor shall submit, into ODIN or other platforms as directed, any other known sensor non-availability periods when the information is available.
- f. The contractor shall notify RTS SOPS, 18th SDS, NSDC, CSpOC, and USASMDC Operations Center of any unplanned maintenance resulting in mission failure within 60 minutes via available secure communications (i.e. SIPR over voice call, email, Chatsurfer, etc.)

3.2.1.4 Sensor Recall

The contractor shall maintain all sensors on a recall status.

Standard:

- a. The contractor shall maintain a 15-minute recall of the operating metric sensor if access to the restricted area is required during operational hours.
- b. The contractor shall maintain a 90-minute recall of any unscheduled sensor when coordinated in advance with RTS SOPS.
- c. The contractor shall always maintain a 3-hour recall of an imaging sensor.
- d. The contractor shall maintain an 8-hour recall unless otherwise coordinated with RTS Space Operations and RTS Range Operations for sensors in routine maintenance.
- e. Sensors in extended maintenance periods shall not have a recall.

3.2.1.5 Sensor Availability Waivers

The contractor shall submit waiver requests to the availability requirements due to circumstances outside of the contractor's control, including, but not limited to, weather events, safety, network outages, or unplanned facility maintenance.

Standard: The contractor shall submit waiver requests via CDRL A025 and shall notify RTS SOPS within 24 hours of any event likely to trigger a waiver. All submissions will be dispositioned within 30 days of the occurrence to ensure timely and coordinated review and

approval. Waivers shall not be effective until approved in writing by RTS SOPS.

3.2.1.6 Operator Training and Certification

The contractor shall develop, implement, and maintain an Operator Training and Certification Plan for all radar operators within the SOPS organization.

Standard:

- a. The contractor shall include instruction on, at a minimum: metric and MASINT data collection, New Foreign Launch identification, TIP/TOP utilization, and proper data transmission protocols to designated customers. Operators shall demonstrate proficiency in the above task to be considered certified.
- b. The contractor shall ensure new operators are certified within 45 days of start dates and annually thereafter.
- c. The contractor shall maintain training and certification records of all operators.
- d. The contractor shall include the creation and execution of simulations to exercise space threat scenarios.
- e. The contractor shall provide the SOPS Contracting Officer Representative (COR) with documentation certifying the completion of training for each radar operator IAW the training and certification plan IAW CDRL A077 and A078.
- f. The contractor shall implement a method for measuring training effectiveness based on customer feedback and report the results to the SOPS COR monthly.

The contractor shall ensure operators obtain/maintain required credentials and network access.

Standard: The contractor shall ensure all operators maintain NIPR and SIPR Defense Research and Engineering (DREN) RTS accounts and logins for all Automated Information Systems (IS).

3.2.1.7 Operational Planning

The contractor shall conduct and participate in launch and event planning.

Standard:

- a. The contractor Space Operations Manager shall maintain an active Top Secret (TS)/Secured Compartmented Information (SCI) clearance to access and discuss classified information related to operational planning.
- b. All contractor operators shall maintain a Secret clearance. The contractor shall provide a minimum viable bench of operators and a trainer with TS/SCI and Special Access Program (SAP) eligibility.
- c. The contractor shall coordinate with 18th SDS, Lincoln Space Surveillance Center

(LSSC), NSDC, National Space Intelligence Center (NSIC) and other government agencies as required for pre-launch information and special taskings.

- d. The contractor shall maintain the ability to assess and/or create launch nominals, assess and plan launch coverage and determine object visibility to ensure proper operational coverage.

3.2.2 Radar Extended Maintenance Periods

The contractor shall maintain all RTS range radars and associated equipment, to meet the required capabilities IAW the RTS ISFM and individual radar/sensor maintenance manual.

The contractor shall coordinate with RTS for Extended Maintenance Period (EMP) for each sensor.

Standard:

- a. The contractor shall ensure each sensor conducts **1 (one), three-week extended maintenance period per quarter**.
- b. The contractor shall provide sensor schedules to RTS SOPS and CSpOC through the ODIN during extended maintenance periods.
- c. The contractor shall schedule the operational sensor, during EMPs, for **3 to 4, 8-hour weekly maintenance days** to be conducted at the contractor's discretion through coordination with the SOPS.
- d. The contractor shall ensure EMPs are scheduled no less than 6 months prior to execution. The contractor shall coordinate alternate EMP dates within 10 business days and seek GOV approval.
- e. The contractor shall ensure all EMPs are deconflicted between Test Mission Operations (MOPs), SOPS, and Range Operations.
- f. The contractor shall ensure all EMPs of imaging sensors be coordinated with the LSSC imaging sensors.
- g. The contractor shall submit a detailed EMP work package including scope, parts, tools, staffing, expected system availability impact, and post EMP verification testing to demonstrate the system meets ISFM performance parameters.
- h. The contractor shall provide an EMP completion report IAW CDRL A025 documenting tasks performed, parts replaced, test results, and residual risks.

The contractor shall coordinate extensions to EMPs and receive approval by SOPS, MOPs, and Range Operations Division Chiefs.

3.2.3 Optical Sensor

The contractor shall maintain all RTS range optical sensors and associated equipment to meet the required capabilities IAW the RTS ISFM.

3.2.4 Telemetry

The contractor shall maintain all RTS range telemetry systems and associated equipment to meet the required capabilities IAW the RTS ISFM.

3.2.5 Command and Control Center

The contractor shall maintain all required Command and Control Centers (HMCC, KMCC, HSSC and the SCIF User Bay) and associated equipment to meet the required capabilities IAW the RTS ISFM.

3.2.6 Range Safety System

The contractor shall maintain all RTS range safety systems (RSCC and Kwajalein and Roi Namur Command Destruct sub-systems), and the KMRSS to the required capabilities IAW the RTS ISFM to meet mission test plans and schedules.

3.2.7 U.S. Army Vessel Worthy

The contractor shall conduct USAV Worthy maintenance IAW AR 750-1, American Bureau of Shipping, CFR Title 46, and International Convention for Safety of Life at Sea to maintain the load-line documentation in a current, ATP 4-15, and AR 56-9. The contractor shall ensure USAV Worthy annual inspections are conducted as per AR 750-1, American Bureau of Shipping, CFR Title 46, and International Convention for Safety of Life at Sea to maintain the load-line documentation in a current, AR 56-9. The contractor shall ensure that licensed / certified ship pilots and crews are maintained for USAV-Worthy IAW CFR Title 46, ATP 4-15, AR 56-9, and U.S. Coast Guard Pamphlet (CG) 191.

Worthy vessel is required to perform an 8-hour sea trial every 90 days unless underway for mission support within the previous 90 calendar days. Sea trial shall fully test all marine ship systems in an 8-hour trial. The contractor shall utilize ABS preventative maintenance software to track and record Worthy vessel preventative maintenance in support of American Bureau of Shipping load line requirements (IAW 46 CFR).

Contractor is responsible to maintain Worthy operation and seaworthiness at all times, to include all routine, scheduled, and unscheduled maintenance. Scheduled maintenance shall include periodic dry-docking (OCCM, or On-Condition Cyclic Maintenance) of the ship in compliance with Army and Coast Guard Regulations. At the time of this writing, Coast Guard required dry dock every 3-5 years and Army every 4 years, but current regulations shall govern.

3.2.8 Kwajalein Missile Impact Scoring System Maintenance

The contractor shall maintain the KMISS to the required capabilities IAW the RTS ISFM to meet mission test plans and schedules.

3.2.9 Ground Release of Calibration Spheres

The contractor shall prepare and release calibration spheres in support of requirements. The contractor shall ensure calibration spheres operations/launches are successfully conducted to meet schedule and requirements.

3.2.10 Update Instrumentation and Support Facilities Manual

The ISFM describes the capabilities (RTS sensor characteristics) of the current RTS Instrumentation systems. The purpose of the manual is to provide a ready reference for RTS capabilities, planning, and operational elements for Range customers. The manual shall contain a description of all mission support and instrumentation systems available at RTS. The contractor shall provide suggested changes to the RTS ISFM IAW CDRL A036 and A040 during the execution of this PWS if/when required. Government will maintain configuration control of the ISFM.

3.2.11 Mean Time Between Failures and To Repair Record Maintenance

The contractor shall maintain Mean Time Between Failures (MTBF) and Mean Time To Repair (MTTR) records and remain knowledgeable of instrumentation/systems capabilities, performance, reliability, availability, integrity, and maintainability. The contractor shall calculate MTBF and MTTR for each instrumentation/system, including range safety systems. Operational Availability (Ao) is to be calculated as $(MTBF / (MTBF + MTTR))$. This data (MTBF, MTTR, Ao) shall be used to prepare the Sensor, Instrumentation, and Network Availability Plan IAW CDRL A024. The contractor shall notify the COR within 1 hour when a sensor becomes non-operational for any reason.

Standard: Maintain Ao based on published capabilities of not less than 95% for each designated RTS Space Operations instrumentation/system, and 99% of designated mandatory RTS sensors during RTS Test Mission Windows (from each system defined in the ISFM and TFEM) including range safety system.

3.2.12 Sensor and Network Availability Report

The contractor shall deliver the RTS Sensor and Network Availability report IAW CDRL A023 requirements. Systems will not be considered available if they are not compliant with security/compliance requirements.

3.2.13 Geodetic Coordinate Surveys and Coordinates Manual

The contractor shall provide support to geodetic coordinate surveys and maintain and update the RTS Geodetic Coordinates Manual IAW CDRL A026 and A040 during the execution of this PWS if/when required. Government will maintain configuration control of the manual. This manual specifies the geodetic coordinates of RTS instrumentation sites

and of calibration sites used by instrumentation and sensors.

3.2.14 Instrumentation Status and Weekly Activity Reports

The contractor shall provide a weekly activity report summarizing operational, maintenance and/or testing activities; performance metrics; and significant events conducted in the instrumentation/system areas on a weekly basis IAW CDRL A025. The weekly activity report can also include any information that the contractor deems may be of interest to the Government. The contract shall develop and maintain a dashboard accessible to Government personnel that shows real time status of all instrumentation and sensors.

3.2.15 Wake Island and Off Atoll Missions

The contractor shall operate, maintain, and calibrate RTS systems and associated equipment at Wake Island and other locations within the RTS AOR outside Kwajalein Atoll as part of RTS range readiness and instrumentation maintenance. The contractor shall provide a report for extended areas of support IAW CDRL A032.

3.3 Improvements & Modernization Engineering Services

The contractor shall provide RTS Improvements and Modernization (I&M) Division Systems Engineering, Integration, and Test (SEI&T) services to support the execution of Instrumentation upgrades, modifications, and new capabilities; new instrumentation, special studies, special projects/tasks as required and approved by RTS. Support periodic projects at RTS that require additional engineering and/or technician staffing.

3.3.1 I&M Studies / Task / Engineering Projects / Instrumentation Modifications

The contractor shall ensure all Study/Task/Project/Instrumentation Modifications:

- a. Are completed on schedule
- b. Material cost and labor within baseline estimate
- c. Effort passes requirements, design, and test readiness reviews (if they were planned and scheduled)
- d. Effort passes all acceptance tests before IOC and performs as expected after the modification is complete
- e. Documentation and drawings completed on schedule.

3.3.1.1 Studies / Tasks

The contractor shall conduct studies and perform defined tasks as described by a RTS approved TD. Studies and tasks are specific in scope, require less effort than that required for an Engineering project and cost less than \$1M.

The contractor shall provide studies as a written document IAW CDRL A041 that covers specific topics requested by RTS to include determination of current capabilities; analyzing requirements for new or enhanced capabilities; Design / Engineering enhancements; evaluating alternatives, cost analysis and making recommendations on the use of Commercial off-the-Shelf (COTS)/Government off-the-Shelf (GOTS) equipment/software.

Tasks are directives in which scope requires minimal engineering management to execute. Tasks may be procurements in support of new or on-going efforts at RTS both internal and external. Tasks may be labor only in support of internal and external RTS efforts.

3.3.1.2 Engineering Projects

The contractor shall conduct Engineering Projects directed by the Government through a TD. Engineering Projects include a higher level of engineering required to execute the scope of the TD. This type of TD is reserved for projects that require the contractor to manage cost more than \$1M but less than \$5M and manage a schedule that spans no more than two calendar years for execution.

The ceiling and execution timeline can be increased and/or extended at the discretion of RTS I&M Division. Engineering projects shall include project management, systems engineering, and engineering expertise for the development, integration, and testing of capabilities delivered to RTS.

3.3.1.3 Instrumentation Modifications

The contractor shall conduct Instrumentation Modifications directed by the Government through a TD. Instrumentation modifications include high level engineering rigor, is reserved for projects where cost exceeds \$3M, and schedule spans multiple calendar years. The ceiling and execution timeline can be increased and/or extended at the discretion of RTS I&M Division.

3.3.2 Systems Engineering Integration and Test Support

3.3.2.1 Systems Engineering Functions

The contractor shall perform all SE functions following ISO/IEC 15288 processes for the SE lifecycle for RTS-approved engineering projects and instrumentation modifications as required to support upgrades, improvements, and modernizations to instrumentation, hardware, software and networks.

The contractor shall perform and apply Systems Engineering (SE) best practices to the execution of projects and instrumentation upgrades and modifications.

Standard: Each project / instrumentation modification must include:

- a. Requirements development, traceability, and/or analysis of alternatives.
- b. Requirements, design, development, and test readiness reviews.
- c. Verification, validation, test plans and procedures approved by RTS.
- d. Deliverables pass all acceptance testing and perform as expected after the modification is complete.
- e. Documentation and drawings updated to reflect the modification in accordance with the CMP.

3.3.2.1.1 Requirements/Analysis of Alternatives Development

The contractor shall develop and maintain requirements and/or analysis of alternatives for each engineering project or instrumentation modification that links requirements to design, verification, and validation activities for RTS approval.

3.3.2.1.2 Documentation and Drawings

The contractor shall ensure that all documentation and drawings are updated after a task/engineering project/instrumentation modification is completed. These updates shall be performed IAW CMP and CDRL A028 and A040 during the execution of this PWS if/when required. Government will maintain configuration control of the documentation and drawings.

3.3.2.1.3 Model-Based Systems Engineering

The Contractor shall employ Model-Based Systems Engineering (MBSE) methods and tools to develop, analyze, and communicate system architectures, interfaces, and behavior. MBSE products shall support design reviews and be maintained as living artifacts. The contractor shall be able to ingest and integrate with external stakeholder MBSE models.

3.3.2.2 Integration and Test

The contractor shall develop and execute integration and test plans and provide support to internal and external stakeholders in the development and installation of engineering project and instrumentation modifications.

The contractor shall perform integration and testing of all I&M projects and instrumentation upgrades and modifications.

Standard: Integration and test plans are developed, submitted, and approved by RTS Gov for all engineering projects and instrumentation modifications.

3.3.2.2.1 Integration and Test Plan Development

The contractor shall develop integration plans, test plans and procedures, hold document and readiness reviews, and ensure the project meets the requirements/objectives of the TD.

3.3.2.2.2 Verification and Validation

The contractor shall develop and execute verification and validation plans and procedures in support of acceptance testing developed in section 3.3.2.2.1.

3.3.3 Project Management

The contractor shall provide project management and general engineering duties in support of TDs, reporting, subcontracting, and procurement.

3.3.3.1 TD Development and Execution

The contractor shall manage the process for the development of TDs which includes the development and submission for approval of a Work Breakdown Structure (WBS), Project Plan, Engineering Project Estimates (EPE), and/or Engineering Change Proposal (ECP) for Instrumentation Modifications.

The contractor shall lead, or support studies/task/projects/instrumentation modifications IAW approved Technical Directive using Project Management best practices.

The contractor shall develop a project plan for each engineering project/instrumentation modification and deliver to RTS for approval. This plan should include requirements, analysis, cost estimates, and proposed schedules.

The contractor shall maintain cost and schedule to the approved baselines (original or approved) within the latest approved project plan. The contractor shall submit a final report IAW CDRL A028 for final closure and approval.

3.3.3.2 Project Planning / WBS

The contractor shall develop a Project Plan IAW CDRL A041 that includes the contractor's recommendation on the TD level of complexity for each Instrumentation Modification. Level of complexity shall be described in terms of engineering effort such as concept design, preliminary design, and critical design. Each project plan shall identify the need for the project/modification, identification of the requirement(s) for the project/modification, the technical approach, testing methodology, Initial Operational Capability (IOC)/Full Operational Capability (FOC) criteria and proposed date, identification of documentation/drawings affected, a proposed budget, and a preliminary schedule with milestones for progress tracking. These design levels drive system engineering, configuration management, non-recurring engineering, integration and test effort required to complete a project. The contractor shall develop, maintain, update, and deliver a WBS to level 6 and all schedules per CDRL A059.

3.3.3.3 Engineering Project Estimate

The contractor shall develop an EPE IAW CDRL A031 in response to each TD. Each EPE shall contain the contractor's high-level approach to the technical directive scope and an estimate of effort for labor and non-labor costs by fiscal year. EPEs are required for tasks, studies, and engineering projects.

3.3.3.4 Engineering Change Proposal

The contractor shall develop an ECP IAW CDRL A040 for efforts larger than 3 million dollars. Each ECP shall contain justification for the modification, identification of the requirement(s) that the modification will satisfy, the modification approach, testing methodology, Initial Operational Capability (IOC)/Full Operational Capability (FOC) criteria and date, identification of documentation/drawings affected, a proposed detailed budget, and a resource loaded schedule with milestones for progress tracking. ECPs are required for instrumentation modifications.

3.3.3.5 Monthly Status

The contractor shall present the status of Task Directives monthly.

3.3.3.6 Schedule

The contractor shall include each of the TDs identified by RTS in the IMS IAW CDRL A059. The contractor shall update and report the monthly progress of each TD to level 6 of the project WBS in the IMS.

3.3.3.7 Expenditures

The contractor shall deliver monthly funding expenditures by TD. This shall include Budgeted Cost of Work Scheduled, Actual Cost of work Performed, amount funded to date,

remaining funds, and labor vs non-labor.

3.3.3.8 Studies / Tasks

The contractor shall prepare studies / tasks reports at the completion of efforts that describe the results of the effort IAW CDRL A041.

3.3.3.9 Final Report

The contractor shall prepare a final report for each completed ECP IAW CDRL A028 that describes the results of the engineering project or the instrumentation modification scope, effort, associated cost and performance.

3.3.3.10 Subcontracting / Procurement

The contractor shall be responsible for any subcontractor management necessary to integrate work performed on this requirement and shall be responsible and accountable for subcontractor's performance and reporting on this requirement. The prime contractor will manage work distribution to ensure there are no OCI considerations. This effort shall be implemented IAW section 4.2 Subcontract Management and Procurements.

3.3.4 Engineering Support

The contractor shall provide project engineering support and expertise in support of the execution of projects and instrumentation modifications for internal and external stakeholders.

3.3.4.1 Project Execution

The contractor shall execute engineering projects and instrumentation modifications from cradle to grave, managing all efforts associated with the execution of the effort.

3.3.4.2 Technical Interchange Meetings

The contractor shall participate in technical interchange meetings and discussions with government personnel in support of existing and future TD with both internal and external stakeholders.

3.3.4.3 Subject Matter Expertise

The contractor shall provide subject matter expertise to internal and external stakeholders with respect to all RTS sensors, instrumentation, and networks.

3.4 Cyber Security General Requirements

The contractor shall implement a Cybersecurity (CS) program. Implementation of the CS program requires maintaining security expertise in providing input to all levels of systems engineering, software design and integration, testing, and training of the RTS end items. The contractor shall comply with all applicable regulations, policies and guidance in field support of the RTS end items. The contractor shall meet all the hardware and firmware compatibility requirements of the hardware design required for compatibility and interoperability with all maintained RTS mission software. In the event there are discrepancies in a specific requirement as outlined between two or more of the documents specified in applicable documents, then the more stringent guidance shall be implemented.

The contractor shall adhere to Cyber Security regulations and policies including (but not limited to) the National Security Act; Clinger-Cohen Act; Federal Information Security Management Act (FISMA); AR 25-1 Army Information Technology; AR 25-2 Army Cyber Security; any applicable DoD 8500 series directives and/or instructions and their associated references; CNSSI No. 1253 Security Categorization and Control Selection for National Security Systems; Chairman of the Joint Chiefs of Staff Instruction (CJCSI) and Chairman of the Joint Chiefs of Staff Manual (CJCSM) 6000 series; National Security Agency (NSA) Guides; Defense Information System Agency (DISA) Security Technical Implementation Guides (STIGS).

The following additional DoD and Army directives and instructions apply for CS:

- a. DFARS Provision 252.204-7008, Compliance with Safeguarding Covered Defense Information Controls
- b. DFARS Clause 252.204-7012, Safeguarding Covered Defense Information and Cyber Incident Reporting
- c. DFARS provision 252.204-7019, Notice of National Institute of Standards and Technology (NIST) SP 800-171
- d. DFARS clause 252.204-7020, NIST SP 800-171
- e. FAR Clause 52.204-2, Security Requirements
- f. CNSS 4009
- g. AR 380-5, AR380-3
- h. ISO/IEC15408
- i. Program Protection Plan Outline & Guidance, version 1.0
- h. Army Directive 2024-02 Army Modernization Strategy

The contractor shall apply appropriate STIGs and DoD Security Requirements Guides (SRGs) to all developed and delivered IT components. The Government Information System Security Manager (ISSM) must approve all deviations from the STIGs and SRGs.

The contractor shall support in the development and execution of the Critical Functional Analysis, Critical Program Information Assessment, Threat Assessment, Vulnerability Assessment, Risks Assessment, and Program Protection Plan documentation.

The contractor shall support the Government and their evaluation authorities in all Cybersecurity, Assessment and Authorization (A&A), Authority to Operate (ATO), and Authority to Connect (ATC) or Interconnectivity activities, including providing A&A documentation, upon request, in a format acceptable to DoD Cybersecurity, and A&A activity authorities.

The contractor shall implement a security-focused configuration management program for all RTS Configuration Items and system components. The security-focused configuration management program must use industry best practices defined within NIST 800-128 when implementing security-focused configuration management processes and procedures.

The contractor shall submit a request to support the use of any third-party, public domain, Free and Open-Source Software (FOSS) products to include software libraries, components, and applications within the System for Government O-ISSM email approval prior to implementation. The contractor shall obtain Government O-ISSM email approval at either Engineering or CS TIM or CCB prior to use of Mobile Code within the System IAW NIST SP 800-53.

The contractor shall implement a software assurance program for all software applications created on behalf of the Government. The software assurance program must use industry best practices when developing software.

The contractor shall implement a security-focused configuration management program for all RTS Configuration Items and system components. The security-focused configuration management program must use industry best practices defined within NIST 800-128 when implementing security-focused configuration management processes and procedures.

The contractor shall support recovery and reconstitution (i.e. contingency operations) of RTS systems to a known state after a system disruption, compromise, or failure.

The contractor shall support incident response operations for RTS systems.

The contractor shall procure technologies for RTS in accordance with CNSSP No. 11, "National Policy Governing the Acquisition of Cybersecurity and Cybersecurity-Enabled Information Technology Products and Services." In addition, technologies shall be procured which have been validated by Common Criteria Testing Labs, in accordance with the National Information Assurance Partnership (NIAP) Protection Profiles (PPs), and/or DoD Information Network (DODIN) Approved Products List (APL). For NSS, where classified data is being protected at rest or in transit by commercial products, technologies from the CSfC Components List shall be used, in accordance with NSA's published CSfC Capability Packages. Capability Packages and the CSfC Components List can be found by visiting the CSfC Components List page. NIAP-validated products can be found on the NIAP website on the CCEVS Product Compliant List page.

3.4.1 Cyber Operations

The following sections and subsections detail specific Contractor responsibilities pertaining to Cybersecurity Operations and Sustainment Support within RTS Mission Networks and Communication Circuits. These subsections consist of user account management and analysis, audit records review, analysis, and archiving, configuration item baselines, incident response and contingency planning support, backup of user-level and system-level information, password management, multifactor authentication, vulnerability management and scans, incident monitoring and event management, assessment and authorization, and continuous monitoring support. The RTS Mission Networks and Communication Circuits are comprised of the following IT systems and assets owned by RTS: (1) RTS-Classified Network (2) RTS-Unclassified Network, (3) KMISS, (4) Starshield, (5) Mission DOTCOM, (6) RTS Voice, Video, and Mission Communications, (7) Optical Transport Communications to include battery power plant equipment (RTS Mission Networks and Communications Circuits backbone), (8) RTS Development Network, (9) RTS Top Secret Network, (10) Satellite Communications (SATCOM) Facility, (11) Radio and Pagers, (12) Monitoring and minor troubleshooting for Long-haul undersea fiber optic cable system(s) including support facilities and infrastructure, and (13) associated voice and data circuits. The contractor shall be responsible for end-to-end cybersecurity operations and sustainment of RTS Mission Networks and Communications Circuits including but not limited to day-to-day cybersecurity operations. From this point forward all mention of RTS Mission Networks and Communication Circuits refers to the IT systems, assets, and capabilities described in this paragraph.

The contractor shall operate RTS CS capabilities and tools throughout the cybersecurity architecture IAW all applicable NIST 800 series and CNSS controls and overlays to address a system categorization of High-Confidentiality, High-Integrity, and High-Availability. The contractor shall operate RTS CS capabilities and tools throughout the system cybersecurity architecture IAW AR 25-2.

Standard: RTS Mission Networks and Communication Circuits have a control risk or overall system risk that is =< Moderate, which allows the Government to maintain an ATO.

3.4.1.1 User Account Management

The contractor shall support directory services (i.e. Lightweight Directory Access Protocol, Authentication, Authorization, and Accounting (AAA) servers, active directory, etc.) to authenticate and authorize all RTS users and computing resources in the domain and to assign and enforce security policies. The contractor shall provide account management with multifactor authentication. The contractor shall create, enable, disable, modify, or remove RTS user accounts within 24 hours of written notification from the ISSM/Information System Security Officer (ISSO) and/or receiving a service request notification. The contractor shall ensure authorized users are assigned appropriate account types, group

memberships and grant access based on access control policies and a valid need to know. The contractor shall review for compliance and analyze the need for RTS individual user, application, and service accounts quarterly and report analysis activities in the Cybersecurity Activity Report IAW CDRL A061.

Standard: (1) 100% compliance: only authorized users are allowed to access RTS applications, systems, and networks; (2) 100% compliance: RTS individual user, application, and service accounts are disabled or removed within 24 hours.

3.4.1.2 User Account Analysis

The contractor shall ensure no unauthorized access to RTS IT environment and networks. The contractor shall ensure account permissions are granted and created based on user account procedures and least privileged access. The contractor shall review and analyze RTS user accounts for atypical usage (e.g. unusual account activity, rapid logon/logoff, multiple logons, failed access, expired accounts and elevated rights usage) quarterly. The contractor shall ensure privileged accounts have Government approval prior to creation. The contractor shall review for compliance and analyze the need for RTS privileged user accounts every 30 days and report analysis activities in the Cybersecurity Activity Report IAW CDRL A061. The contractor shall provide an Atypical Usage Analysis Report and Privilege User Analysis Report IAW CDRL A062.

Standard: (1) 100% compliance: RTS Privileged users accounts are removed or disabled within 72 hours; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

3.4.1.3 Audit Records Analysis and Archive

The contractor shall review and analyze RTS system audit records daily for indications and the potential impact of inappropriate or unusual activity. If analysis identifies potential threats and/or inappropriate or unusual activity or connection, then the contractor shall notify the ISSM immediately (within 15 mins). The contractor shall ensure audit processing failures are resolved when alerted. Audit review activities shall be reported in the Cybersecurity Activity Report IAW CDRL A061. The contractor shall report audit review and analysis findings quarterly to the RTS ISSM in an Audit Review and Analysis Report IAW CDRL A062. The contractor shall verify and validate that audit records are archived daily and retained for 1 calendar year after archiving.

Standard: (1) 100% compliance: Audit record archive file exists for every day of the calendar year (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

3.4.1.4 Configuration Item Baseline

The contractor shall develop, document, and maintain a secure baseline configuration for each RTS Configuration Item (CI) and CI system components under configuration control annually. The contractor shall use technical and configuration documentation and logical configuration management tools IAW CDRL A042 to create a secure baseline configuration that at a minimum contains:

- a. STIG/SRG configuration settings.
- b. STIG/SRG/Security Patch deviation sheet.
- c. Software loads.
- d. Vendor released software and security patch levels.
- e. Security control implementation.
- f. Cryptography employed.
- g. Network protocols and services.
- h. Cybersecurity technical specifications and design documentation.

The contractor shall develop a CI Recommendation Plan IAW CDRL A074 that designates and identifies CIs and identify the types of configuration tools and documentation that will be used to maintain the secure baseline configuration. The contractor shall update the secure baseline configuration for the RTS CI when a CI system component is installed or upgraded. The contractor shall review RTS CI's baselines monthly for changes or when changes are applied to a CI baseline, to determine whether unauthorized changes occurred. RTS CI review activities shall be reported in the Cybersecurity Activity Report IAW CDRL A061. The contractor shall analyze all changes to a RTS CI's baseline and provide a Security Impact Analysis Report IAW CDRL A063 for all changes prior to change implementation. The Security Impact analysis should be included in all submitted configuration control board technical data packages. The contractor shall identify and document STIG/SRG/Security Patch deviations, from configuration settings, in a STIG/SRG/Security Patch Deviation Sheet IAW CDRL A063 when STIG/SRGs/Security Patches are newly applied or updated on RTS CI's or CI system components. The STIG/SRG/Security Patch Deviation sheet should be included in the configuration control board technical data package, if required. The contractor shall develop, document, and coordinate inventory management for IT equipment, software, firmware and services annually IAW CDRL A055 and CDRL A056.

Standard: 1) 100% review of all changes to the CI baseline; (2) 100% inventory accounting for IT equipment, software, firmware and services; (3) CDRL delivered with: (a) 100% content and data accuracy, b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

3.4.1.5 Incident Response and Contingency Plan Support

The contractor shall adopt, execute, and provide security assessment support for RTS Contingency Plans for all RTS systems. The contractor shall integrate and coordinate RTS Contingency Plan activities with outside organization Contingency Plans when contingency operations impact the outside organization. The contractor shall perform emergency preparedness activities to include disaster recovery planning and execution of contingency planning operations. The contractor shall support Contingency Plan tabletop testing annually.

Standard: 100% contractor support to Government Contingency Plan support activities and testing during scheduled dates and milestones

The contractor shall provide and perform Incident Response support such as responding to a compromise, continuous and ongoing monitoring activities to identify Indicators of Compromise, determine the extent of the incident, remove the adversary from RTS systems, assess threat intelligence, restore RTS networks to a more secure state, and assist with responding to crisis or urgent situations within the RTS domain to mitigate immediate and potential threats. The Contractor shall use information and threat intelligence specifically focused on incidents to identify undiscovered attacks and/or indicators of compromise and shall investigate and analyze all relevant response activities. The Contractor shall coordinate with Cyber Security Service Provider and provide technical support to network defense technicians to resolve network defense incidents. The Contractor shall correlate incident data, with the necessary technical resources, to identify specific vulnerabilities, provide and execute recommendations resulting from correlated incident information, enable expeditious incident remediation, and monitor resulting activities through resolution. The contractor shall support Incident Response Plan tabletop testing annually.

Standard: 100% contractor support to Government Incident Response Plan support activities and testing during scheduled dates and milestones.

3.4.1.6 Backup of User and System Level Information

The contractor shall conduct backups of user-level information contained in directory data servers (e.g. active directory, LDAP, and AAA servers) daily. The integrity of user-level information shall be protected using digital signatures, cryptographic hashes or other DoD approved cryptographic mechanisms. The contractor shall conduct backups of DoD/RTS Certificate Revocation List (CRLs) and DoD/RTS trust anchors (i.e. root certificates) on RTS systems daily. The integrity of certificate information shall be protected using digital signatures, cryptographic hashes or other DoD approved cryptographic mechanisms. The contractor shall conduct backups of system-level information (e.g. operating system, firmware, patch levels, and application software) RTS system components monthly. The integrity of system-level information shall be protected using digital signatures, cryptographic hashes or other DoD approved cryptographic mechanisms. The contractor shall transfer user-level, DoD/RTS certificate revocation list (CRLs) and DoD/RTS trust anchors (i.e. root certificates) and system-level backup information to alternate storage

monthly. The contractor shall demonstrate and test back-ups to ensure restoration reliability. Backup of User and System-Level Information activities shall be reported in the Cybersecurity Activity Report IAW CDRL A061.

Standard: 100% Compliance – Restore requests can be completed within 4 hours.

3.4.1.7 Password Management

The contractor shall issue and manage passwords for user, group, application, and service accounts in accordance with DOD and/or Army password management standards. The contractor shall ensure passwords emergency accounts or accounts of last resort are only shared with system administrators and passwords are changed annually, when a system administrator terminates employment, or the actual or suspected password compromise. The contractor shall ensure user password accounts that are not used for 90 calendar days are disabled and reviewed for possible deletion. The contractor shall ensure that disabled password accounts are removed after 60 calendar days. The contractor shall update application and/or service account passwords annually. The contractor shall utilize a password manager to securely store and manage credentials for privileged, group, and application/service accounts. The master password for this tool shall be treated with the highest level of security and not stored in an insecure manner. The contractor shall immediately notify the Government ISSM of any confirmed or suspected password compromises, unauthorized account access, or other security incidents related to password credentials. The contractor shall provide a report detailing the scope of the incident, mitigation actions taken, and a plan for preventing future occurrences. Password management activities shall be reported in the Cybersecurity Activity Report IAW CDRL A061.

Standard: 100% Compliance – All account passwords are valid, password accounts are disabled after 90 calendar days of inactivity, and passwords have changed within last year.

3.4.1.8 Multifactor Authentication

The contractor shall establish and manage cryptographic keys for required cryptography employed within the IS in accordance with centralized management of key generation, distribution, storage, access and destruction in accordance with NIST SP 800-57, Recommendation for Key Management. The contractor shall install and update DoD/RTS CRLs and DoD/RTS trust anchors (i.e. root certificates) on the RTS IT networks within 24 hours of certificate expiration. The contractor shall disable RTS PKI tokens that have more than 90 days of inactivity. The contractor shall archive RTS PKI tokens that have been disabled/inactive for more than 365 calendar days. The contractor shall immediately notify the Government ISSM of any confirmed or suspected password compromises, unauthorized account access, or other security incidents related to PKI credentials. Multifactor Authentication activities shall be reported in the Cybersecurity Activity Report IAW CDRL A061.

Standard: (1) 100% Compliance –CRLs and DoD/RTS trust anchors (i.e. root certificates) are within the period of validity; (2) 100% Compliance – Only authorized users and/or devices are allowed access.

3.4.1.9 Vulnerability Management Plan and Scans

The contractor shall develop and deliver a Vulnerability Management Plan (VMP) IAW CDRL A064. The contractor shall update the VMP annually. The contractor shall remediate, mitigate, and test security patches and STIG non-compliance findings within 90 calendar days of discovery. The contractor shall identify proposed remediation(s) for all identified risks, with an Impact Code of Critical, High, and Moderate, discovered through vulnerability scanning or testing. The contractor shall test and implement remediation(s) for all identified risks, with an Impact Code of Critical, High, and Moderate, discovered through vulnerability scanning or testing. The contractor shall provide mitigation recommendations to all identified risks, with an Impact Code of Critical, High, and Moderate that cannot be remediated and implement the mitigation recommendations upon ISSM email approval. The contractor shall perform network vulnerability scans of all RTS system components monthly or when changes are applied to the system. The contractor shall analyze network vulnerability scans and document the results of non-compliant security patch analysis and STIG findings (i.e. Information Assurance (IA) Vulnerability Alerts and Bulletins and CVEs) in a Plan of Action and Milestones IAW CDRL A065. The contractor shall deliver and upload the results of network vulnerability scans IAW CDRL A066 to Enterprise Mission Assurance Support Services (eMASS) within 72 hours of scan completion. The contractor shall report remediation, mitigation, and test activities in the Cybersecurity Vulnerability Report IAW CDRL A071. The contractor shall update the Plan of Action and Milestones IAW CDRL A065 with all remediation and mitigation activities for non-compliant security patches.

Standard: (1) 100% Compliance – Security Patches and STIG Non-compliance findings are remediated, mitigated, and tested within 90 calendar days of discovery; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

3.4.1.10 Application Static Code Testing and Analysis

The contractor shall perform application static code testing and analysis on RTS hosted applications after the delivery of custom applications or custom application updates as part of the contract. The contractor shall analyze application static code analysis test results and document the analysis of non-compliant security patches (i.e., IA Vulnerability Alerts and Bulletins, CWE and CVEs) and/or security findings in a Plan of Action and Milestones IAW CDRL A065 and Software Assurance Evaluation Report CDRL A068. The contractor shall deliver and update the results of application static code testing and analysis IAW CDRL A067 to eMASS within 72 hours of scan completion.

Standard: CDRL delivered with: (1) 100% content and data accuracy, (2) 100% data clarity and understandable diagrams, (3) 100% satisfies requirements stated in the contract, (4) 100% delivered on or before contract deadlines.

3.4.1.11 STIG/SRG Scans

The contractor shall perform automated Security Technical Implementation Guide (STIG) scans (i.e. SCAP, Eval STIG) on RTS systems quarterly. The contractor shall analyze automated/manual STIG scan data and document the results of non-compliant STIG findings analysis in a Plan of Action and Milestones IAW CDRL A065 and STIG Report CDRL A069. The contractor shall deliver and upload the checklists of automated/manual STIG scans, with false positives and Not Applicable STIG checks identified, IAW CDRL A070 to eMASS within 30 days of scan completion. The contractor shall perform manual STIG and SRG checklists on RTS systems semi-annually.

Standard: CDRL delivered with: (1) 100% content and data accuracy, (2) 100% data clarity and understandable diagrams, (3) 100% satisfies requirements stated in the contract, (4) 100% delivered on or before contract deadlines.

3.4.1.12 Incident Monitoring and Event Management

The Contractor shall address incidents within established DoD and Army guidelines and shall provide incident monitoring & event management support. The contractor shall participate in the incident management process to resolve and close escalated incidents within tasked timeframes and to regularly update the Government with troubleshooting actions, diagnosis of problem, and resolution actions. The contractor shall oversee response(s) to all incidents involving malicious or suspicious code to include, but not limited to, viruses, Trojan horses, worms, and macros. The contractor shall respond to malicious attacks, provide technical advice when required, and collect incident tracking information. The contractor shall maintain a database of all pertinent information relating to malicious code encounters and incidents including, but not be limited to viruses, users, organization, location, source, affected media, and whether the incident was internal or external to RTS. The contractor shall ensure applications, systems, and networks are continuously monitored and to enable rapid service restoration of IT services resulting from negative changes in the environment, thereby minimizing mission disruptions, and addressing security risks. Incident monitoring and event management activities shall be reported in the Cybersecurity Activity Report IAW CDRL A061. The contractor shall notify the ISSM immediately (within 15 mins) and complete an After-Action Report documenting the incident within 24 hours when the contractor identifies potential cybersecurity incidents. The contractor shall audit and analyze incoming and outgoing data communication traffic weekly to detect potential threats to internal and external systems and to identify users associated with denied communications. If analysis identifies potential threats or internal threats to external connections, the contractor shall notify the ISSM immediately (within 15 mins). The contractor shall report data communication analysis results in the Data

Communications Analysis Report IAW CDRL A062.

Standard: 100% compliance: (1) Incidents are resolved within tasked timeframes; (2) Audit and analysis of incoming and outgoing data communication traffic weekly.

3.4.1.13 Antivirus / Antispyware / Intrusion Detection and Prevention

The contractor shall implement anti-viral, intrusion detection and prevention tools as necessary including, but not limited to, configurations and dissemination mechanisms, filtering, blocking, and auditing. The contractor shall provide senior expertise, guidance, resources, and analysis for organizational and enterprise virus protection audits as required. The contractor shall provide reports of virus and/or malicious code encounters to RTS ISSM immediately. The contractor shall update malicious code signatures (i.e. Antivirus, Antispyware, Intrusion Detection and Intrusion) weekly. The contractor shall perform malicious code scans (i.e. Antivirus and Antispyware) and review the results of the scans weekly. The contractor shall report update and malicious code scan activities in the Cybersecurity Activity Report IAW CDRL A061.

Standard: 100% Compliance – Signature files are <= 7 days old.

3.4.1.14 Assessment and Authorization

The contractor shall develop and maintain up to date (quarterly or within 90 calendar days of system changes/modifications) assessment and authorization documentation using eMASS. Documentation shall include system categorization, security plan, appointment documentation, system authorization boundary diagram, hardware/software/firmware list, system enterprise and information security architecture, data and information flow diagram and paths, network connection rules, interconnected ISs and identifiers, encryption techniques, STIG/SRG applicability information, security control and inheritance information, overlay information, external connection information, Ports, Protocols and Services Management information (PPSM), and Plan of Action and Milestones (POA&M). The contractor shall update eMASS test results and security control artifacts when artifacts are updated, scan results are collected and/or changes are made to the environment. The contractor shall collaborate directly with the Government ISSM to deliver quality artifacts that are aligned with the Risk Management Framework and industry best practices for mapping documented security requirements to both planned and realized security controls. The contractor shall complete self-assessments 60 calendar days before independent security assessments are conducted. The contractor shall update and deliver the system categorization, security plan, system authorization boundary diagram, hardware/software/firmware list, data and information flow diagram and paths, PPSM, risk assessment report, and POA&M semi-annually IAW CDRL A065.

Standard: (1) 100% compliance with PWS task requirements; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before

contract deadlines; (3) System Level Baselines are $\leq$ 30 days old.

3.4.1.15 Continuous Monitoring

The contractor shall update, deliver and maintain a Continuous Monitoring Program and Plan IAW CDRL A072. The Continuous Monitoring Plan shall document continuous monitoring mechanisms, strategies, and approaches, provide and maintain a catalog of all continuously monitored system security controls, monitoring frequency, and monitoring approach (manual, automated etc.), a plan for maintaining all ATOs, updating assessment and authorization packages, identifying missing required cybersecurity documentation, updates with progress on the overall cybersecurity hygiene improvements and incorporating mitigation and/or remediation activities for new security assessment findings.

The Contractor shall provide a quarterly POA&M report IAW CDRL A065 to track the status of all POA&Ms for RTS systems and networks. If POA&Ms are due within the month, the Contractor shall work with the ISSM to get resolution on the POA&M item or if the POA&M date needs to be extended, the Contractor shall provide valid justification to the ISSM. The Contractor shall track the history of each POA&M to include the initial POA&M information and any modifications to the POA&M (i.e., justification(s) to the POA&M date).

Standard: (1) 100% compliance with PWS task requirements; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines; (3) System Level Baselines are $\leq$ 30 days old.

3.4.1.16 Data Calls and Mandate Responses

The Contractor shall take the appropriate actions to assist with response to Government data calls such as evaluating, applying, and responding to IA Vulnerability Alert (IAVA) Certification (CERTS), Bulletins, Cybersecurity Tasking Order (CTOs), and advisories within identified tasked timeframes. These data calls can originate from numerous sources, both internal and external to RTS. The Contractor shall be prepared to support IT-related data calls and assist in responding to data calls and mandates as directed by the Government. The contractor shall support the compilation and maintenance of information to develop responses to cybersecurity related data calls or investigations.

Standard: 100% adherence to PWS Task Statement.

3.5 Communications/Networks/Information Technology Services

The following sections detail specific services support responsibilities pertaining to operations and sustainment support for RTS Mission Communications, Networks and Information Technology services.

The contractor shall provide end-to-end operations, sustainment, and maintenance of RTS

Mission Networks and Communication Circuits, ensuring high availability, performance, and reliability for mission-critical and day-to-day operations.

The RTS mission communications, networks and information technology services consists of the following IT systems and assets owned by RTS: (1) RTS-Classified Network (2) RTS-Unclassified Network, (3) KMISS, (4) Starshield, (5) Mission DOTCOM, (6) RTS Voice, Video, and Mission Communications, (7) Optical Transport Communications to include battery power plant equipment (RTS Mission Networks and Communications Circuits backbone), (8) RTS Development Network, (9) RTS Top Secret Network, (10) Satellite Communications (SATCOM) Facility, (11) Radio and Pagers, (12) Monitoring and minor troubleshooting for Long-haul undersea fiber optic cable system(s) including support facilities and infrastructure, and (13) associated voice and data circuits.

The contractor shall provide IT services in the form of system administrators (desktop and server), service desk technicians, software application administration and maintenance support, database administrators, network engineers, firewall administrators and telecommunications and connectivity administration, support and maintenance to the Government in accordance with the requirements specified in this PWS.

All work as described and performed in the Mission Communications, Networks, and IT Services shall adhere to the ITIL established framework, as well as, DoDI 8100.04 DoD Unified Capabilities, DoDI 8310.01 Information Technology Standards in the DoD, DoDI 8320.02 Sharing Data, IT Services in the Department of Defense, DoDI 8320.03 Unique Identification (UID) Standards for Supporting the DOD Information Enterprise, DoDI 8320.07 Implementing the Sharing of Data, Information, and IT services in the Department of Defense, DoDI 8330.01 Interoperability of IT, Including National Security Systems (NSS), DoDI 8410.01 Internet Domain Name and Internet Protocol Address Space Use and Approval, DoDI 8440.01 DoD IT Service Management (ITSM), Army Regulation 25-1 IT.

NOTE: These services are separate and distinct from U.S. Army Network Enterprise Technology Command (NETCOM) and Defense Research and Engineering Network (DREN) SMDC G6 enterprise services. NETCOM provided baseline (Installation wide) C4IM services.

The contractor shall provide user and help desk support, system level administration, network engineering, applications and database administration, telecommunications and network administration support for all IT operations and sustainment contract line-item numbers (CLINs) while adhering to RTS established processes, procedures, and tools. Note: All applications, systems and networks covered by this contract are owned by the federal government; therefore, the Government shall be the final approving body for granting privileged access. The Government has the right to access all IT systems supported by the contractor.

The contractor shall provide software engineering and administration support for all applications and operating systems used on RTS unclassified and classified systems. The

contractor shall follow the current Software Capability Maturity Model Integration (CMMI) framework and apply the best industry software development practices that contain iterative and incremental project management techniques like agile software development lifecycle standards. The contractor shall establish and implement criteria for lifecycle software development and ensure all software applications remain compatible with RTS system and mission requirements.

The contractor shall develop, maintain, and periodically update Supply Chain Risk Management (SCRM) plans. SCRM plans are intended to reduce performance and security risks of the products sold, installed, and maintained throughout the lifecycle. SCRM plans shall include sufficient detail for the Government to determine that the contractor reasonably understands its supply chain and the associated risks. The contractor shall ensure that genuine Information and Communication Technology (ICT) will be available under the contract and shall manage the risk to ensure counterfeit or illegally modified products are not shipped. SCRM plans shall describe the processes and practices ensuring contractors can deliver genuine ICT. Within thirty (30) days of contract award, unless otherwise notified by the Government, the contractor shall make the SCRM plans available for review by the Government.

It is the intention of RTS to use Knowledge Management (KM) capabilities such as Microsoft Teams and SharePoint to develop and improve IT control, efficiency, and effectiveness. The contractor shall be responsible for curating, analyzing, creating, organizing, storing, using, and sharing all data generated as part of performing tasks identified within this PWS. The contractor shall make available and manage information for RTS IT services, applications, platforms, databases, and infrastructure, including but not limited to troubleshooting guides, user instructions, training materials, project management artifacts, architectural diagrams and documentation, system and network diagrams, configuration management tools, change management records, methodologies, processes, developed SOPs, system and network statistics, solutions, and decision-making rationale. The contractor shall ensure the Government has system and visual access to all Knowledge Management capabilities (IT applications, databases and collaboration tools) used to manage data as part of the contract and upon request provide the Government with reports or logs generated as output from Knowledge Management capabilities.

3.5.1 Communications/Networks/IT Operations and Sustainment

The contractor shall provide technical monitoring and assistance, managing operations and maintenance to ensure high Ao for mission communication, networks, and IT, including the backbone, power systems, the SATCOM facility, Optical Transport Network (OTN), long-haul fiber optic cable, and associated equipment. The contractor shall ensure network availability to pass the required metric and imaging data. The contractor shall provide operational and sustainment support to ensure RTS mission communications, networks and IT services achieve an overall network Operational Availability (Ao) of 95% during

normal operations and Ao of 99% during test and space mission support windows. The contractor shall proactively perform preventive maintenance on equipment quarterly to address potential failures IAW CDRL A024.

Note: Long-haul fiber optic cable shall not be included in calculations of network availability, but OTN or intra-atoll-island fiber optic cable shall be.

Standard: Network operational availability (Ao) of 95% during normal operations shall have at Total Downtime Per Year ≤ 18 days; Network availability Ao of 99% during test and space mission support windows shall have Total Downtime Per Year ≤ 90 hrs.

- a. The contractor shall ensure data rates remain available to transfer data without inhibiting other operations.
- b. The contractor shall ensure network connections to external networks are available to respond to Machine-to-Machine (M2M) taskings.
- c. The contractor shall actively maintain inter-network connectivity to support M2M interfaces and other cross-domain communications and solutions.
- d. The contractor shall ensure network connections remain available to provide near-real-time TIP data when producing imaging data.
- e. The contractor shall ensure network connections are available to publish data products and reports.
- f. The contractor shall utilize available, authorized data pathways IAW Government direction and with due regard to bandwidth capacity and requirements.

3.5.1.1 IT Support Service Desk

The contractor shall provide IT service desk support during core work hours for RTS Mission Communications, Networks and IT services. IT service desk support shall consist of telephonic, or email technical assistance, to include, but not limited to, remote technical support, emails for service requests, walk-in service requests, online portal service requests. The contractor shall provide on-site technical support during test mission support windows. The contractor shall provide on-call technical support including holidays during space mission support windows. The contractor shall provide immediate (service request first contact) service desk technical support, such as resetting passwords, diagnosing issues for supported hardware, software, applications, and services, and providing basic hardware/software support while utilizing the knowledge base of documented fixed actions, lessons learned, industry practices, and standards of troubleshooting. The contractor shall perform IT tasks such as setup, moving, delivery, and collection of IT equipment (within 24 hours) when users are onboarding/offboarding/moving office and/or workstation locations. The contractor shall immediately (within 15 minutes) create and triage service requests to properly determine actual need, prioritize the issue accurately, and flag the request for escalation based on test and space mission operations. The contractor shall regularly (every 24-48 hours) update IT service requests with description of the fixed action, diagnosis of problem, time of completion, the responsible point of contact, troubleshooting actions, and/or resolution actions. The contractor shall assign a service request priority

based on the following conditions:

- a. High Priority Outage: Complete disruption and/or outage of mission communications, networks and IT services required for supporting normal operations or test mission and space mission operations processing
- b. Medium Priority Outage: Moderate (50% or more) disruption and/or outage of mission communications, networks and IT services required for normal operations or test mission and space mission operations processing.
- c. Low Priority Outage: Minimal (25% or less) disruption and/or outage of mission communications, networks and IT services required for normal operations or test mission and space mission operations processing.

Standard: (1) $\geq 99\%$ of accounting for created service requests are updated and triaged appropriately; (2) PWS IT tasks (e.g. setup, moving, delivery, and collection) are completed within 24 hours.

The contractor shall record, analyze, maintain, prepare and submit an IT Service Desk Trend Analysis Report quarterly IAW CDRL A043 to identify recurring services incidents, avg service request response and resolution times, mission disruption and/or outage times, service request volume per IT service support areas, service request backlogs, and service request first contact resolution volume.

Standard: CDRL delivered with: (1) 100% content and data accuracy, (2) 100% data clarity and understandable diagrams, (3) 100% satisfies requirements stated in the contract, (4) 100% delivered on or before contract deadlines.

3.5.1.2 Restoration of IT Services and Response Time

Task: The contractor shall respond to high priority outages within 30 minutes of notification, 24 hours per day, seven days per week, including holidays. The contractor shall respond to medium priority outages within 1 hours of notification, 24 hours per day, seven days per week, including holidays. The contractor shall respond to low priority outages within 2 hours of notification, 24 hours per day, seven days per week, including holidays. During test and space mission operations, the contractor shall respond to all outages immediately (within 15 minutes) after notification. The contractor shall restore and/or resolve reported high and medium priority disruptions and/or outage of IT services, identified in service request, within 4 hours of service request, first contact and within 24 hours of service request first contract for low priority service disruption and/or outages. Exceptions will be allowed if a service request resolution requires assistance from outside Government agencies, third-party commercial vendors, and/or at a time stated within contracted/service level agreements. If issues are not completely resolved within the contracted (PWS) timeframes or resolution requires recovery and restoration of IT equipment, the contractor shall provide an explanation with a plan and timeframe for resolving the service request, close the service request and create a problem report that references the service request identifier, for tracking and monitoring.

Standard: $\geq 99\%$ of all service requests are resolved and closed within the contracted time as determined by priority.

Task: In the event the contractor anticipates not being able to restore and/or respond to disruptions and/or outages of essential IT services during test and space mission operations, the contractor shall notify the Government as expeditiously as possible and use its best efforts to cooperate with the Government in the Government's efforts to maintain the continuity of operations.

Standard: 100% adherence to PWS Task Statement.

3.5.1.3 Scheduled and Unscheduled Outages

The contractor shall notify the Government 30 calendar days prior to any scheduled IT service outage that will have an impact on any IT services. The contractor shall provide a detailed explanation for the outage to include the reason for the interruption, start and stop times, the duration of downtime and the IT services affected. The contractor shall perform all non-emergency disruptive maintenance, repairs, and upgrades (i.e., network operating system upgrades, server refresh, etc.) during non-business or mission hours or during agreed upon timeframes defined by the Government.

Standard: 100% adherence to PWS Task Statement.

The contractor shall provide immediate (within 15 minutes) notification (verbally or via email) to the Government when an unscheduled IT service outage occurs. If one or multiple IT services are impacted by an unscheduled IT service outage, the contractor shall notify the Government and assign a priority to the outage within a service request and establish a repair priority. After notification of unscheduled IT service outage verbally or via email, the contractor shall provide a documented follow-up notification that includes

detailed explanation for the outage to include the reason for the interruption, start and stop times, the duration of downtime and the IT services affected.

Standard: 100% adherence to PWS Task Statement.

3.5.1.4 Problem Reporting, Tracking, and Monitoring

The contractor shall implement and maintain a structured process and centralized database for logging, tracking, resolving, and reporting IT hardware, software, and performance issues to ensure 100% accountability and timely (within 5 business days) resolution. The contractor shall ensure problem reports are created immediately (within 15 minutes) and accurately (95% of the time) categorizes problem report severity and documents first-time and/or recurring IT service performance-related issues that caused mission disruptions and/or outages. The contractor shall notify Government Mission or Space Operations via

email (within 15 mins) of all problem reports causing mission disruptions or outages. The contractor shall maintain up to date (within 1 business day) problem report tracking information for all created problem reports. Created problem reports shall contain effective resolution methods which consist of but are not limited to root cause analysis, identified corrective actions, potential resolutions, and Service Level Agreement (SLA) agreed-upon resolutions. The contractor shall ensure that problem report resolutions that changes the configuration baseline of IT equipment are presented and approved by the Government CCB before implementation. The contractor shall create and deliver a quarterly Problem Report Tracking Log IAW CDRL A054 that documents average problem report resolution time, resolution times per problem report, SLA compliance, and recurring problems reported in a quarter.

Standard: (1) $\geq 99\%$ accountability of Problem Reports created, contain a resolution and received approval by Government CCB before change implementation; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

The contractor shall have weekly Problem Report review meeting with the Government to assist with categorizing, assigning severity and priority, re-establishing priority, reviewing present and potential problem resolutions for authority to proceed or close out approval from the Government.

Standard: Problem Report updates provided weekly during CLIN Period of Performance (PoP).

3.5.1.5 Mission Specific Knowledge Management via Government Enterprise

The contractor shall utilize the Government-provided enterprise information management system (Army365 SharePoint) as a knowledge base to manage, store, and organize all RTS mission technical and administrative data, including CDRL deliverables, maintain an IT activity log, and document problem resolutions that are related to recurring service requests and inquiries on IT services and software applications. The contractor shall ensure that all mission documentation is accurately maintained, regularly updated (within 2 business days), and readily accessible to authorized personnel. This shall include ensuring proper version control, timely uploads of new or revised documents, and adherence to Government-provided guidelines for data organization and security. The contractor shall support operational and administrative requirements by ensuring the enterprise information management system is used effectively to facilitate collaboration, knowledge sharing, and compliance with mission needs. It is expected that the contractor shall utilize the knowledge base as a guide to resolve reported issues.

Standard: Knowledge base updates are no later than 5 calendar days old.

3.5.1.6 Endpoint Management Operations and Sustainment

The contractor shall provide IT endpoint operations and sustainment support for RTS mission communications, networks and IT 24 hours per day, seven days per week, including holidays. IT endpoint management support shall include, but not limited to installation and checkout, deployment, administration, maintenance, repair, change modification, final disposition, account termination, telecommunication services (i.e. data, voice, and video), backup/restoration and upgrade to operating systems, software applications, peripheral equipment, printers and hardware utilized on user workstations (thin client, thick clients, laptops, workstations, video conferencing equipment). The contractor shall perform a weekly review of RTS user workstations (thin client, thick clients, laptops, workstations, video conferencing equipment) and endpoint devices to ensure (1) only approved applications and software is installed, (2) unsanctioned software installations are detected and blocked or removed, and (3) standardized application deployments are documented including application whitelists, (4) anti-malware protection software is operating and up to date and (5) operating systems are up to date with all vendor provided security patches and updates. The contractor shall record all IT endpoint management actions within an IT Activity Report IAW CDRL A053. The contractor shall install and test endpoint devices operating system upgrades and security patches for RTS IT networks, voice and video communication endpoint devices weekly or in a timeframe consistent with contracted third-party service agreements and/or change control requirements.

Standard: (1) $\geq 99\%$ of Endpoints are reviewed and scanned weekly and updated within 30 days of patch release; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

The contractor shall create and maintain system-level (e.g., operating system, firmware, patch levels, and application software) baseline images of RTS IT networks, voice and video communication endpoint devices monthly for recovery and restoration support. The contract shall maintain the documented and established configuration baselines and baseline exceptions for RTS Mission Networks and Communication Circuits endpoint devices. The contractor shall maintain the availability of endpoint monitoring and diagnostic tools to ensure health and status monitoring of RTS Mission Networks and Communication Circuits endpoint devices are conducted daily.

Standard: (1) System Level Baselines are ≤ 30 days old; (2) Endpoint monitoring and diagnostic tools and configuration baseline integrity tools operate at 99% Availability or Downtime Per Year ≤ 90 hrs.

3.5.1.7 IT Server Management Operations and Sustainment

The contractor shall provide IT server management support for RTS Mission Networks and Communication servers 24 hours per day, seven days per week, including holidays. Responsibilities include installation, deployment, administration, maintenance, repair,

change modification, account termination, backup/restoration, storage management, and upgrades to operating systems, virtualization, software applications, and hardware. The contractor shall perform weekly reviews of servers to apply security patches, anti-malware updates, and service packs, ensuring corrections are applied within 90 days of patch release. The contractor shall maintain system-level baseline images monthly for recovery and restoration support and monitor server health daily for signs of failure, network connectivity issues, and automated alerts. The contractor shall install and test server operating system upgrades and security patches for RTS mission networks and communication servers weekly or in a timeframe consistent with contracted third-party service agreements and/or change control requirements. The contractor shall record all IT server management operations and sustainment actions within an IT Activity Report IAW CDRL A053.

Standard: 100% of servers are reviewed weekly and corrected within 7 days; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

The contractor shall maintain the availability of server monitoring and diagnostic tools to ensure health and status monitoring of RTS IT networks, voice and video communication servers are conducted daily.

Standard: Server monitoring and diagnostic tools operate at 99% Availability or Downtime Per Year $\leq$ 90 hrs.

3.5.1.8 Application Engineering Support for Networks

The contractor shall provide IT software application management support for RTS Mission Networks and Communication software applications. IT software application management support shall include, but not limited to application installation and checkout, deployment, administration services (application, database, and software containers), application maintenance, troubleshooting support, fault isolation, change modification, final disposition, account termination, telecommunication services (i.e. data, voice, and video), backup/restoration, storage management, upgrade to operating systems, software applications, and hardware hosting the application and ancillary software on servers. The contractor shall perform a weekly review of software applications, databases, software containers and ancillary software to ensure application servers, database systems, software containers (1) are installed with only approved applications and software (2) unsanctioned software installations are detected, blocked and/or removed, (3) standardized application deployments are documented including application whitelists and (4) are up to date with all vendor provided security patches and the latest CM-approved version. The contractor shall record all IT software application management actions within an IT Activity Report IAW CDRL A053. The contractor shall install and test software applications, ancillary software, database, and software container upgrades and security patches for RTS IT networks, voice and video communication servers in a timeframe consistent with

contracted vendor service agreements and/or change control requirements. The contract shall coordinate and validate changes and upgrade with software application owners.

Standard: (1) $\geq 99\%$ of Software Applications, Databases, and Application Servers are reviewed weekly and corrected within 7 days; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines; (3) 100% CCB Technical Package contains application test results.

The contractor shall conduct a performance management analysis quarterly to ensure performance optimizations of software applications, services, systems, and networks while maintaining efficient and effective operations to maximize user productivity. The contractor shall perform utilization trend analysis quarterly to provide recommendations for equipment and service replacements and identify upgrades or enhancements to prevent the delivery of services from falling below acceptable levels as identified through resource utilization trends. The contractor shall document the results of the performance management and utilization trend analysis in the Network Utilization and Performance Trend Analysis Report IAW CDRL A054.

Standard: CDRL delivered with: (1) 100% content and data accuracy, (2) 100% data clarity and understandable diagrams, (3) 100% satisfies requirements stated in the contract, (4) 100% delivered on or before contract deadlines.

The contractor shall monitor and maintain the availability of software applications installed on RTS IT networks, voice and video communication servers and application server monitoring and diagnostic tools to ensure health and status monitoring of RTS IT networks, voice and video communication servers are conducted daily. The contractor shall maintain the documented and established configuration baselines and exceptions for software applications, ancillary software, databases, and software containers installed and hosted on RTS Mission Networks and Communication Circuits servers. The contractor shall record all IT software applications maintenance and monitoring actions within the IT Activity Report IAW (CDRL A053).

Standard: Server monitoring and diagnostic tools and configuration baseline integrity tools operate at 99% Availability or Downtime Per Year ≤ 90 hrs.

3.5.1.9 Warranty/Software License Compliance

The contractor shall track, maintain, and monitor warranties, maintenance/service agreements, and software licenses/agreement coverage periods to ensure 100% adequate coverage periods, adhere to applicable cybersecurity standards, copyright regulations, vendor requirements and to inform the Government when support levels are in jeopardy of being exhausted. The contractor shall notify the Government that warranties, maintenance/service agreements, and software licenses/agreement coverage will expire in 180 or fewer calendar days and submit purchase requisition(s) to the Government for

renewal at least 120 calendar days prior to the expiration date. The contractor shall provide a Warranties/Software License/Service Agreement Report that identifies coverage periods, costs, acquisition/procurement information and planned and actual purchase dates semi-annually IAW CDRL A073.

Standard: (1) 100% warranties, maintenance/service agreements, and software licenses/agreement renewals purchased before coverage expires.; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

The contractor shall manage a warranty program for failed IT equipment and repair IT equipment no longer covered by warranty/service agreement coverage periods. For IT equipment no longer covered by warranty/ service agreement, the contractor shall perform repairs and rework on assemblies and spare parts while adhering to cost-effective practices and established reliability and maintainability requirements. If repair costs exceed 60% of the current acquisition cost and have negative impacts on system performance and/or operational requirements, the contractor shall notify the Government to determine the appropriate course of action. The contractor shall coordinate maintenance/repair provided by external providers, including site visits and returns for repairs in accordance with terms defined within warranties, maintenance/service agreements, and software licenses/agreement.

Standard: >=99% of issues requiring maintenance/repair by the contractor or external providers are supported during schedule timeframes and/or milestones.

The contractor shall manage the disposal of obsolete and/or damaged IT equipment and GFE within 15 working days using the Government property management process. The contractor shall perform disposal actions IAW with DoD and/or Army regulation. The contractor shall manage, maintain up to date (within 1 business day), and deliver maintenance logs IAW CDRL A054 which track the supply, repair, disposal and inventory of all IT equipment undergoing rework and/or repair.

Standard: (1) >98% of IT equipment disposed of within 15 days; (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines.

3.5.1.10 Networks Operation, Maintenance, and Sustainment

The contractor shall provide network operations, maintenance, and sustainment support for RTS MDN and Communication Circuits, ensuring reliable performance, security, and availability. Responsibilities shall include, but not limited to installation and checkout, deployment, network device and firewall administration, AAA server administration, network

device and firewall maintenance, troubleshooting support, fault isolation, change modification, final disposition, account termination, network capacity planning, load balancing, network optimization and tuning, telecommunication services (i.e. data, voice, and video), backup/restoration, upgrade to operating systems, and cabling systems installation, maintenance, and upgrades. The contractor shall conduct weekly network utilization and performance trend analysis reviews, to ensure network devices are up to date, document infrastructure failure conditions including environmental issues, network conditions, bandwidth utilization, latency, packet loss and generated automated alerts.

The contractor shall ensure network data rates are available to transfer data without inhibiting other operations. The contractor shall utilize available, authorized data pathways with due regard to bandwidth capacity and requirements. The contractor shall ensure network connections are available to provide near-real-time TIP data when conducting imaging data and to publish data products and reports

The contractor shall document the results of the network utilization trend analysis in a Network Utilization and Performance Trend Analysis Report IAW CDRL A054. The contractor shall create and maintain system-level baselines maintained monthly for recovery and restoration.

Standard: (1) $\geq 99\%$ of Network devices are reviewed weekly (2) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines; (3) System Level Baselines are ≤ 30 days old.

The contractor shall monitor RTS Mission Networks firewalls daily for penetration attempts to evade security and maintain incident reports of such events. The contractor shall maintain the availability of network monitoring and diagnostic tools to ensure health and status monitoring of RTS Mission Networks and Communication Circuit network devices and infrastructure are conducted daily. The contractor shall record all IT network operation, maintenance, and sustainment activities and monitoring actions within the IT Activity Report IAW CDRL A053.

Standard: (1) Firewalls are always monitored via monitoring tools; (2) Server monitoring and diagnostic tools operate at 99% Availability or Downtime Per Year ≤ 90 hrs.

3.5.1.11 Voice, Video and Mission Communications Operations and Maintenance

The contractor shall provide comprehensive support for mission-critical voice and video communication systems, ensuring reliable and secure communication capabilities for RTS operations. Responsibilities shall include (but not limited to) operating, maintaining, troubleshooting, and repairing mission voice equipment to include associated channels, end-to-end communications operation, cable systems (analog/digital, fiber optic, and copper for voice and data) and video communication systems to include mission video displays and the mission video wall. The contractor shall review and inspect data, voice,

video mission communication circuits weekly for potential abnormalities, tampering, cybersecurity threats, and verify circuit remediation issues are being addressed and/or resolved. The contractor shall monitor the signal quality of data, voice, and video mission communication circuits daily, using diagnostic tools to identify and resolve potential issues, and record all monitoring actions in the IT Activity Report CDRL A053.

Standard: Circuit monitoring and diagnostic tools operate at 99% Operational Availability (Ao) or Downtime Per Year $\leq$ 90 hrs.

The contractor shall maintain an accurate and up-to-date inventory of all telecommunications equipment, including Service Delivery Points, Domain-based Message Authentication, Reporting and Conformance, circuit termination equipment, point of presence, and other pertinent information. The contractor shall develop and maintain a documented configuration baseline for all voice and video communication circuits, including as-built diagrams, schematics, rack layouts, and cable plant records. All cabling and components shall be installed, tested, and labeled in accordance with industry standards or Government-approved specifications, with deviations requiring prior approval from the Government Configuration Control Board (CCB).

Standard: $\geq$ 99% accountability of all configuration items and documentation, and all activities shall be conducted to ensure operational readiness, mission success, and compliance with applicable standards.

The contractor shall assist with outside Government organizations and/or telecommunication service providers to address and resolve backbone, wide area networks, satellite, hardware, and protocol issues and/or outages experienced on data, voice, video mission communication circuits. Responsibilities shall include monitoring system performance, addressing and resolving issues within the contractor's control, and promptly reporting failures caused by external factors, such as telecommunications link outages or scheduled maintenance downtimes. The contractor shall create service requests with the outside Government organizations and/or commercial vendors to assist with addressing and resolving issues and/or outages experienced on data, voice, video mission communication circuits. The contractor shall monitor network telecom service provider's circuits (used to support RTS data, voice, and video communications) daily to identify and resolve potential issues and to provide performance feedback based on contract SLAs. The contractor shall record all circuit monitoring actions within the IT Activity Report IAW CDRL A053.

Standard: (1) $>$ 90% of issues requiring outside Government organizations and/or telecommunication service providers are closed within the time specified in SLAs. (2) Telecom service provider's circuits are always monitored via monitoring tools.

3.5.1.12 Optical Transport Communications System

The contractor shall operate, maintain, troubleshoot, and repair OTN systems supporting

long-haul communications off Kwajalein and the intra-atoll communications ring connecting the six active defense sites (Kwajalein, Meck, Gagan, Illegini, Legan, and Roi-Namur). Responsibilities shall include ensuring reliable data transport, monitoring system performance, addressing outages, and performing preventive and corrective maintenance to minimize disruptions. The contractor shall ensure all transport systems are properly documented in accordance with the Outside Plant and Inside Plant Requirements defined within this PWS. This includes but is not limited to Circuit Layout Records (CLRs) for existing and new circuits and maintain compliance with applicable DoD and Army standards. This contractor shall monitor the optical transport communications system to identify and implement alternative routing during primary circuit outages, network performance monitoring to track bandwidth, latency, and packet loss, signal quality, diagnosing and repairing mission circuit faults, implementing alternate routing during outages, and intra-atoll transport circuit monitoring to maintain seamless connectivity. The contractor shall create and maintain system-level baselines maintained monthly for recovery and restoration. The contractor shall maintain the availability of network monitoring and diagnostic tools to ensure health and status monitoring of optical transport communication system devices and infrastructure. The contractor shall proactively detect and address issues, mitigate vulnerabilities, and respond to incidents promptly to minimize disruptions. All monitoring activities, incidents, and resolutions shall be documented and reported in the IT Activity Report IAW CDRL A053. The contractor shall conduct weekly network utilization and performance trend analysis reviews, to ensure network devices are up to date, document infrastructure failure conditions including environmental issues, network conditions, bandwidth utilization, latency, packet loss and generated, automated alerts. The contractor shall document the results of the network utilization trend analysis in a Network Utilization and Performance Trend Analysis Report IAW CDRL A054.

Standard: (1) $\geq 99\%$ compliance with PWS contract requirements (2) $\geq 99\%$ of Optical Transport Communication devices are reviewed weekly (3) CDRL delivered with: (a) 100% content and data accuracy, (b) 100% data clarity and understandable diagrams, (c) 100% satisfies requirements stated in the contract, (d) 100% delivered on or before contract deadlines; (4) System Level Baselines are ≤ 30 days old; (5) Circuit monitoring and diagnostic tools operate at 99% Operational Availability (Ao) or Downtime Per Year ≤ 90 hrs.

3.5.1.13 Outside Plant, Inside Plant, and Fiber Optics

The contractor shall provide comprehensive Inside and Outside Plant support to ensure reliable operation of mission-specific conduit systems, terrestrial long-haul cable infrastructure on Kwajalein, and the intra-atoll submarine fiber optic cable communications ring connecting the six active defense sites (Kwajalein, Meck, Gagan, Illegini, Legan, and Roi-Namur). The contractor shall configure components and install- fiber /copper cabling for inside plant (e.g., power, groundings, HVAC, racks, fiber optic distribution panels, equipment, internal cabling, comm. closet, etc.) as required by the Government or to repair communications. The contractor shall configure all components and install fiber/copper cabling between RTS facilities and assets for outside plant (e.g., fiber, manholes, duct,

building entries, trenching, digging, constructions, external cabling, etc.) as required by the Government or to repair communications. The contractor shall test all installed cable IAW accepted industry standards (unless superseded by a Government COR approval), remove dead-in-place infrastructure and perform repair and maintenance of damaged cabling. The contractor shall provide attached labels that are durable and legible, ensure proper labeling of all cabling and conduit systems, and create and maintain accurate Circuit Layout Records (CLRs) for existing and new circuits. The contractor shall create configuration baseline versions and maintain backup copies for all CLRs. The contractor shall install, maintain, and repair cabling within mission-specific conduit systems, ensuring compliance with industry standards and best practices. For any deviations to the specific installation specification, the contractor shall submit an engineering change proposal to the Government CCB for approval. The contractor shall provide continuous monitoring of Inside and Outside plant task requirements and repair services shall be performed within Government tasked timelines and milestones. The contractor shall document all outside plant, inside plant, and fiber optics activities in the IT Activity Report IAW CDRL A053.

Standard: 100% compliance with PWS task requirements.

3.5.1.14 Technical Control Facility

The contractor shall operate and maintain the RTS Technical Control Facility to provide 24 hours per day and 7 days per week monitoring, troubleshooting, and support for long-haul circuits, ensuring continuous and reliable operations within the RTS authorization boundary. The contractor shall provide continuous monitoring and maintenance of RTS operational systems, networks, and circuits to ensure optimal performance and reliability. This includes tech control-level monitoring for long-haul circuits, SATCOM-level monitoring to identify and implement alternative routing during primary circuit outages, network performance monitoring to track bandwidth, latency, and packet loss, systems monitoring to ensure software and operating system configurations are updated and tracked, and intra-atoll transport circuit monitoring to maintain seamless connectivity. The contractor shall proactively detect and address issues, mitigate vulnerabilities, and respond to incidents promptly to minimize disruptions. The contractor shall ensure network connections to external networks (i.e. SDIN, SDANet, BCN, SIPR and other cross-domain communications) are available to respond to machine-to-machine taskings. The contractor shall actively maintain inter-network connectivity to support M2M interfaces to include but not limited to, SDIN, SDANet, BCN, SIPR, and other cross-domain communications and solutions. All monitoring activities, incidents, and resolutions shall be documented and reported in the IT Activity Report IAW CDRL A053. The Technical Control Facility shall be always manned to provide immediate technical assistance, address Requests for Information, and support the Government across all geographical locations (ROC-H in Huntsville, Alabama and the Kwajalein Atoll), ensuring seamless communication and coordination regardless of location. Responsibilities shall include monitoring signal quality, diagnosing and repairing mission circuit faults, implementing alternate routing during outages, and maintaining accurate documentation of all circuit activities in accordance with RTS Instrumentation and Communications standards, Plant-In-Place Drawings, and CLRs

IAW CDRL A027. All activities shall adhere to industry's best practices and ensure operational readiness and mission success.

Standard: 100% compliance with PWS task requirements.

3.5.1.15 End User Video Telecommunications Support

The contractor shall provide classified and unclassified Video Telecommunication (VTC) support, including processing requests, scheduling sessions, resolving conflicts, and monitoring Video Telecommunication (VTC) events to ensure seamless operations. The contractor shall operate, maintain, repair, and modernize Government-furnished VTC systems as required, ensuring all equipment functions optimally and meets mission needs. This shall include cable management and maintaining a professional, presentable environment suitable for all levels of Government visibility. The contractor shall ensure all VTC sessions are connected and completed at the maximum available bandwidth. Requests shall be accepted via paper, email, phone, or in person, with confirmation of scheduling provided to the requestor within one working day. The contractor shall maintain up to date (within 1 business day) VTC Action Logs and Schedule Logs to document all scheduling and support activities, ensuring accountability, operational readiness, and a high standard of service. The contractor shall document end user video telecommunications support activities in the IT Activity Report IAW CDRL A053.

Standard: 100% Compliance with PWS task requirements.

3.5.1.16 Satellite Communications Facility

The contractor shall operate, maintain, troubleshoot, and repair the RTS SATCOM Facility 24/7 in compliance with DISA requirements, including the Wideband Satellite Communications Operations Security Classification Guide (SCG), DISAC 800-70-1, DISAC 310-55-1 (Status Reporting and Outages), and Wideband Standards and Operations Procedures. The contractor shall perform monthly SATCOM tests to verify system functionality, identify potential issues, and ensure readiness for mission requirements. All operations shall adhere to security protocols, reporting requirements, and standardized procedures to ensure reliable and secure satellite communication support for RTS missions. The contractor shall ensure the RTS Wideband SATCOM terminal maintains an Operational Availability (Ao) of 95% during normal operations and 99.97% during RTS Space Operations and RTS Test Mission Support windows when SATCOM is the primary path for instrumentation command and control (CDRL A024, CDRL A023). The contractor shall document satellite communications facility activities in the IT Activity Report IAW CDRL A053.

Standard: Standard: Network availability (Ao) of 95% during normal operations shall have at Total Downtime Per Year <= 40 days; Network availability Ao of 99% during test and

space mission support windows shall have Total Downtime Per Year $\leq$ 90 hrs.

3.5.1.17 Radio and Pagers

The contractor shall provide comprehensive radio services to support test missions and customer requirements, ensuring operational availability and reliable communication across RTS operations. Responsibilities shall include operating, maintaining, and repairing Government-provided radio systems to ensure optimal performance and uninterrupted service during mission-critical activities. The contractor shall operate, maintain, and manage the Government pager system, including programming customer-procured pagers and ensuring proper functionality of all pagers and associated infrastructure. All activities shall adhere to industry's best practices and be documented to ensure accountability, operational readiness, and mission success. The contractor shall document radio and pager activities in the IT Activity Report IAW CDRL A053.

Standard: 100% Compliance with PWS task requirements.

3.5.2 Network Support for Joint Worldwide Intelligence Communications System

The contractor shall provide lifecycle support for end-user hardware and peripherals utilized on RTS Joint Worldwide Intelligence Communications System (JWICS), including desktop computers, laptops, monitors, printers, plotters, and approved peripherals and accessories within 24 hours of notification from the Government. Lifecycle support shall encompass IT equipment installation and functionality testing of IT hardware, software, and peripherals to meet operational requirements and installation, for RTS Civilians, Military, and Contractors. The contractor shall ensure JWICS is operated and maintained in accordance with SLAs/MoUs/MoAs, applicable DoD and Army policies. Responsibilities shall include monitoring system availability, addressing and resolving issues within the contractor's control, and promptly reporting failures caused by external factors, such as telecommunications link outages or scheduled maintenance downtimes. The contractor shall assist outside Government organizations and/or telecommunication service providers to address and resolve JWICS issues and/or outages. The contractor shall create problem reports within RTS and help desk support requests with outside Government organizations and/or commercial vendors to assist with addressing and resolving issues and/or outages experienced on JWICS data, voice, video mission communication circuits. The contractor shall generate and maintain property passes for assigned JWICS owned equipment and ensure DA FORM 2062 (hand receipt/annex number) records are maintained in accordance with DA PAM 710-2-1. The contractor shall maintain a 100% hardware and software inventory list and identify lifecycle replacement for all hardware and software supporting RTS personnel using JWICS network. The contractor shall document network support activities for JWICS in the IT Activity Report IAW CDRL A053.

Standard: 100% compliance with PWS task requirements.

3.5.3 IT Support for NIPR and SIPR

The contractor shall provide lifecycle support for end-user hardware and peripherals utilized on NETCOM NIPR and SIPR DRENs, including desktop computers, laptops, monitors, printers, plotters, and NETCOM-approved peripherals and accessories within 24 hours of notification from the Government. Lifecycle support shall encompass IT equipment installation (e.g. relocation, delivery, user onboarding, offboarding, office or workstation moves, and tech refresh planning, inventory management, and property accountability), configuration, and functionality testing of IT hardware, software, and peripherals to meet installation and operational requirements. The contractor shall address and resolve issues promptly report failures caused by external factors, such as telecommunications link outages or scheduled maintenance downtime. The contractor shall ensure NIPR and SIPR are operated and maintained in accordance with SLAs/MOUs/MOAs, applicable DoD and Army policies, including security and access control requirements and monitoring system performance. The contractor shall assist outside Government organizations and/or telecommunication service providers in identifying and implementing an effective resolution (e.g. identifying root cause analysis, implementing corrective actions, verifying potential resolutions) or an SLA agreed-upon resolution for all issues and/or outages reported on NIPR and SIPR. The contractor shall generate and maintain property passes for assigned NIPR and SIPR owned equipment and ensure DA FORM 2062 (hand receipt/annex number) records are maintained in accordance with DA PAM 710-2-1. The contractor shall provide a 100% hardware and software inventory list and identify lifecycle replacement for all hardware and software supporting RTS personnel using NIPR and SIPR networks, as required by CDRL A056 and CDRL A057. The contractor shall document IT support for NIPR and SIPR in the IT Activity Report IAW CDRL A053.

Standard: 100% compliance with PWS task requirements.

3.5.4 Mission Business Network

The contractor shall provide comprehensive operations and maintenance support for the RTS Mission Business Network, as outlined in the RTS TFEM. The contractor shall ensure the availability and optimal performance of commercial internet services to meet the needs of RTS, its contractors, and mission customers and visitors at operational locations. The network shall operate and maintain on a non-DoD, commercially provided circuit in accordance with identified SLAs. Responsibilities shall include monitoring system performance, addressing and resolving issues within the contractor's control, and promptly reporting failures caused by external factors, such as telecommunications link outages or scheduled maintenance downtimes. The contractor shall implement cybersecurity best practices, assessment and authorization procedures to include TAA-compliant hardware, to protect the RTS Mission Business Network. The contractor shall assist outside Government organizations and/or telecommunication service providers to address and resolve Mission Business Network issues and/or outages. The contractor shall create problem reports within RTS and help desk support requests with outside Government organizations and/or commercial vendors to assist with addressing and resolving issues and/or outages experienced on Mission Business Network data, voice, video mission communication circuits. The contractor shall assist in

identifying and implementing an effective resolution (e.g. identifying root cause analysis, implementing corrective actions, verifying potential resolutions) and/or an SLA agreed-upon resolution for all issues and/or outages reported on the Mission Business Network. The contractor shall implement network inspection technologies to filter and inspect internet access, maintaining logs of access and traffic for a minimum of three months. The RTS Mission Business Network shall be maintained as a commercial internet connection without a required ATO. The contractor shall control and manage access to the network, ensuring that all users are aware of their responsibilities and the potential risks associated with using a non-accredited network. The contractor shall document Mission Business Network activities in the IT Activity Report IAW CDRL A053.

Standard: 100% compliance with PWS task requirements.

3.5.5 Development Network Maintenance (RTS-D)

The contractor shall operate, maintain, and sustain the RTS Development Network as a dedicated environment for hardware and software development, acceptance testing, and operational testing, ensuring no testing is conducted in production environments. The contractor shall ensure the development network provides a secure, reliable, and isolated platform for validating software and hardware functionality prior to deployment on RTS Classified and Unclassified networks.

Maintenance shall include ensuring availability, reliability, integrity, supportability, compatibility, and ease of use for all testing activities. The contractor shall conduct necessary testing to collect data and verify that maintenance requirements are met, presenting this data as required to demonstrate compliance. The contractor shall ensure the development network maintains authorization and adheres to all applicable standards, providing a robust environment to support RTS developmental efforts and mission readiness. The contractor shall assist outside Government organizations and/or telecommunication service providers to address and resolve RTS-D issues and/or outages. The contractor shall create problem reports within RTS and help-desk support requests with outside Government organizations and/or commercial vendors to assist with addressing and resolving issues and/or outages experienced on RTS-D. The contractor shall assist in identifying and implementing an effective resolution (e.g. identifying root cause analysis, implementing corrective actions, verifying potential resolutions) and/or an SLA agreed-upon resolution for all issues and/or outages reported on RTS-D. The contractor shall document RTS-D activities in the IT Activity Report IAW CDRL A053.

Standard: 100% compliance with PWS task requirements.

3.5.6 Network/IT Performance, Reliability and Maintainability

The contractor shall record, analyze, and maintain a quarterly IT R&M Data Analysis Report that captures reliability, availability, and maintainability data elements to allow

sufficient identification of IT and/or telecommunication equipment causing mission data processing and network communication disruptions and/or outages. The IT R&M Data Analysis report shall include the following data elements: IT and Telecommunication Equipment Identification, Serial Number, Material Number or National Stock Number, as applicable, disruption(s) and/or outages by network asset/component, circuit, and/or network services, categorized conditions during which the disruption or outage occurred (e.g., during mission windows, during normal operations, during maintenance and/or inspection, during replacement), repair costs, repair turn-around times for Third-Party/outside vendor repairs, disruption analysis/outage trends, network uptime, network downtime (based on the total period the IT network or a particular IT service is disrupted or not accessible), mean time between failure (MTBF), and Mean Time To Repair (MTTR). The IT R&M Data Analysis Report shall be delivered quarterly, IAW CDRL A054.

Standard: CDRL delivered with: (1) 100% content and data accuracy, (2) 100% data clarity and understandable diagrams, (3) 100% satisfies requirements stated in the contract, (4) 100% delivered on or before contract deadlines.

3.5.7 5G Support

The contractor shall monitor, operate, maintain, and document all 5G support services and technology required to meet the mission needs of RTS. The contractor shall provide engineering design support, technician support, and project estimation support for repairs and upgrades required to maintain 5G support services. The contractor shall ensure 5G support services are operated and maintained in accordance with Service Level Agreements (SLAs)/Memorandum of Understanding/Memorandum of Agreements, applicable DoD and Army policies, including security and access control requirements. Responsibilities shall include monitoring system performance, addressing and resolving issues within the contractor's control, and promptly reporting failures caused by external factors, such as telecommunications link outages or scheduled maintenance downtimes. The contractor shall assist outside Government organizations and/or telecommunication service providers to address and resolve 5G support issues and/or outages. The contractor shall provide a 100% hardware and software inventory list and identify lifecycle replacement for all hardware and software supporting 5G support services, as required by CDRL A056 and CDRL A057. The contractor shall document IT support for 5G in the IT Activity Report IAW with CDRL A053.

Standard: 100% Compliance with PWS task requirements.

3.6 Support Services Functions

The contractor shall provide support to RTS in the following areas.

3.6.1 Work Center for Technical Facility Maintenance

RTS is a tenant on the USAG-KA. As such the Garrison is responsible for maintenance and repair of facilities including normal/baseline creature comfort HVAC, plumbing, electrical wiring, building maintenance, etc. RTS is responsible for all above baseline systems due to RTS sensors and equipment requirements located inside or on Garrison facilities. The contractor shall provide above baseline facility maintenance and repair (such as air conditioning/equipment cooling/electrical/ machine shop mission critical work/antenna maintenance) for the facilities that enclose range instrumentation/systems as identified in the ISFM. The contractor shall also provide and track the status of RTS technical facilities projects for inclusion into the USAG-KA Annual Work Plan (AWP) and Installation real property management process. The focus of this effort will be to ensure Range Facilities/critical facility systems are secured, maintained, and repaired to support the success of mission objectives by range instrumentation/system assets.

3.6.1.1 Projects and USAG-KA Annual Work Plan Process

The contractor shall support RTS projects by providing relevant information to the USAG-KA Annual Work Plan (AWP) and installation real property management process. The contractor shall compile and prioritize RTS technical facilities projects for inclusion in the USAG-KA AWP. The contractor shall develop and submit facility plans necessary to accomplish RTS facility projects IAW CDRL A044.

Standard: Input into the USAG-KA installation and real property management process must be delivered to RTS for review/approval in time to meet those process input deadlines. Time metrics are defined in the CDRL.

3.6.1.2 Design and Engineering Support

The contractor shall provide design and engineering support for specialized electrical/mechanical systems at range technical facilities.

3.6.1.3 Air Conditioning Systems Supporting Instrumentation

The contractor shall provide experienced air conditioning and refrigeration mechanics to install and maintain all types of air conditioning equipment at RTS range facilities, including all RTS equipment and not limited to 190-ton chillers providing cooling to radar transmitters.

Standard: All air conditioning systems are 100% operational during mission support times and no damage occurs to mission support instrumentation/systems or equipment because of Air Conditioning and Refrigeration (AC&R) failures during non-mission periods.

3.6.1.4 Electrical Systems at All Range Technical Facilities

The contractor shall provide experienced maintenance personnel necessary to perform preventive maintenance and repair of electrical equipment and wiring at the range facilities, including but not limited to specialized control systems, motor generators, servo systems,

digital encoders, Uninterruptable Power Supply (UPS) systems, electrical distribution panels, transformers, and large switching equipment connected to high voltage feeders from the on-island power generating station.

Standard: All electrical systems remain 100% operational during mission support times and no damage occurs to mission support instrumentation/systems or equipment because of facility electrical failures.

3.6.1.5 Technical Facilities and Equipment Maintenance

The contractor shall provide experienced maintenance personnel necessary to perform preventive maintenance and repair tasks on instrumentation, IT equipment, networks, and facilities as listed in the RTS TFEM. The contractor shall create, update, and deliver an Annual Preventative Maintenance Plan IAW CDRL A058. The Annual Preventative Maintenance Plan shall include at a minimum a preventative maintenance plan, schedule and a list of planned preventative maintenance activities for each service support area listed above. NOTE: Preventive maintenance and repairs may be required more frequently due to the corrosive environment. Specialized maintenance is required for RTS Instrumentation such as antennas, towers, radomes, and transmitters.

Standard: All instrumentation, IT equipment, networks, and facilities are 100% operational during mission support times and no damage occurs to mission support instrumentation/systems or equipment because of facility electrical failures.

3.6.1.6 Technical Facilities and Equipment Manual

The contractor shall propose changes as needed to the RTS TFEM IAW CDRL A037 and A040 during the execution of this PWS if/when required. Government will maintain configuration control of the TFEM.

3.6.1.7 Obsolescence Surveillance

The contractor shall conduct an obsolescence surveillance (End-of-Life/End of Support) of RTS Mission Networks and Communication Circuits to include but not limited to endpoint devices, workstations, printers, peripheral devices, servers, application servers, software applications, ancillary software, databases, software containers, network and infrastructure devices, and telecommunications equipment annually. The contractor shall provide an Obsolescence Surveillance Report IAW CDRL A075. Obsolescence Surveillance Report shall identify obsolete hardware, software, firmware and acceptable substitutes/replacements that meet or exceed the system's reliability, availability, and maintainability requirements defined within the TFEM and ISFM.

Standard: Obsolescence surveillance of RTS Mission Networks and Communication Circuits is conducted within 60 days at the start of a new calendar year and report is submitted to Government within CDRL defined timeframes.

3.6.2 Calibration Lab Support

The contractor shall calibrate and repair Army TMDE using the guidelines in AR 750-43 for all entities across the Kwajalein Atoll.

3.6.2.1 TMDE Calibration and Repair

The contractor shall calibrate and repair all TMDE in accordance with the current Army TMDE program. TMDE consists of electronic measurement equipment, avionics, communication, temperature and pressure gauges, physical measurement tools and other general purpose precision equipment. TMDE also includes support to USAG-KA and other Garrison tenants' equipment in addition to the items used to support the RTS. The contractor shall maintain traceability to the NIST through compliance with the Army TMDE support program per AR 750-43, U.S. Army TB 750-25, and U.S. Army TB 43-180 and provide report IAW CDRL A039.

Standard: The contractor shall ensure RTS TMDE calibration delinquency is less than 2%.

3.6.2.2 TMDE Shipment

The contractor shall ship items which cannot be supported locally IAW local Standard Operating Procedures (SOPs) to designated locations for calibrations according to the current Support Agreement Between U.S. Army TMDE Activity and the USASMDC. This includes shipment and tracking of TMDE items that belong to the USAG-KA as well as other Garrison tenants. The contractor shall pick up and deliver items as required and maintain transfer logs recording items, times, and dates. The Government will have unlimited access to these logs.

3.6.3 RF Safety for Instrumentation

The contractor shall monitor and support RF safety with respect to RTS emitters on Kwajalein, Roi-Namur, Meck Island and elsewhere as required within the RTS Operational Area IAW DoDI 6055.11, AR 385-10 Chapter 7, DA PAM 385-24, and IEEE C 95.1, whichever is more restrictive.

3.6.3.1 Safety Engineering Support

The contractor shall provide RF Safety Engineering support with respect to RTS RF emitters on Kwajalein, Roi Namur, Meck Island and elsewhere as required within the RTS Operational Area IAW DoDI 6055.11.

3.6.3.2 Risk Assessments and Safety Evaluations

The contractor shall conduct RTS safety assessments and prepare and deliver RF safety assessment reports to the RTS Mission Safety Division. Contractor assessments shall include RF safety evaluation and/or review for RTS Instrumentation and visiting range sensors (RTS mission customer instrumentation). Safety assessments shall be coordinated with the USAG-KA and provided IAW CDRL A035.

3.6.3.3 RF Surveys

The contractor shall perform RF surveys and maintain databases for RF surveys and emitters IAW CDRL A033.

3.6.3.4 RF Emitters Procedures

The contractor shall coordinate real-time approval to operate RF emitters related to RTS operations with other local operators of RF emitters. The contractor shall be responsible for issuing their own real-time approval to operate RF emitters related to RTS operations.

3.6.3.5 RF Exposure Safety Training

The contractor shall provide training as required for local personnel and visitors with respect to RF safety and exposure. The contractor shall report on the status of this training IAW CDRL A010.

3.6.3.6 RF Sensor Safety Oversight

The contractor shall evaluate and monitor RTS range and RTS mission related emitters including visiting tactical weapon systems / assets. Evaluate designs, modifications and upgrades, monitor testing, approve test plans and provide reports. The contractor shall ensure that emitter designs are in compliance with standards and regulations as part of RF Safety Assessment reports IAW CDRL A035.

3.6.3.7 RF Radiation Monitoring

The contractor shall implement the RTS RF Radiation Monitoring (RADMON) program to include evaluation of changes to sensor systems affecting RF Safety, and monitoring of RF radiation of RTS Instrumentation testing / operations. The contractor shall maintain a daily (weekdays) 10-hour per day manned presence in the RADMON control office. The contractor shall notify the Government of all anomalies detected in the use of RADMON.

3.6.3.8 RF Incident Analysis

The contractor shall conduct initial RF incident analysis. The contractor shall determine whether RF safety exposure has or has not occurred. The contractor shall ensure all suspected RF exposure incidents are reported to RTS Mission Safety Division within two

hours IAW CDRL A034.

3.6.3.9 RF Incident Investigation

The contractor shall investigate RF incidents and report its findings to the Government. Contractor investigations shall include identification, evaluation, and reporting of incidents of suspected RF exposure. The contractor shall utilize RADMON and other technical sensor information to determine if the incident is or is not an incident of exposure. The contractor shall provide recommended corrective actions as warranted by the results and implement within 60 calendar days of incident IAW CDRL A034.

3.6.4 Environmental Support

The contractor shall support environmental compliance for all government activities on the Kwajalein Atoll. The USAG-KA manages environmental compliance for the Installation and its activities. The contractor shall support environmental services operations relating to RTS activities within the RTS Operational Area. The UES is a government-provided supporting document to this PWS.

3.6.4.1 Spill Response Material / Kit Provision for RTS Missions

Relative to RTS missions, the contractor shall provide spill response materials and spill kits, conduct spill response actions, and assist and coordinate with USAG-KA base operations

contractor in responding to larger spills. The contractor shall ensure spill response is performed in accordance with UES and the KEEP.

3.6.4.2 National Environmental Policy and Records

The contractor shall assist in the preparation of National Environmental Policy Act (NEPA) documentation and assist in preparation of Records of Environmental Consideration (REC) documentation.

3.6.4.3 Environmental Protection Compliance

The contractor shall ensure compliance with UES, AR 200-1, HPP, and KEEP with Documents of Environmental Protection (DEP) specific to RTS range activities.

3.6.4.4 Environmental Management System/Quality Control Committee

The contractor shall participate in both the USAG-KA Environmental Management System (EMS) and the USAG-KA Environmental Quality Control Committee (EQCC). Participation entails attendance at meetings (currently quarterly – frequency subject to change) and provision of RTS mission related inputs into Garrison EMS and EQCC planning activities.

3.6.4.5 Solid and Hazardous Waste

The contractor shall store, collect and transfer to USAG-KA, range operations-specific (within the RTS AOR) solid and hazardous waste IAW UES, KEEP, and Garrison SOPs.

3.6.4.6 Hazardous Material Use and Storage Oversight

The contractor shall maintain an inventory of Polychlorinated Biphenyl (PCB) equipment, asbestos and lead containing materials, as well as Ozone Depleting Chemicals (ODC) used for RTS range mission specific purposes IAW UES, KEEP, and Garrison SOPs.

3.6.5 Mission Specific Kwajalein Installation Logistics Support

The contractor shall serve as liaison to USAG-KA installation support elements for RTS Mission specific logistics support requirement identification, coordination and consolidation.

The contractor shall coordinate with USAG-KA for required Garrison logistical support of Range instrumentation and technical facility maintenance.

3.6.6 Government Property Management

The contractor shall provide RTS Government Property Management.

3.6.6.1 Accountability Records Maintenance

The contractor shall maintain property accountability records in accordance with FAR 52.245-1 (and associated clauses 52.245-2 Government Property Installation Operation Services, 52.251-1, Government Supply Sources, DFAR 252.211.7003 Item Unique Identification Valuation, 252.242-7005 Contractor Business Systems, 52.237-2 Protection of Government Buildings, Equipment and Vegetation, 252.245-7003 Contractor Property Management Administration).

3.6.6.2 Government Property Investigation and Reporting (Issued to the Contractor)

The contractor shall report and investigate all loss and or damage of Government property as prescribed in FAR 52.245-1.

3.6.6.3 Repair or Replacement of Government Property (Issued to Contractor)

The contractor shall repair or replace damaged, destroyed, or lost Government furnished property IAW the liability provisions of Contract Section I clause FAR 52.245-1 and requirements of DFARS 252.245-7002. This repair or replacement shall be accomplished without significant disruption of normal operations and executed through established supply and service channels. Property shall be considered acceptable unless it is Beyond Economical Repair (BER). BER is when the estimated cost of parts, labor, and potentially

data recovery surpasses the current market value of the property.

3.6.6.4 Excess Property Identification and Turn-In

The contractor shall report excess in the Procurement Integrated Enterprise Environment (PIEE) GFP Module in accordance with DFAR 252.245-7005 Management and Reporting of Government Property and FAR 45.6.

Excess property that is working condition will be screened on Island by tenant DOD/Government organizations for reutilization. The screening will be coordinated by the Cognizant Plant Clearance Officer (PLCO) or the Contracting Officer.

Property that is in scrap condition will not be screened and the PLCO will provide the following disposition instructions.

- a. All electronic devices will be sent to Defense Logistics Agency (DLA). Computers/Servers will have the hard drives removed (the removal must be witnessed by a US Government representative) and then destroyed (this destruction must also be witnessed). The carcass of the computer/server will be sent to DLA.
- b. All GP that requires a DEMIL code of B, C, D, E or F will be disposed of in accordance with the PLCO's instructions.
- c. For those assets in working conditions that are not needed by any of the tenant US Government organizations; do not require DEMIL, or to be sent to DLA; The following document Title 43-Foreign Affairs, Citizenship and Immigration, Chapter 11- Right of First Refusal (Granting and Facilitation Act 2023) shall be followed.
- d. The Kwajalein Atoll Local Government Authority and the Kwajalein Atoll Development Authority will be given the opportunity to acquire the excess property.

3.6.7 Range User's Manual

The contractor shall maintain and update the RTS Range User's Manual (RUM) IAW CDRL A015 and A040 during the execution of this PWS if/when required. Government will maintain configuration control of the RUM. The RUM provides a comprehensive document delineating how a new or existing customer interfaces with the RTS.

3.6.8 Data Item Description Manual

The contractor shall provide recommended updates to the RTS Data Item Description Manual IAW CDRL A016 and A040 during the execution of this PWS if/when required. Government will maintain configuration control of the manual. This manual delineates standard data products that are available from RTS. This document is given to existing and potential range users/customers for program/mission planning purposes. The data from this manual includes information received as input from other technical contractors supporting RTS as provided by the Government.

3.6.9 State of the Range Briefing

The contractor shall provide periodic State of the Range briefings that: (1) provide RTS management with more specific insight into the overall status of RTS sensors and systems; (2) compare current performance metrics to metric baselines in the ISFM; (3) identify sensor and system risks/issues and recommended mitigations; (4) provide opportunities and needs to improve RTS sensors, systems and personnel performance, and/or reliability; (5) and assess sensor and system obsolescence and future capabilities for known needs IAW CDRL A060.

3.7 Transition

3.7.1 Phase-In

The contractor shall execute the Government-approved Transition/Phase-In plan (developed as part of contractor proposal) with minimal disruption to incumbent contractor operations.

Standard: The contractor shall ensure Transition/Phase-In is executed during a 90 calendar day period with no more than five Government validated client complaints per month regarding smooth transition of functions.

3.7.2 Phase-Out

The contractor shall fully support and cooperate with any successor contractor in the phase-out of performance under this contract. All phase-out transition support shall be executed within the prescribed phase-out period. At a minimum the contractor shall provide the following:

- a. Identify all accounts and subscription services to be terminated or transferred to the successor contractor within the first two weeks of Phase Out start. If accounts/subscription services are to be transferred, ensure that such transfers are executed in a timely manner prior to performance end date.
- b. Execute a full joint inventory (with Government and successor contractor) and transference of all Government Furnished Material inventory (including all records).
- c. Provide status of all items in inventory as well as the items that have been inducted into repair facilities, being modified or being purchased.
- d. Identify and execute the removal of all contractor's owned items.
- e. Submit to the Government a critical path schedule for risk mitigation of the transition period. This schedule shall be included in the Transition Phase-Out Plan.
- f. Make available training and certification records of incumbent personnel who are considering employment with the succeeding contractor.
- g. Fully cooperate with and support the successor contractor in assuming all on-

going RTS customer mission requirements in a manner that minimizes any disruption to mission schedules.

The contractor shall develop and submit a Transition Phase-Out Plan IAW CDRL A011 detailing the specific plans, processes, and procedures for executing the above identified phase-out requirements. The Government will specify the delivery date of this Plan via Contracting Officer's Representative or Contracting Officer direction.

4. GENERAL REQUIREMENTS

The following section specifies the performance objectives and performance elements for this contract.

4.1 Contract Administration and Management

The contractor shall establish clear organizational lines of authority and responsibility to ensure effective management of the resources assigned to the requirement. The contractor must maintain continuity between the support operations at Kwajalein Atoll and Huntsville, Alabama, and the contractor's corporate offices. The contractor will manage work distribution to ensure there are no Organizational Conflict of Interest (OCI) considerations.

The contractor shall be responsive to occasional requests for special data/reports by the contracting officer or the COR and provide reports as required.

4.2 Subcontract Management and Procurements

The contractor shall be responsible for any subcontract management necessary to integrate work performed on this requirement and shall be responsible and accountable for subcontractor's performance on this requirement. The prime contractor will manage work distribution to ensure there are no OCI considerations.

The contractor shall implement a procurement approval process for execution for all Purchase Requests (PRs) submitted to the Government Contracting Officer's Representative (COR). All PRs shall include the supporting documentation to include associated CCB number and approval documentation for PR approval. The PR approval process shall be applied to all procurements that have been approved by the CCB for testing and/or deployment into the operational baseline and/or on operational networks, both maintenance and sustainment, I&M efforts/projects, and other Government requested equipment and services. All items requested in the annual operating budget to be installed or deployed on RTS systems and networks shall be procured by the contractor's Cyber, IT, or Communications group, have prior CCB approval for procurement for testing and/or deployment, and shall include the CCB number with approval documentation.

4.3 Key Contractor Personnel, Disciplines, Specialties

The contractor shall accomplish the assigned work by employing and utilizing qualified personnel with appropriate combinations of education, training, and experience, as well as all required certifications, licenses, etc.

The contractor shall match personnel skills to the work or task with a minimum of under/over employment of resources. The contractor shall provide the necessary resources and infrastructure to manage, perform, and administer the contract.

Key contractor personnel are discussed below.

4.3.1 Program Manager

Serves as contractor's single contract manager / director and representative authorized to interface with the Contracting Officer, COR, and Government customer representatives concerning the ROSIE contract. Shall have a minimum of twelve years of relevant senior leadership and program manager experience with DoD programs and a bachelor's degree from an accredited college or university or nine years of relevant senior leadership and program manager experience with a masters' degree from an accredited college or university.

Demonstrated experience must include capability to lead project development from inception to deployment completion, expertise in the management and control of funds and resources using complex reporting mechanisms and demonstrated capability in managing multi-task contracts and/or subcontracts of various types and complexity.

4.3.2 Instrumentation Operations Manager

Serves as the RTS operations manager working within and managing RTS Instrumentation Operations & Maintenance. Prepares and coordinates the development of RTS Instrumentation maintenance and Improvement and Modernization plans for contract scope of work. Creates and implements measurement systems for established processes, analyzes performance metric results, and identifies improvements.

Must have a Master of Science degree from an accredited college or university in Engineering or scientific field of study and fifteen (15) years of experience in progressively more complex work experience in engineering or scientific disciplines. Must have prior experience in operations, planning or ballistic missile test mission planning; familiarity with radar, telemetry, and/or optics concepts, operations, and maintenance. A PhD in a related science and engineering field may be substituted for three (3) years of relevant experience.

4.3.3 Chief Engineer

Serves as the RTS Instrumentation /Improvements and Modernizations Chief Engineer managing efforts in a research and development environment.

Ensures that a performance baseline is maintained in configuration control using approved change control procedures. Establishes format and content of summaries and reconciliations, including segregation of cost according to level of confidence and identification of areas of risk. Plan, coordinate, and establish a level of quantification appropriate to the objectives of the effort and to the level of scope definition for each discipline or specialty. Measures the execution and completion of all RTS instrumentation related ECPs and related deliverables within budget. Creates and implements measurement systems for established processes, analyzes performance metric results, and identifies improvements. Supports project organizational and administrative activities.

Executes simultaneous tasks within broadly defined parameters.

Must have a Master of Science degree from an accredited college or university in Engineering or scientific field of study and fifteen (15) years of experience in progressively more complex work experience in engineering or scientific disciplines. Must have prior experience in operations, planning or ballistic missile test mission planning; familiarity with radar, telemetry, and/or optics concepts, operations, and maintenance. A PhD in a related science and engineering field may be substituted for three (3) years of relevant experience.

4.3.4 Safety Manager

Manages RTS safety system operations and maintenance in support of RTS operations during research, development, test, and/or deployment of RTS systems. Provides safety analyses, safety engineering assessments, reviews system acquisition and program planning documents, engineering change proposals, deviations, waivers, mishap reports, and failure data for safety impact in relation to execution of RTS Range Mission operations.

Must have a bachelor's degree in engineering from an accredited college or university. Eight years of experience performing or managing system safety engineering activities during the development of complex systems and a minimum of five (5) years safety managerial experience. Must have knowledge of engineering techniques and system safety practices to identify, assess and resolve hazards. Shall have the ability to communicate effectively, both orally and in writing, in working solutions to problems or questions relating to work.

4.3.5 Information Systems Manager

The contractor shall provide an Information Systems Manager to oversee all activities related to RTS Mission Communications, IT support, and cybersecurity services.

Shall interpret Government, corporate, and user requirements, provide contractor direction for developing, installing, and maintaining information systems, and ensure contract-wide compliance with Army and DoD cybersecurity standards. Responsibilities include managing systems analysis, computer operations, and cybersecurity compliance, as well as ensuring alignment with industry best practices and emerging technologies.

Must have a bachelor's degree in a relevant field (e.g., Information Technology, Computer Science, or Cybersecurity); a master's degree is preferred. The manager shall have at least ten (10) years of experience in communications, IT, and cyber-security operations, including three (3) years of experience managing a cybersecurity program. Current IAM Level II certification (e.g., CAP, CISSP, CISM, or equivalent) is required to meet DoD 8570.01-M standards.

4.4 Location and Hours of Work

The primary locations for work are Kwajalein Atoll, RMI, and Huntsville, Alabama. Secondary locations may include various contractors, subcontractor, and Government facilities. Normal workdays are Monday through Friday in CONUS and Tuesday through Saturday on Kwajalein Atoll except U.S. Federal Holidays. The standard work week will be based on working eight hours per day, 40 hours per week. Surge hours will be required in support of RTS range customer missions on a case-by-case basis. RTS Mission windows will be coordinated with both the RTS customer and the contractor through the RTS mission planning process and delineated / defined within mission support

documentation as coordinated through the contracting officer and contracting officer's representatives.

4.5 Travel / Temporary Duty

Travel to other Government facilities or other contractor facilities (known as Temporary Duty (TDY)) may be required subject to oversight by and approval of the cognizant COR. The contractor shall manage travel within preapproved contractor travel budget limitations. All travel requirements are on a cost reimbursable basis only. Costs for travel shall be billed in accordance with the regulatory implementation of Public Law 99-234 and FAR 31.205-46 travel costs (subject to local policy and procedures).

4.6 Program Management

The contractor shall provide program management for the full range of work under this PWS. The contractor shall plan, organize, staff, control, and direct an organization that proactively promotes minimal levels of management, maximum effective supervisory span of control, and the empowerment of employees by the delegation of authority and responsibility to the lowest practicable level. The contractor shall align organizational goals with RTS's evolving goals, strategy, and objectives.

4.6.1 Integrated Master Schedule

The contractor shall develop, maintain, update, and deliver monthly to RTS a resource-loaded Integrated Master Schedule (IMS) in Microsoft Project to level 6 following the Government approved WBS to be used in execution and reporting of tasks planned, completed, and used to deconflict missions and sustainment activities. The IMS shall be IAW CDRL A059.

4.7 Financial Resource Management

The contractor shall provide Financial Resource Management IAW the Government-provided Cost and Software Data Reporting Plan (CSDR) DD Form 2794.

4.7.1 Annual Operating Budget

The contractor shall develop and maintain a contractor Annual Operating Budget (AOB) based upon Fiscal Year (FY) funding allocated IAW the Government's notification CDRL A001.

4.7.2 Contract Work Breakdown Structure

The contractor shall develop and maintain a Contract Work Breakdown Structure (CWBS) IAW the purpose and intent of MIL-STD-881F (or current version) to be used as the reporting structure for tasks and cost accounting data IAW CDRL A002.

4.7.3 Budget Cost Variance Reporting

The contractor shall provide budget cost variance reports IAW CDRL A003.

4.7.4 Contract Funds Status Report

The contractor shall provide a Contract Funds Status Report (CFSR) IAW CDRL A052.

4.7.5 Cost Reporting for RTS Customer Reimbursable Support

The contractor shall track actual mission support costs for all activities for each RTS test mission program IAW FAR Part 31. The contractor shall document and deliver the actual mission support costs using the Customer Reimbursable Cost Report IAW CDRL A004.

4.8 Configuration Management

The contractor shall implement and maintain a comprehensive Configuration Management process to ensure all changes to RTS systems, including hardware, software, and documentation, are systematically managed and tracked. The contractor shall provide a Configuration Management Plan (CMP) IAW CDRL A042. All changes impacting configuration baselines or items shall be submitted to the Government Configuration Control Board (CCB) for approval via Engineering Change Proposals (ECPs), Engineering Deviations, or Firewall Exception Requests, with supporting documentation that includes test results, cybersecurity impact assessments, and vulnerability scans. The contractor shall ensure all Purchase Requests (PRs) submitted to the Government Contracting Officer's Representative (COR) include items that have been approved by the CCB for testing and/or deployment on operational networks, along with the associated CCB number and approval documentation. All items requested in the annual operating budget to be installed or deployed on RTS systems and networks shall be procured by the contractor's Cyber, IT, or Communications group, have prior CCB approval for procurement for testing and/or deployment, and shall include the CCB number with approval documentation. The contractor shall maintain a Configuration Management Database (CMDB) to track and log all configuration items and baselines. Configuration audits shall be conducted to verify

compliance, and all activities shall adhere to established change management processes to ensure system integrity, security, and operational readiness. Management.

Standard: The contractor shall ensure 99% Operational Availability (Ao) of tools and repositories used for configuration.

4.8.1 Configuration Identification

The contractor shall perform configuration Identification processes IAW the CMP to accurately and completely identify all RTS hardware and software items at the appropriate level.

4.8.2 Change Control and Status Accounting

The contractor shall implement change control and status accounting processes IAW the CMP. Change Control ensures that all proposed configuration changes are documented, processed, coordinated, and approved in accordance with established policies and procedures.

4.8.3 Configuration Audits

The contractor shall perform scheduled and unscheduled audits to confirm the 'as-built' equipment matches the 'as-designed' documentation. The contractor shall establish procedures for configuration audits IAW the Government-approved CMP. The contractor shall keep internal records of audits that shall be available for periodic Government inspection.

4.8.4 Configuration Control

The contractor shall implement a Configuration Control Board (CCB) process IAW the CMP to ensure no changes are made to RTS sensors without prior CCB approval.

Configuration control processes ensure that coordinated engineering evaluations are performed, including a cost and technical impact, for each proposed change(s) processed for submittal to the CCB. The contractor shall ensure all ECPs/EPEs are processed by CCB IAW the CMP and that ECP/EPE disposition requires no more than ten working days to complete the CCB process once submitted for CM.

4.8.5 Technical Library

The contractor shall maintain an RTS Technical Library which contains cataloged reference books and periodicals. This technical library shall provide access to all program documentation (UDS), mission test documentation, technical manuals, technical documentation, CDRLs and vendor literature used/developed in support of the mission and/or technical aspects of this contract. The contractor shall prepare an index of

documentation via an electronic folder structure. These documents shall be kept in electronic form. The contractor shall ensure the RTS Technical Library is maintained and contains current versions of all Program Documentation, Mission Test Documentation, technical documentation, manuals and CDRLs developed and/or used in support of this contract. The contractor shall provide a Data Accession List (DAL), an index of documentation produced under this contract but not required for delivery as a CDRL. Access to the DAL shall be provided to Government and other Government-approved agencies as required. The Government may request data delivery of any document on the DAL IAW CDRL A013.

4.9 Agile Software Development and Management

The contractor shall demonstrate compliance with Army Directive 2024-02, including modern Agile software development methodologies, Development, Security, and Operations (DevSecOps) practices, and Continuous Integration / Continuous Delivery (CI/CD) principles. These principles shall ensure iterative delivery and continuous improvement of secure, high quality, user-centered solutions modularity, open standards, and modernized practices. These teams shall work iteratively to define, prioritize, and deliver solutions that meet operational needs while ensuring compliance with Army cybersecurity standards. All software development plans shall be submitted IAW CDRL A076 and software assurance reports IAW A068. A sample list of development methodologies to consider (but not limited to):

- a. Agile methodologies (sprint planning, daily standups, backlog grooming)
- b. Engage users (user personas, problem space definitions, wireframes)
- c. Multi-Disciplinary teams (configuration management, DevSecOps, user interface)
- d. Personalized solutions
- e. Accessibility compliance
- f. User information protection
- g. Automate testing
- h. Continuous user/developmental teaming
- i. Continuous Integration/Continuous Delivery pipeline

4.10 Risk Management

The contractor shall establish a Risk Management (RM) process and Risk Management Plan (RMP) IAW CDRL A005 for safety as required in AR 385-10 and in accordance with SMDC Tech Center Risk Management Plan Rev 1.0 to identify, assess and mitigate risks to program cost, schedule, performance objectives, and to support evaluation of alternative approaches if required. The contractor shall state risk in terms of uncertainty relative to specific cost, schedule, and/or performance parameters of the program or plan whenever possible. The contractor shall incorporate the elements of the risk management process into all projects and weekly activities and status meetings.

4.10.1 Identification

The contractor shall identify and capture those uncertainties known as risks that have the potential to result in significant safety, cost, schedule, or performance deviations from the program plan. The contractor shall use risk identification as part of an organized assessment process aimed at establishing a total program risk baseline. Risk shall be continuously assessed thereafter based on the results of developmental activities and progress relative to the program plan, and periodically to support major program events and decisions, such as technical reviews.

4.10.2 Risk Assessment

The contractor shall use a risk analysis and ranking process to quantify the likelihood and impact of a risk to determine the risk level (high, medium, or low) and to determine if formal management is required to support mitigation and control planning.

4.10.2.1 Risk Analysis

The contractor shall use a risk ranking approach to provide a prioritization of the risk by quantifying, to the degree practical, the probability/likelihood and severity/consequence of the risk. The risk analysis shall consider safety, cost, schedule, performance and/or supportability impacts and provide an estimate of the degree of uncertainty or likelihood.

The consequence analysis shall identify safety, cost, schedule, and/or performance impacts of the risk event, and when the risk impact will occur. The impact shall be determined and understood well in advance of its occurrence.

4.10.2.2 Risk Ranking

The risk shall be ranked or assigned a risk rating that defines its risk level (high, medium or low) or its potential to adversely affect the program. High and medium risks shall be managed more formally with documented mitigation plans and regular progress reviews provided for leadership. Low level risks may be managed at team level.

4.10.3 Risk Mitigation

The contractor shall use risk mitigation to reduce the likelihood and/or consequence of a risk. Risk mitigation may require compromise by making cost, schedule, and/or performance trades to reduce risk. Consequently, it is important that both the risk and the proposed mitigation are understood to ensure the mitigation is worth the effort.

4.10.3.1 Risk Mitigation Planning

The contractor shall develop and implement risk mitigation plans for medium and high-level risks. Initial mitigation planning shall be developed when the risk is identified with the following requirements:

- a. Reduce High risks to Moderate risks.
- b. Reduce Moderate risks to Low risks.
- c. There shall be one or more contingency plans identified if the recommended mitigation plan fails to reduce the risk to the acceptable level at the required date.
- d. Mitigation plans shall demonstrate a methodical process and include measurable events and expected dates for reducing and/or controlling risk to acceptable levels.

4.10.4 Risk Monitoring

The contractor shall track the progress of mitigation plans by coordinating with risk owners or mitigation plan leads, providing regular reports of mitigation progress relative to plan, and supporting assessment of mitigation effectiveness throughout the mitigation plan process until plans are complete.

4.10.5 Risk Reporting

The contractor shall report on risks and risk management status using the figures provided in AR 385-10 and SMDC Tech Center Risk Management Plan Rev 1.0 as guidance. The reporting frequency and data shall be provided IAW CDRL A010.

4.11 Supply Services

The contractor shall develop, maintain, and periodically update Supply Chain Risk Management (SCRM) plans (incorporate this planning in CDRL A005). SCRM plans are intended to reduce performance and security risks of the products sold, installed, and maintained throughout the life-cycle. SCRM plans shall include sufficient detail for the Government to determine that the contractor reasonably understands its supply chain and the associated risks. The contractor shall ensure that genuine Information and Communication Technology (ICT) will be available under the contract and shall manage the risk to ensure counterfeit or illegally modified products are not shipped. SCRM plans shall describe the processes and practices ensuring contractors can deliver genuine ICT.

4.11.1 Mission Specific Stock Management

The contractor shall manage and maintain all RTS materials (spare parts, raw materials, consumables, and other general supplies) held at central warehouse or storage areas for subsequent issue to users. These material assets shall be controlled by the contractor's business system requirements driven by DFARS 252.242-7005 and the approved Material Management and Accounting System (MMAS) required by DFARS 252.242-7004. The contractor shall establish a system agreed to by the RTS Property Administrator on how to

track Government property items issued to individuals for use in contract performance. The contractor shall establish signature authority IAW AR 710-2 CDRL A038. The contractor shall provide personnel to support Freight Forwarding at Government directed locations, currently San Leandro, CA.

4.11.1.1 Annual Supply Budget Management

The contractor shall manage and control supply budget levels for procurement of stock and non-stock supplies. The contractor shall ensure that expenditures do not exceed authorized budget levels while optimizing the support provided IAW FAR 52.245-1. The contractor shall prepare and submit an annual RTS supply budget for Government approval.

4.11.1.2 Military Standard Requisitioning and Issuances Management

The contractor shall properly utilize Military Standard Requisitioning and Issue Procedures (MILSTRIP) IAW AR 725-50 for requisitions from/to Government sources through the Installation Supply Support Activity.

4.11.1.3 Military Standard Requisitioning and Issuances Priority

The contractor shall establish MILSTRIP requisition priority based on the Uniform Material Movement and Issue Priority System (UMMIPS) IAW AR 725-50 and designated priority designators 01, 04, and 11 and minimize use of priority 01 designator. The contractor

shall utilize appropriate priority designators. Priority designator 01 will not be used as a substitute for improper material management.

4.11.2 Government-Provided Warehouse(s) Supplies Management

The contractor shall manage Supplies within Government-provided warehouse(s) to include tool rooms on Kwajalein and Roi-Namur. All materials held at a Government-provided central warehouse or storage area for subsequent issue to users shall be managed for proper storage. The contractor shall coordinate with USAG-KA for supply movement support within RTS warehouses. Contractor Warehouse Supply manager(s) shall be responsible for RTS supply stock level management (to include ordering and issuance) and inventory of RTS supplies within Government-provided warehouse(s).

4.11.2.1 Warehouse Stock Level Surveys

The contractor shall perform warehouse stock level surveys as part of warehouse management for RTS supplies within Government-provided warehouse(s). The contractor shall meet applicable performance standards contained in AR 710-2, AR 710-2 (paragraph 4-28b), and AR 735-5.

4.11.2.2 Sensitive Items Inventories

The contractor shall complete a 100% inventory of Controlled Cryptographic Items (CCI) at least quarterly and submit results IAW AR 710-2 CDRL A010.

4.11.2.3 Shelf-Life Program Management

The contractor shall manage shelf-life for RTS supply commodities including (but not limited to): batteries, packaged petroleum, oils and lubricants, chemicals, hoses, O-rings, paints, sealants, and adhesives. The contractor shall inspect supplies to ensure sufficient shelf-life remains for supplies requiring shelf-life management IAW DoDM 4140.27. The contractor shall inspect all shelf-life items to ensure 85% of shelf-life remains for Shelf Ready Packaging (SRP) items IAW DoDM 4140.27.

4.11.3 Hazardous Materials Management

The contractor shall manage Hazardous Materials (HAZMAT) using the most current OSHA standard, OSHA Globally Harmonized System (GHS), and IAW AR 710-2 and the U.S. Army Logistics Information System (LIS).

4.11.3.1 Material Safety Data Sheet Maintenance

The contractor shall maintain Material Safety Data Sheet (MSDS)s for all RTS related HAZMAT on Kwajalein Atoll. The contractor shall ensure that MSDSs are available to support all operations involving hazardous materials IAW AR 710-2.

4.11.3.2 Shelf-Life Management in Storage

The contractor shall manage the shelf life for HAZMAT in storage IAW AR 710-2.

4.11.3.3 Turn-In

The contractor shall initiate processes for turn-in of HAZMAT IAW AR 710-2.

4.12 Quality

The contractor shall develop a Quality Control Plan (QCP) IAW CDRL A012 and maintain an effective quality control program to ensure services are performed in accordance with the contract and this PWS. The contractor shall develop and implement procedures to identify, prevent, and ensure non-recurrence of defective services. The contractor QCP assures the Government that work complies with the requirement of the contract.

5. SPECIAL REQUIREMENTS

5.1 Security

5.1.1 Management

5.1.1.1 Common Access Card Issuance

The contractor employee will be issued a Common Access Card (CAC) only if duties involve one of the following:

- a. Both physical access to a DoD facility and access, via logon, to DoD networks onsite or remotely.
- b. Remote access, via logon, to a DoD network using DoD-approved remote access procedures.
- c. Physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of six (6) months or more.

Before CAC issuance, the contractor's employee requires, at a minimum, a favorably adjudicated National Agency Check with Inquiries (NACI) or an equivalent or higher investigation in accordance with Army Directive ARMY (DIR) 2014-05.

At the discretion of the sponsoring activity, an initial CAC may be issued based on a favorable review of the Federal Bureau of Investigation (FBI) fingerprint check and a successfully scheduled NACI at the Office of Personnel Management.

Upon completion of the contractor's employment or termination of the contract itself, and to prevent delay in payment to the contractor, the contractor's program manager will ensure CACs are turned into the COR. Revoked or terminated Government credentials must immediately be returned to Government personnel as Government property.

5.1.1.2 Non-Common Access Card Access to DoD Facility or Installation

For contractors and subcontractors who do not meet the duty requirements for issuance of a CAC, but require access to a DoD facility or installation, the contractor and all associated subcontractor employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index and Terrorist Screening Database (TSDB) (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by Government representative), or at Outside Continental United States (OCONUS) locations, IAW status of forces agreements and other theater regulations.

5.1.1.3 Formal Operations Security Program

The contractor shall develop an Operations Security (OPSEC) Standing Operating Procedure (SOP)/Plan within 90 calendar days of contract award, to be reviewed and approved by the Government OPSEC officer responsible. This plan shall include a process to identify critical information, where it is located, who is responsible for it, how to protect it, and why it needs to be protected. The contractor shall implement OPSEC measures as ordered by the commander. In addition, the contractor shall have an identified certified level II OPSEC coordinator per AT 530-1.

5.1.1.4 General Protection/Security Policies and Procedures

The contractor and all associated sub-contractor employees shall, upon request, provide all information required for background checks to meet installation access requirements to be accomplished by installation Provost Marshall Office, Director of Emergency Services or Security Office. The contractor workforce shall comply with all personal identification verification requirements (FAR 52.204-9) as directed by DoD, HQDA and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

5.1.1.5 Information Assurance/Information Technology Certification

Per DoD 8570.01-M, DFARS 252.239-7001, and AR 25-2, the contractor employees supporting Cyber IA/IT functions shall be appropriately certified upon contract award. The baseline certification as stipulated in DoD 8570.01-M shall be completed upon contract award.

5.1.1.6 Contractor Access to Classified Information

The contractor shall comply with FAR 52.204-2, Security Requirements. This clause involves access to information classified "Confidential," "Secret," or "Top Secret" and requires contractors to comply with the Security Agreement (DD Form 441), including the National Industrial Security Program Operating Manual (DoD 5220.22-M); and any revisions to DoD 5220.22-M.

5.1.2 Classification

5.1.2.1 Facility Classification

The contractor shall comply with the requirements of the DoD Contract Security Classification Specification (DD Form 254) and shall utilize the SCGs provided by the Government for classification guidance. SCGs will be provided after contract award. The contractor shall maintain a TOP SECRET facility clearance.

5.1.2.2 Personnel Classification

The contractor shall have enough employees with a Personnel Security Clearance at the SECRET level to meet the requirements of tasks delineated in this PWS. All contractor personnel with access to unclassified information systems, including e-mail, shall have at a minimum a favorable National Agency Check with Inquiries (NACI).

5.1.2.3 Documentation Markings

The contractor shall mark documentation IAW DoDI 5230.24 for Distribution Statements and Export Control Warning. The Government will supply the required markings.

5.1.3 Operations

The contractor shall provide support for security management of RTS Operations.

5.1.3.1 Command Security Procedures

The contractor shall comply with all RTS Kwajalein Atoll, ROC-H, and USASMDC Command Security Procedures as required under this contract, IAW all applicable regulations, and under the direction and oversight of the cognizant RTS Security Manager and/or ROC-H Security Manager. Any questions or disputes regarding these procedures and the contractor's role in implementing them shall be reviewed by the Contracting Officer Representative (COR) and shall be forwarded to the contracting officer who will consult with the cognizant USASMDC G-2 official for resolution. The contractor shall assist and support USASMDC inspections as required.

The contractor shall ensure that the RTS Security Program receives a "pass" on all Department of the Army Inspector General (DAIG) and DoD IA (DIA) security inspections and Command Security Assistance Visits (SAVs).

5.1.3.2 Physical Security Assets and Infrastructure Security Policies

The contractor shall protect physical security assets IAW AR 380-5, AR 190-13,

AR 190-51, DoD Manual (DoDM) 5200.01 Volume 3, ATP 3-39.32, and local security requirements.

5.1.3.3 Security-Related Mandatory Training

The contractor shall ensure security-related mandatory training is accomplished for all contractor employees IAW applicable DoD and Army regulations. The contractor shall ensure mandatory training results are 100% compliance. The contractor shall report status of this training IAW CDRL A010.

5.1.3.4 Kwajalein Atoll Facility Access Control

The contractor shall provide access control for RTS Kwajalein Atoll facilities. Access control shall comply with the Federal Acquisition Regulation (FAR), the Security Agreement (DD Form 441), AR 380-5, AR 190-13, AR 190-51, and DoDM 5200.01 Volumes 1-4. The contractor shall ensure that the RTS Access control program receives a “pass” on all inspections.

5.1.3.5 Document Control

The contractor shall ensure all classified information / materials are properly accounted for, contain applicable markings and protected IAW AR 380-5, DoDM 5200.01 Volumes 1-4, and RTS Kwajalein Atoll and ROC-H Command Security Procedures. The contractor shall provide document control for RTS Government and contractor operations and process and prepare classified material for transportation / transmission as required by applicable Army regulations and policies.

The contractor shall ensure that the RTS Security Program receives a “pass” on all DAIG and DIA security inspections and Command Security Assistance Visits (SAVs).

5.1.3.6 Courier Authorizations

The contractor shall issue courier authorizations as required IAW the National Industrial Security Program Operating Manual (NISPOM) and AR 380-5. The contractor shall brief personnel on proper procedures for couriating classified information (local, CONUS, Outside CONUS). The contractor shall provide courier authorizations for contractor personnel only.

5.1.3.7 Security Self-inspections

The contractor shall implement a Government-approved self-inspection program to ensure

compliance with physical security requirements. The contractor shall conduct security self-inspections IAW applicable DoD, Army, USASMDC, USAG-KA, and RTS Kwajalein Atoll and ROC-H Security Procedures. The Government will conduct official inspections with assistance and support from the contractor.

5.1.3.8 Security Container Management

The contractor shall conduct security container management subject to the direction and oversight of the cognizant Security Manager. The contractor shall ensure that all security containers utilized to store classified information are General Service Administration (GSA) approved and certified. The contractor shall maintain/self-inspect all security containers IAW AR 380-5, DoDM 5200.01 Volumes 1-4, and RTS/USASMDC Security Procedures. The contractor shall conduct security container management at RTS Kwajalein Atoll and at ROC-H, subject to the direction and oversight of the RTS Kwajalein Atoll and ROC-H Security Managers. The contractor shall conduct combination changes as required under AR 380-5 at RTS Kwajalein Atoll and at ROC-H subject to the review and approval of the RTS and ROC-H Security Managers respectively.

The contractor shall ensure that the RTS Security Program receives a “pass” on all DAIG and DIA security inspections and Command Security Assistance Visits (SAVs).

5.1.3.9 Sensitive Compartmented Information / Special Access Program Management

The contractor shall operate and maintain a Sensitive Compartmented Information Facility (SCIF) on Kwajalein Atoll. The contractor shall designate a Special Security Representative (SSR) who will oversee the operations and maintenance of the RTS SCIF and the protection of SCI/ Special Access Program (SAP) information IAW the requirements of DoD 5105.21-M-1 and any other applicable regulations subject to the review and approval of the RTS Security Manager. The contractor shall provide SCIF/SAP operations and maintenance for RTS Kwajalein Atoll only.

The contractor shall ensure that the RTS SCIF is operated and maintained, and SCI/SAP is managed IAW applicable policies/ procedures on Kwajalein Atoll.

The contractor shall ensure that the RTS Security Program receives a “pass” on all DAIG and DIA security inspections and Command Security Assistance Visits (SAVs).

5.1.3.10 Contractor Personnel Security Management

The contractor shall ensure all contractor personnel (including visitors) who do not require a security clearance possess the appropriate adjudication IAW Homeland Security Presidential Directive (HSPD) 12.

The contractor shall initiate contractor investigations and contractor re-investigations as required, maintain contractor personnel security records and process outgoing contractor visit requests, and process incoming visitor requests for all RTS Kwajalein Atoll visitors and obtain Government need-to-know verification IAW the RTS Command Security Procedures.

5.1.3.11 Operations Security

The contractor shall designate an Operations Security (OPSEC) representative to serve as a member of the RTS OPSEC Working Group, conduct OPSEC for contract operations, and prepare and deliver an OPSEC plan IAW CDRL A009.

The contractor shall ensure that the RTS Security Program receives a “pass” on all DAIG and DIA security inspections and Command Security Assistance Visits (SAVs).

5.1.3.12 Safe and Vault Technician and Locksmith (RTS Kwajalein Atoll Only)

The contractor shall:

- a. Provide a GSA-certified safe and vault technician/inspector to perform servicing and inspection functions of GSA-approved security containers (provided to the contractor as GFE) and vault doors within RTS areas of responsibility. These services include installation, repairing, and removal of security containers and vault door locking mechanisms, certifying and de-certifying security containers and vault doors, and servicing of cipher locks on restricted areas.
- b. Provide 24 hours a day, seven days a week U.S. cleared locksmith service; maintain locks, panic hardware, and mechanical ciphers; maintain a master key system and provide keying services.
- c. Ensure that the RTS Security Program receives a “pass” on all DAIG and DIA security inspections and Command Security Assistance Visits (SAVs).

5.1.3.13 Destruction of Information

The contractor shall prepare, document and conduct the destruction of Controlled Unclassified Information (CUI), For Official Use Only (FOUO) and classified material as needed for RTS Kwajalein Atoll IAW AR 380-5, SMDC-Instruction Policy No. 380-30 and RTS Command Security Procedures, subject to the direction, control, review and/approval of the RTS Security Manager. NOTE: The current and future designation of documents and/or material as FOUO no longer exists and has been superseded by CUI. However, the Government acknowledges there are documents/materials still in circulation that were marked as FOUO in the past.

The contractor shall prepare and document ROC-H classified material for destruction in

accordance with the ROC-H Command Security.

The contractor shall ensure that the RTS Security Program receives a “pass” on all DAIG and DIA security inspections and Command Security Assistance Visits (SAVs).

5.1.3.14 Communications Security Management

The contractor shall operate and maintain a Communications Security (COMSEC) account and provide technical / administrative support for COMSEC equipment and materials, ensuring compliance with NSA, DoD 5220.22-M, and Army policies, including AR 380-40, NISPOM, and NSA/CSS Policy Manual No. 3-16. The contractor shall safeguard COMSEC facilities, materials and CCI IAW CNSSI No. 4005 and CNSSI No. 4001. Responsibilities include maintaining 100% accountability of COMSEC equipment and Controlled Cryptographic Items (CCI), managing inventories, re-keying circuits, distributing and transferring COMSEC keys, and supporting emergency circuit repairs. The contractor shall utilize tools such as the Key Management Infrastructure (KMI) and Simple Key Loader (SKL) to manage keying material and ensure proper storage and handling of COMSEC assets. The contractor shall deliver a Key Management Plan annually and participate in periodic and annual audits to verify compliance and accuracy. All COMSEC activities shall be documented and reported to ensure operational readiness and adherence to performance standards.

5.2 Training

The contractor shall ensure all contractor employees and subcontractor employees complete required training within this section of the PWS. The contractor shall report on training status of all employees and subcontractor employees monthly.

5.2.1 Anti-Terrorism Level I

All contractor employees, including subcontractor employees, requiring access to Army installations, facilities and controlled access areas shall complete Anti-Terrorism (AT) Level I Awareness training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable and within 5 calendar days of new employees commencing performance. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee to the COR or to the Procuring Contracting Officer (PCO), if a COR is not assigned, within 5 calendar days after completion of training by all employees and subcontractor personnel. AT Level I Awareness training is available at <https://jko.jten.mil>.

5.2.2 Anti-Terrorism Awareness for Contractor Personnel Traveling Overseas

All U.S.-based contractor employees, including subcontractor employees, required to travel overseas as a condition of this contract, shall receive AT Awareness training specific to the

destination countries as directed by AR 525-13. Additionally, all contractor employees, including subcontractor employees, stationed or deployed in OCONUS contingency operations areas or designated combat zones shall receive Government-provided AT Awareness training specific to the AOR.

5.2.3 iWATCH

The contractor and all associated subcontractors shall brief all employees on the local iWATCH program (training standards provided by the requiring activity Anti-Terrorism Officer (ATO)). This locally-developed training shall be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within 30 calendar days of contract award and within 5 calendar days of new employees commencing performance with the results reported to the COR no later than 30 calendar days after contract award.

5.2.4 Army Training Certification Tracking

All contractor employees with access to a Government Information System (IS) must be registered in an Army Training Tracking System. The Army Training Certification Tracking System (ATCTS) is currently being replaced; however, this requirement will exist at commencement of services. All IS access requests shall successfully complete the DoD Cyber Awareness Challenge training prior to access to the IS and then annually thereafter.

5.2.5 Operations Security

All contractor employees shall complete Level I OPSEC Awareness training IAW AR 530-1. New employees shall be trained within 30 calendar days of their reporting for duty and annually thereafter. The contractor shall report on the status of this training. OPSEC Awareness for military members, DoD employees, and contractors is available at the following website: <https://securityawareness.dcsa.mil/opsec/index.htm>

5.2.6 Information Assurance/Information Technology

All contractor employees and associated sub-contractor employees must complete the DoD IA /Cyber Awareness training before issuance of network access and annually thereafter. All contractor employees working IA/IT functions shall comply with DoD and Army training requirements in DoDD 8140.01, DoD 8570.01-M, and AR 25-2 within six months of appointment to IA/IT functions. DoD Cyber Awareness Challenge Training is available at the following website: <https://cs.signal.army.mil/login.asp>

5.2.7 Threat Awareness Reporting Program

All contractors with security clearances shall receive annual Threat Awareness and Reporting Program (TARP) training by a Counterintelligence (CI) agent or other trainer IAW AR 381-12, para. 2-4b.

5.3 Industrial Safety

5.3.1 Contractor Compliance

The contractor and its subcontractors shall comply with Public Law 91-596, DoDD 4715.1E, 29 CFR Chapter XVII Part 1925, and applicable DoD and Army regulations and policies and shall ensure these requirements are incorporated into the contractor's safety and health program. The contractor shall ensure that managerial actions/decisions are timely in resolving and responding to Safety and Occupational Health issues.

The contractor shall ensure total injuries, illness rates, and lost workday rates are at or below the latest yearly Bureau of Labor Statistics (BLS) Private Industry Rate.

The contractor shall ensure that any Class A or B accidents are not the result of contractor industrial safety program inadequacies of standards, training, and /or supervision.

5.3.2 Accident/Incident Investigations Coordination

The contractor shall perform investigations, in coordination with the RTS Director, on all accidents/incidents related to the contractor's operations which result in personal injury, illness, death or property damage, or other incidents with high command visibility, to include, but not limited to, failure review boards and safety review boards. The contractor shall submit Accident/Incident Report IAW CDRL A008. The contractor shall also fully cooperate with U.S. Government investigating officers and/or criminal investigators at the direction of the COR or PCO.

5.3.3 Mishap Notification and Investigation

The contractor and its subcontractors (if applicable) shall promptly report pertinent facts regarding mishaps involving U.S. Government property damage or injury to U.S. Government personnel and will cooperate in any resulting safety investigation. The contractor shall notify (via telephone) the cognizant contracting officer, COR, and/or other applicable members within four hours of all mishaps or incidents. The U.S. Government person notified by the contractor will in-turn notify the RTS Mission Safety Division. Contractor notifications made after duty hours shall be reported to the appropriate PCO/ COR, USAG-KA Command Post. If requested by the cognizant PCO, COR, and/or the cognizant program manager, the contractor shall immediately secure the mishap scene/damaged property and impound pertinent maintenance and training records until released by the investigating safety office. If the U.S. Government investigates the mishap, the contractor and the subcontractors shall cooperate fully and assist the U.S. Government personnel until the investigation is completed. The U.S. Government reserves the right to investigate any accidents that cause damage to U.S. Government property, equipment, materials, cause injury to U.S. Government employees, or have potential for negative

publicity. The contractor shall ensure all accidents are reported and properly logged IAW OSHA requirements. A copy of the OSHA Form 300 log shall be provided annually to the RTS Safety Office.

5.3.4 Inspections

The contractor shall allow U.S. Government safety representatives access to workplaces to conduct compliance inspections at frequent, regular and irregular intervals, with and without prior coordination. On the spot corrections for compliance failures do not require PCO/COR approval before corrections are implemented, the PCO/COR shall be notified by the contractor anytime there is a compliance issue identified and a correction made. The cognizant Safety Office will also contact the PCO/COR within 24 hours of making any on the spot corrections to ensure continued compliance and ensure PCO/COR awareness of safety and health requirements. The contractor shall annotate all compliance issues on a violation inventory log and track through completion of repair, correction, or compliance.

5.3.5 Safety Plans, Reports and Procedures

The contractor shall provide an Industrial Safety plan that addresses every area and safety requirement including, but not limited to, safety program elements for operations in: Accident Investigation, Marine, Calibration, Explosives, Electrical, Hazard Communication, Hazardous Materials Management, Industrial, Inspections, Job Hazard Analysis, Lock Out-Tag out, Material Handling, Personal Protective Clothing and Equipment, Radiation Safety, Respiratory Protection, Risk Management, Safety Awareness, and any other areas included in this contract IAW CDRL A006.

The contractor shall develop and maintain a hazard tracking system.

The contractor shall develop hazard abatement plans for issues that will take over 30 calendar days to resolve. The contractor shall submit a Hazard Tracking, Abatement Violation Inventory Report IAW CDRL A007.

The contractor shall perform job safety/hazard analyses for all contractor operations IAW OSHA Publication 3071.

5.3.6 Rescue Services for RTS

The contractor shall provide tower climber rescue support in order to safely perform PWS required maintenance tasks at heights per OSHA and ANSI Z359.2/ANSI Z359.4 standards.

5.4 Environmental Compliance

The contractor shall comply with the current USAG-KA Environmental Standards (which are updated annually) and all documents referenced under paragraph 3.0 of this

PWS as mandatory. Compliance with documents listed as non-mandatory is at the contractor's discretion.

5.5 Data Rights Assertions / Inventions / Proprietary Information

5.5.1 Contract Data Requirement List

The contractor shall prepare and deliver to the Government technical data and computer software in accordance with the requirements and schedules set forth in the CDRL.

5.5.2 Assertions

The contractor shall identify and assert any restrictions or limitations on the Government's use, release, or disclosure of technical data or computer software pertaining to the requirements of this PWS. These assertions shall be included in any proposals, including responses to each technical directive received from the Government. The contractor shall comply with the Instructions Pertaining to Assertions Regarding Data Rights and Intellectual Property as provided in Exhibit B in Section J of the contract. The contractor's assertions, including the assertions of its subcontractors or suppliers, shall be submitted to the Contracting Officer for review. If no assertions are made, the contractor shall state "None", sign and date.

5.5.3 Reporting Subject Inventions

In accordance with DFARS 252.227-7038 (large business) and FAR 52.227-11 / DFARS 252.227-7039 (small business and nonprofit), the contractor shall disclose and report via DD Form 882 any "subject inventions" (as defined in the clauses) to the administrative contracting officer / procuring contracting officer (ACO/PCO). Additional instructions are provided in Exhibit C in Section J of the contract. Annual and final reports (including negative reports) are required of the contractor and its subcontractors.

5.5.4 Proprietary Markings on Non-Technical Data

The contractor shall notify the contracting officer of any deliverables / CDRLs that may contain "non-technical data" information ("technical data" is defined in DFARS 252.227-7013(a)(15)) that the contractor intends to assert and mark as "proprietary." The contractor shall identify the specific information and provide justification for its "proprietary" claim. The contractor and the contracting officer shall agree upon permitted uses and disclosure of such information and the contractor agrees to mark such information with the following legend: "Proprietary – Special License: Use and disclosure is permitted within the Government for Government Purposes (as defined in DFARS 252.227-7013(a)(12)). All other use and disclosure require a non-disclosure agreement.

5.5.5 Special Works

As identified by the COR for each technical directive, software developed incidental to the performance of this contract shall be delivered to the Government as both source and executable code in accordance with DFARS 252.227-7014, Rights in Noncommercial Computer Software and Noncommercial Computer Software Documentation; as appropriate, all other deliverables where the Government must own the copyright shall be considered a "Special Work" pursuant to DFARS 252.227-7020, Rights in Special Works.

5.5.6 Deferred Ordering

The contractor shall maintain a list of all technical data and computer software deliverables and other items generated in the performance of this contract or any subcontract. This data shall be available upon request in accordance with the deferred ordering clause (DFARS 252.227-7027, Deferred Ordering of Technical Data or Computer Software).

5.6 Government Access/Use

The contractor shall ensure that any non-Government provided Information System used for the management of contract assets, and in direct support of PWS requirements, can either be hosted on a Government provided .mil network or accessible by internet via a Government provided .mil network. The contractor shall ensure that all contractor data systems operating on Government provided .mil networks are compliant IAW DODI 8500.01, AR 25-2, and DISA standards.

The contractor shall ensure that applications that perform corporate functions, including, but not limited to, corporate processes which process or store contractor employee Personally Identifiable Information (PII), Protected Health Information (PHI), timekeeping, payroll, recruiting, and Human Resources functions, are hosted on corporate networks. Hosting these functions on any Government provided .mil network will not be allowed.

The contractor shall provide use of, and access to, their contract asset management system, to the Government and to other entities as designated by the Government. Upon request of the Government, the contractor shall provide copies of the licenses of any commercial data software and/or open-source software which were acquired for use on this contract. The contractor will ensure that software licenses are transferrable to the Government or to other entities as designated by the Government. The contractor shall not affix its corporate logos or any unauthorized and nonconforming markings on any deliverables made to the Government.

5.7 Contract Performance or Delivery in a Foreign Country

The contractor and its subcontractors, if performing or traveling outside the United States under this contract, shall comply with the requirements of DFARS 252.225-7043, with the

exceptions for certain subcontractors as noted in that clause.

6. DELIVERABLES

The contractor shall provide deliverables as outlined below. Format and delivery schedule for deliverables shall be outlined in the indicated CDRL.

Number	Name	Frequency
A001	Annual Operating Budget (AOB) Report	Annually
A002	Contract Work Breakdown Structure / Dictionary	Initial/Update AS REQ
A003	Budget, Cost and Variance Report	Monthly
A004	Customer Reimbursable Cost Report	Monthly
A005	Contractor's Risk Management Plan	Initial/Update AS REQ
A006	Industrial Safety Plan	Initial/Update AS REQ
A007	Hazard Tracking & Abatement and Violation Inventory Report	Monthly
A008	Accident / Incident Report	As required
A009	Operations Security (OPSEC) Plan	Initial/Update AS REQ
A010	Monthly Activities Report	Monthly
A011	Transition/Phase-Out Plan	Initial/Update AS REQ
A012	Quality Assurance Program Plan	Initial/Update AS REQ
A013	Technical Document Library Index	As required
A014	Mission Briefing Package	Per mission, Update AS REQ
A015	Range User's Manual (RUM)	Annually
A016	Data Item Description Manual	Annually
A017	Operations Supported Document	Annually
A018	Program Cost Estimates Report	As required
A019	Planning and Operations Schedule	Weekly
A020	Space Surveillance Data Report	Weekly
A021	Quicklook Report	As required
A022	Mission Data Report	As required

A023	Instrumentation and Network Availability Report	Annually
A024	Sensor and Network Availability Plan	Bi-Annual
A025	Instrumentation Status and Weekly Activity Report	Weekly
A026	Geodetic Coordinates Manual	Annually
A027	Plant-In-Place Drawings and Circuit Layout Record	Annually
A028	Instrumentation Modification Final Report	One per Modification
A029	Studies and Task Report	As Required
A030	Ordnance Inventory Report	Monthly
A031	Engineering Project Estimate	One per Project
A032	Extended Areas of Support Report	One per Support Activity
A033	RF Survey Database Report	Annually
A034	RF Incident Analysis Report	Within 24 hrs. of incident, 7 day follow up
A035	RF Safety Assessment Report	Within 7 days completion of Mission Support
A036	Instrumentation and Support Facilities Manual (ISFM)	First submission 12 months ACA
A037	Technical Facilities and Equipment Manual (TFEM)	Annually
A038	Instrumentation Supply Plan	Annually
A039	TMDE Delinquency Report	Monthly
A040	Engineering Change Proposal	One per Project
A041	Project / Study / Task Final Report	One per Project`

A042	Configuration Management Plan	One time with revisions
A043	IT Service Desk Trend Analysis Report	Quarterly
A044	Annual Work Plan List	Annually
A045	Mission Test Plan	One per Mission
A046	Range Safety Mission Support Plan	One per Mission
A047	Range Safety Readiness Report	One per Mission
A048	Range Safety Countdown Procedures	One per Mission
A049	Mission Assessment Report	One per Mission
A050	Range Safety Assessment Report	One per Mission
A051	Space Operations Maintenance Report	Monthly
A052	Contract Funds Status Report (CFSR)	Quarterly
A053	Information Technology Activity Report	Monthly
A054	Information Technology Analysis Report	Monthly
A055	Information Management System (IMGTS) Software List	Annually
A056	Information Management System (IMGTS) Hardware List	Annually
A057	Information Management System (IMGTS) Tech Refresh Plan	Annually
A058	Annual Preventative Maintenance Plan	Annually
A059	Integrated Master Schedule	Monthly
A060	State of the Range Briefing	Annually
A061	Cybersecurity Activity Report	Monthly

A062	Cybersecurity Analysis Report	Quarterly
A063	Security Impact Analysis Report	As Required
A064	Vulnerability Management Plan	Annually
A065	RMF Authorization Package	Bi-Annually
A066	Network Vulnerability Scans	Monthly
A067	Application Static Code Analysis Scans	As Required
A068	Software Assurance Evaluation Report	As Required
A069	STIG Report	Quarterly
A070	STIG Scans	Quarterly
A071	Cybersecurity Vulnerability Report	Quarterly
A072	Continuous Monitoring Plan	Annually
A073	Warranties/Software License/Service Agreement Report	Bi-Annually
A074	Configuration Item Recommendation Plan	Annually
A075	Obsolescence Surveillance Report	Annually
A076	Software Development Plan	As Required
A077	Space Operations Operator Training Plan	Annually
A078	Space Operations Operator Certifications	Semi-Annual

7. GOVERNMENT FURNISHED ASSETS

Government property that is incidental to the place of performance such as office space, desks, chairs, telephones, computers, and fax machines, as authorized by FAR 45.000, shall be made available for contractor use and will not be considered GFP. Provided Incidental assets will be made available on the dates specified (see Attachment 13) for use by the contractor in the performance of this contract.

8. CONTRACT MANPOWER REPORTING

8.1 Reporting Process

The contractor shall provide all Contractor Manpower Reporting (CMR) labor hours (including subcontractor labor hours) required for performance of services provided under this contract for RTS via a secure data collection site, if applicable. The contractor is required to completely fill in all required data fields using the following web address: <https://sam.gov>.

8.2 Reporting Details

The contractor's reporting inputs shall be for the labor executed during the period of performance during each Government FY, which runs October 1 through September 30. While the contractor may report inputs at any time during the FY, the contractor shall report all data no later than October 31 of each calendar year, if applicable.

9. APPLICABLE DIRECTIVES

Documents referenced within the paragraphs of the PWS are identified by number, title, date, and category in the Document Summary List (DSL), an attachment to the contract. The document versions specified in the DSL take precedence over the generic references (without revision letters) cited in this PWS. The contractor is required to consult the most current documents available in performing the requirements of this PWS.

10. IDENTIFICATION OF CONTRACTOR EMPLOYEES

All contract personnel attending meetings, answering Government telephones, sending e-mails, and working in other situations where their contractor status is not obvious to third parties, are required to identify themselves as such to avoid creating an impression that they are Government officials. Contractor personnel shall be required to obtain and wear appropriate badges in the performance of this contract.

11. CONTRACTOR TELEWORK

An on-site contractor employee may be authorized to telework or work from his/her company home-based office with prior written approval (e-mail acceptable) from the KO in the case of unexpected closure of a Government facility due to 1) inclement weather or extraneous circumstances or 2) on a recurring or situational basis (such as temporary closure directed by the appropriate Government official due to an act of the Government in its sovereign or contractual capacity or such events as natural disasters, epidemics, quarantine restrictions, etc.). Approval is limited to on-site contractor employees and subject to the contractor's company policies permitting such arrangements. Approval is not applicable in the case of closures due to a lapse in funding appropriations; in such cases the contractor shall seek official guidance from the authorized KO. "On-site contractor employee" is defined as a contract employee whose place of duty performance is a government facility per the terms and conditions of a prime contract or flow down subcontract.

12. ACRONYMS AND ABBREVIATIONS LIST

<u>Acronym/Abbreviation</u>	<u>Description</u>
AAA	Authentication, Authorization, and Accounting
A&A	Assessment and Authorization
AAR	After Action Review
ABM	Anti-Ballistic Missile
ALCOR.....	ARPA/Lincoln C-Band Observable Radar
ALTAIR	ARPA Long Range Tracking and Instrumentation Radar
Ao	Operational Availability
AOB	Annual Operating Budget
AOR	Area of Responsibility
AR.....	Army Regulation
ARPA	Advanced Research Projects Agency
AT	Anti-Terrorism
ATCTS	Army Training Certification Tracking System
ATO	Authority to Operate
ATP.....	Army Techniques Publication
AWP.....	Annual Work Plan
BER	Beyond Economic Repair
BLS	Bureau of Labor Statistics
C4IM	Command, Control, Communications, Computers, and Information Management
CAC	Common Access Card
CCB	Configuration Control Board
CCI.....	Controlled Cryptographic Items
CDRL	Contract Data Requirements List
CERTs.....	Certification
CFR	Code of Federal Regulations
CFSR	Contract Fund Status Report
CG	Coast Guard
CI	Configuration Item
CJCSI	Chairman of the Joint Chiefs of Staff Instruction
CJCSM	Chairman of the Joint Chiefs of Staff Manual
CLR.....	Circuit Layout Record
CMP.....	Configuration Management Plan
CMR.....	Contract Manpower Reporting
COMSEC	Communications Security
CONUS	Continental United States
COR.....	Contracting Officer's Representative
COTS	Commercial Off-The-Shelf
CRL	Certificate Revocation List
CSDR.....	Cost and Software Data Reporting Plan
CS	Cyber Security
CSS	Central Security Service
CSpOC.....	Combined Space Operations Center
CTO	Cybersecurity Tasking Order

CUI.....	Controlled Unclassified Information
CWBS	Contract Work Breakdown Structure
DA.....	Department of the Army
DA PAM	DA Pamphlet
DAIG	Department of the Army Inspector General
DEP	Documents of Environmental Protection
DFARS	Defense Federal Acquisition Regulation Supplement
DIA.....	DoD Information Assurance
DIR.....	Directive
DISA	Defense Information Systems Agency
DISAC	DISA Circular
DLA	Defense Logistics Agency
DoD.....	Department of Defense
DoDD	Department of Defense Directive
DoDI.....	Department of Defense Instruction
DoDM.....	Department of Defense Manual
DREN.....	Defense Research and Engineering
DSL	Document Summary List
ECP	Engineering Change Proposal
eMASS.....	Enterprise Mission Assurance Support Services
EMS	Environmental Management System
EMP	Extended Maintenance Period
EPE.....	Engineering Project Estimate
EQCC.....	Environmental Quality Control Committee
FAR.....	Federal Acquisition Regulation
FBI	Federal Bureau of Investigation
FFRDC.....	Federally Funded Research and Development Center
FISMA	Federal Information Security Management Act
FOC	Full Operational Capability
FPCON	Force Protection Condition
FTSO	Flight Termination System Officer
FY	Fiscal Year
GFE	Government Furnished Equipment
GFI.....	Government Furnished Information
GFP	Government Furnished Property
GHS	Globally Harmonized System
GOTS.....	Government Off-the-Shelf
GSA	General Services Administration
HAZMAT	Hazardous Materials
HAZOP.....	Hazardous Operation
HMCC	Huntsville Mission Control Center
HPP	Historic Preservation Plan
HSPD	Homeland Security Presidential Directive
HSSC.....	Huntsville Space and Surveillance Center
I&M	Improvement & Modernization
IA	Information Assurance
IAVA	Information Assurance Vulnerability Alert

IAW	In Accordance With
IMGTS	Information Management System
IMS	Integrated Master Schedule
IOC	Initial Operational Capability
IPT	Integrated Product Team
IS.....	Information System
ISFM	Instrumentation Support Facilities Manual
ISSM	Government Information System Security Manager
ISSO	Information System Security Officer
IT.....	Information Technology
ITSM	IT Service Management
JWICS.....	Joint World-Wide Intelligence Communications Systems
KEEP	Kwajalein Environmental Emergency Plan
KMCC	Kwajalein Mission Control Center
KMISS.....	Kwajalein Missile Impact Scoring System
KMRSS	Kwajalein Mobile Range Safety System
KREMs.....	Kiernan Reentry Measurements System (KREMS)
LIS	Logistics Information System
LSSC.....	Lincoln Space Surveillance Center
MASINT	Measurement Signature Intelligence
MDN.....	Mission Data Network
MFCO	Missile Flight Control Officer
MILSTRIP	Military Standard Requisitioning and Issue Procedures
MIT/LL.....	Massachusetts Institute of Technology / Lincoln Lab
MMAS	Material Management and Accounting System
MMW.....	Millimeter Wave Radar
MOA.....	Memorandum of Agreement
MOU	Memorandum of Understanding
MRTFB.....	Major Range and Test Facility Base
MSDS.....	Material Safety Data Sheet
MSS	Meteorological Support Service
MTBF	Mean Time Between Failures
MTTR.....	Mean Time To Repair
NACI	National Agency Check with Inquiries
NEPA	National Environmental Policy Act
NETCOM	U.S. Army Network Enterprise Technology Command
NIPR.....	Non-secure Internet Protocol Router
NISPOM	National Industrial Security Program Operating Manual
NIST.....	National Institute of Standards and Technology
NSA	National Security Agency
NSDC.....	National Space Defense Center
NSIC.....	National Space Intelligence Center
NSS	National Security Systems
OCI	Organizational Conflict of Interest
OCONUS	Outside Continental United States
OD	Operations Directive
ODC	Ozone Depleting Chemicals

ODIN.....	Operational Data Integrated Node
OI.....	Operating Instructions
OPSEC	Operations Security
OR	Operations Requirement
OSHA.....	Occupational Safety and Health Administration; Occupational Safety and Health Act
PCB	Polychlorinated Biphenyl
PCO	Procuring Contracting Officer
PHI.....	Protected Health Information
PLCO	Plant Clearance Officer
PIEE	Procurement Integrated Enterprise Environment
PI	Program Introduction
PII.....	Personally Identifiable Information
POA&M	Plan of Action and Milestones
POE	Primary Organizational Element
PoP	Period of Performance
PPSM	Ports, Protocols and Services Management
PRD	Program Requirements Document
PSP.....	Program Support Plan
PWS.....	Performance Work Statement
QASP	Quality Assurance Surveillance Plan
QCP	Quality Control Plan
RADMON	Radiation Monitoring
R&M	Reliability and Maintainability
RCC	Range Commander Council
RCS	Radar Cross Section
RDAC	RTS Data Analysis Center
RDO.....	Range Distributed Operations
RDT&E	Research, Development, Test, and Evaluation
REC	Records of Environmental Consideration
ROSIE	RTS Engineering and Technical Services
RF	Radio Frequency
RMF	Risk Management Framework
RMI	Republic of the Marshall Islands
ROC-H	RTS Operations Center- Huntsville
RSCC.....	Range Safety Control Center
RSS	Range Safety System
RSSS	Range Safety Support Services
RTS.....	Reagan Test Site
RUM.....	Range User's Manual
SAP.....	Special Access Program
SAR.....	Security Assessment Report
SATCOM.....	Satellite Communications
SAV.....	Security Assistance Visit
SC	Statement of Capability
SCG	Security Classification Guidelines
SCI.....	Sensitive Compartmented Information

SCIF.....	Sensitive Compartmented Information Facility
SDA.....	Space Domain Awareness
SDS	Space Defense Squadron
SEI&T.....	Systems Engineering, Integration, and Test
SIPR.....	Secret Internet Protocol Router
SLA	Service Level Agreement
SMDC	United States Army Space and Missile Defense Command
SOP	Standing/Standard Operating Procedure
SRGs	Security Requirements Guides
SRP	Shelf Ready Packaging
SOPS	Space Surveillance Operations
SSR	Special Security Representative
STIGS	Security Technical Implementation Guides
TARP.....	Threat Awareness and Reporting
TD.....	Technical Directive
TDY.....	Temporary Duty
TFEM	Technical Facilities and Equipment Manual
TIP	Tactical Imaging Processor
TMDE.....	Test, Measurement, and Diagnostic Equipment
TMS	Test Mission Support
TOP	Tactical Observation Processor
TS	Top Secret
TRADEX	Target Resolution Discrimination Experiment
TSDB	Terrorist Screening Database
UDS	Universal Documentation System
UES	USAG-KA Environmental Standards
UID	Unique Identification
UMMIPS.....	Uniform Material Movement and Issue Priority System
UPS	Uninterruptable Power Supply
U.S.	United States
USAG-KA.....	U.S. Army Garrison – Kwajalein Atoll
USAKA.....	U.S. Army Kwajalein Atoll
USASMDC	U.S. Army Space and Missile Defense Command
USAV	U.S. Army Vessel
USSPACECOM..	U.S. Space Command
VMP	Vulnerability Management Plan
VTC.....	Video Teleconference
WBS.....	Work Breakdown Structure

- END OF DOCUMENT -